TECHNOLOGY

The City of Domes : a walk with an architect about the courts and palaces of the

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The City of Domes

A Walk with an Architect About the Courts and Palaces of the Panama Pacific International ExposItion with a Discussion of Its Architecture - Its Sculpture - Its Mural Decorations Its Coloring - And Its Lighting - Preceded by a History of Its Growth

by John D. Barry

To the architects, the artists and the artisans and to the men of affairs who sustained them in the cooperative work that created an exposition of surpassing beauty, unique among the expositions of the world.

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Preface

In the main, this volume consists of articles originally published in the San Francisco BULLETIN. It includes material gathered from many visits to the Exposition grounds and from many talks with men concerned in the organization and the building and ornamentation. The brief history that forms the Introduction gives an account of the development. For me, as, I presume, for most people, the thing done, no matter how interesting it may he, is never so interesting as the doing of the thing, the play of the forces behind. Even in the talk with the architect, where the finished Exposition itself is discussed, I have tried to keep in mind those forces, and wherever I could to indicate their play.

The dialogue form I have used for several reasons: it is easy to follow; it gives scope for more than one kind of opinion; and it deals with the subject as we all do, when with one friend or more than one we visit the Exposition grounds. It has been my good fortune to he able to see the Exposition from points of view very different from my own and much better informed and equipped. I am glad to pass on the advantage.

The Exposition is generally acknowledged to be an achievement unprecedented. Merely to write about it and to try to convey a sense of its quality is a privilege. I have valued it all the more because I know that many people, not trained in matters of architecture and art, are striving to relate themselves to the expression here, to understand it and to feel it in all its hearings. If, at times, directly or in indirectly, I have been critical, the reason is that I wished, in so far as I could, to persuade visitors not to swallow the Exposition whole, but to think about it for themselves, and to bear in mind that the men behind it, those of today and those of days remote, were human beings exactly like themselves, and to draw from it all they could in the way of genuine benefit.

Though the volume is mainly devoted to the artistic features associated with the courts and the main palaces, I have included, among the illustrations, pictures of the California Building, both because of its close relation to California and because it is in itself magnificent, and of two notable art features, the mural painting by Bianca in the Italian Building, and "The Thinker", by Rodin, in the court of the French Pavilion.

Introduction

The First Steps

In January, 1904, R. B. Hale of San Francisco wrote to his fellow-directors of the Merchants' Association, that, in 1915, San Francisco ought to hold an exposition to celebrate the opening of the Panama Canal. In the financing of the St. Louis Exposition, soon to begin, Mr. Hale found a model for his plan. Five million dollars should be raised by popular subscription, five million dollars should be asked from the State, and five million dollars should be provided by city bonds.

The idea was promptly endorsed by the business associations.

From their chairmen was formed a board of governors. It was decided that the exposition should be held, and formal notification was given to the world by introducing into Congress a bill that provided for an appropriation of five million dollars. The bill was not acted on, and it was allowed to die at the end of the session.

Soon after formulating the plan for the exposition Mr. Hale changed the date from, 1915 to 1913, to make it coincide with the four hundredth anniversary of the discovery by Balboa of the Pacific.

In 1906 came the earthquake and fire. The next few years San Franciscans were busy clearing away the debris and rebuilding. It was predicted that the city might recover in ten years, and might not recover in less than twenty-five years.

Nevertheless, in December, 1906, within nine months of the disaster, a meeting was held in the shack that served for the St. Francis Hotel, and the Pacific Ocean Exposition Company was incorporated.

In three years the city recovered sufficiently to hold a week's festival, the Portola, and to make it a success.

Two days afterward, in October, 1909, Mr. Hale gave a dinner to a small group of business men, and told of what had been done toward preparing for the Exposition. They agreed to help.

Shortly afterward a meeting was held at the Merchants' Exchange. It was decided that an effort should at once be made to raise the money and to rouse the people of San Francisco to the importance of the project of holding the Panama-Pacific International Exposition in San Francisco in 1915.

As many as twenty-five hundred letters were sent to business men, asking if they favored the idea of holding an exposition. Out of about eight hundred replies only seven were opposed. Presently there were signs of enthusiasm, reflected in the newspapers.

A committee of six representative business men was appointed and the announcement was made that the committee should be glad to hear from anyone in the city who had suggestions or grievances. It was determined that every San Franciscan should have his day in court.

Later the committee of six appointed a foundation committee of two hundred, representing a wide variety of interests.

The committee of two hundred chose a committee of three from outside their number.

The committee of three chose from among the two hundred a directorate of thirty. The thirty became the directorate of a new corporation, made in 1910, the Panama-Pacific International Exposition Company.

Financing

The Panama-Pacific Company two local millionaires, W. H. Crocker and W. B. Bourn, started financially with twenty-five thousand dollars each. They established

the maximum individual subscription. They also secured forty subscriptions of twenty-five thousand dollars each. Then followed the call for a mass meeting. Before the meeting was held the business men of the city were thoroughly canvassed. The Southern Pacific and the Union Pacific together subscribed two hundred and fifty thousand dollars. There were many other large subscriptions from public-service organizations.

On the afternoon of the meeting there was a crowd in the Merchants' Exchange Board Room. The announcement of the subscriptions created enthusiasm. In two hours the amount ran up to more than four million dollars. During the next few years they were increased to about $6,500,000.

Meanwhile, the State voted a tax levy of five million dollars, and San Francisco voted a bond and issue of the same amount, and by an act of the Legislature, in special session, the counties were authorized to levy a small tax for county Participation, amounting, in estimate, to about three million dollars.

Recognition From Congress

Next came the task of securing from Congress official recognition of San Francisco as the site of the International Exposition in celebration of the Panama Canal.

Headquarters were established in Washington. Presently serious opposition developed. Emissaries went from San Francisco to Washington singly and in delegations. Stress was laid on San Francisco's purpose not to ask for an appropriation from the national government. There were several cities in competition - Boston, Washington, Baltimore and New Orleans. New Orleans proved the most formidable rival. It relied on the strength of of a united Democracy and of the solid South.

In the hearings before the Congressional Committee it was made plain that the decision would go to the city with the best financial showing. As soon as the decision was announced New Orleans entered into generous cooperation with San Francisco.

The Exposition was on the way.

Naming the President.

The offer of the presidency of the Exposition Company was made to a well-known business man of San Francisco, C. C. Moore. Besides being able and energetic, he was agreeable to the factions created by the graft prosecution of a half dozen years before. Like the board of directors, he was to serve without salary. He stipulated that in the conduct of the work there should be no patronage. With the directors he entered into an a agreement that all appointments should be made for merit alone.

Choosing the Site

The choice of site was difficult. The sites most favored were Lake Merced, Golden Gate Park and Harbor View. Lake Merced was opposed as inaccessible for the transportation both of building materials and of people, and, through its inland position, as an unwise choice for an Exposition on the Pacific Coast, in its nature supposed to be maritime. The use of the park, it was argued, would desecrate the

peoples recreation ground and entail a heavy cost in leveling and in restoring.

Harbor View and the Presidio had several advantages. It was level. It was within two miles or walking distance of nearly half the city's inhabitants. It stood on the bay, close to the Golden Gate, facing one of the most beautiful harbors in the world, looking across to Mount Tamalpias and backed by the highest San Francisco hills. Of all the proposed sites, it was the most convenient for landing material by water, for arranging the buildings and for maintaining sanitary conditions.

After a somewhat bitter public controversy the Exposition directors, in July, 1911, announced a decision. It caused general surprise. There should be three sites: Harbor View and a strip of the adjoining Presidio, Golden Gate Park and Lincoln Park, connected by a boulevard, specially constructed to skirt the bay from the ferry to the ocean.

That plan proved to be somewhat romantic. The boulevard alone, it was estimated, would cost eighteen million dollars.

Harris D. H. Connick, the assistant city engineer was called on as a representative of the Board of Public Works, and asked to make a preliminary survey of Harbor View. He showed that, of the proposed sites, Harbor View would be the most economical. The cost of transporting lumber would be greatly reduced by having it all come through the Golden Gate and deposited on the Harbor View docks. The expense of filling in the small ponds there would be slight in comparison with the expense of leveling the ground at the park.

A few weeks later Harbor View and the Presidia was definitely decided on as the site, and the only site.

For months agents had been at work securing options on leases of property in Harbor View, covering a little more than three hundred acres, the leases to run into December 1915. Reasonable terms were offered and in one instance only was there resort to condemnation. The suit that followed forced the property owner, who had refused fifteen hundred dollars, to take nine hundred dollars. President Moore was tempted to pay the fifteen hundred dollars, but he decided that this course would only encourage other property owners to be extortionate. Some trouble was experienced with the Vanderbilt properties, part of which happened to be under water. After considerable negotiating and appeals to the public spirit of the owners, it was adjusted. About seven hundred thousand dollars was paid for leases and about three hundred thousand dollars for property bought outright.

The Director of Works

While President Moore was looking for the man he wanted to appoint as head of the board of construction, Harris D. H. Connick called to suggest and to recommend another man. Later the president offered Connick the position as director of works.

Connick had exactly the qualifications needed: experience, youth, energy, skill and executive ability. He hesitated for the reason that he happened to be engaged in public work that he wished to finish. But he was made to see that the new work was more important. He removed all the buildings at Harbor View, about 150, and he

filled in the ponds, using two million cubic yards of mud and sand, and building an elaborate system of sewers. The filling in took about six months. On the last day mules were at work on the new land. And within a year the ground work and the underground work was finished.

The Architects

Meanwhile, President Moore asked for a meeting of the San Francisco Chapter of the American Institute of Architects, with more than 250 members. He explained that his purpose was to have them, select twelve representatives from whom he should himself appoint five to act as an architectural board. When the board was formed with Willis Polk at its head, it included John Galen Howard, Albert Pissis, William Curlett, and Clarence R. Ward. This board was dissolved and an executive council composed of Polk, Ward and W. B. Faville was put in charge. Later it gave way to a commission consisting of W. B. Faville, Arthur Brown, George W. Kelham, Louis Christian Mullgardt, and Clarence R. Ward, of San Francisco; Robert Farquhar, of Los Angeles; Carrere & Hastings, McKim, Mead & White, and Henry Bacon, of New York, When it had completed the preliminary plans the board discontinued its meetings and G. W. Kelham was appointed Chief of Architecture.

The Block Plan

At the first meeting President Moore explained that, at the St. Louis Exposition, according to wide-expressed opinions, the buildings had been too far apart. He favored maximum of space with minimum of distance. The architects first considered the conditions they had to meet, climate and physical surroundings. They were mainly influenced by wind, cold and rain.

The result was that for the Protection of visitors, they agreed to follow what was later to be generally known, as the block plan, the buildings arranged in, four blocks, joined by covered corridors and surrounded by a wall, with three central courts and two half-courts in the south wall. It had been developed in many talks among the architects. Valuable suggestions came from Willis Polk and from E. H. Bennett, of Chicago, active in the earlier consultations. The plan finally accepted was the joint work of the entire commission.

Twelve buildings were put under contract, each designed to illustrate an epoch of architecture, ranging from the severity of the early classic to the ornate French renaissance of to-day.

The Architecture

From the start it was realized that, vast as the Exposition was to be, representing styles of architecture almost sensationally different, it must nevertheless suggest that it was all of a piece. The relation of San Francisco to the Orient provided the clue. It was fitting that on the shores of San Francisco Bay, where ships to and from the Orient were continually plying, there should rise an Oriental city. The idea had a special appeal in providing a reason for extensive color effects. The bay, in spite of the California sunshine, somewhat bleak, needed to be helped out with color. The use of color by the Orientals had abundantly justified itself as an integral part of

architecture. The Greeks and the Romans had accepted it and applied it even in their statuary. It was, moreover, associated with those Spanish and Mexican buildings characteristic of the early days of California history.

The General Arrangement

The general arrangement of the Exposition presented no great difficulties. The lay of the land helped. Interest, of course, had to center in the palaces and the Festival Hall, with their opportunities for architectural display. They naturally took the middle ground. And, of course, they had to be near the State buildings and the foreign pavilions. The amusement concessions, it was felt, ought to be in a district by themselves, at one end. Equally sequestered should be the livestock exhibit and the aviation field and the race track, which were properly placed at the opposite end. There would undoubtedly be many visitors concerned chiefly, if not wholly, with the central buildings. If they chose, they could visit this section without going near the other sections, carrying away in their minds memories of a city ideal in outline and in coloring.

Construction

As soon as the plans were decided on, the architects divided the work and separated. Those who had come from a distance went home and in a few months submitted their designs in detail. A few months later they returned to San Francisco and the meetings of the architectural board were resumed. Soon the modifications were made and the practical construction was ready to begin. Incidentally there were compromises and heartburnings. But limitations of funds had to be considered. Finally came the question of the tower, giving what the architects called "the big accent." There were those who favored the north side for the location. Others favored the south side. After considerable discussion the south side was chosen. At one of the meetings, Thomas Hastings did quick work with his pencil, outlining his idea of what the tower should be. Later, he submitted an elaborate plan. It was rejected. A second plan was rejected, too. The third was accepted. It cost five hundred thousand dollars.

Designs for two magnificent gateways, to be erected at the approaches to the Court of the Ages and the Court of the Four Seasons were considered. They had to be given up to save expense.

Clearing The Land

The task of clearing the land was finished in a few months. In addition to the government reserve, the Exposition had seventy-six city blocks. They represented two hundred parcels of land, with 175 owners, and contained four hundred dwellings, barns and improvements. Most of the buildings were torn down. A few were used elsewhere. Precautions were taken to re-enforce with piles the foundations of the buildings and of the heavy exhibits.

The director of works became responsible for the purchase of all the lumber to be used in building. It was bought wholesale, shipped from the sawmills and delivered to the sites. So there was a big saving here, through the buying in bulk and through reduced cost in handling and hauling. The first contracts given out were for the

construction of the palaces. An estimate was made of the exact number of feet available for exhibits and charts were prepared to keep a close record on the progress of the work. Incidentally, other means of watching progress consisted of the amounts paid out each month. During the earlier months the expenditures went on at the rate of a million a month. Every three weeks a contract for a building would be given out. The same contractors figured on each building. From the start it was understood that the work should be done by union men. The chief exceptions were the Chinese and the Japanese. The exhibitors had the privilege of bringing their own men. In all about five thousand men were employed, working either eight or nine hours a day. During the progress of the work there were few labor troubles.

One wise feature of the planning lay in the economy of space. It succeeded in reaching a compactness that made for convenience without leading to overcrowding. Great as this Exposition was to be, in its range worthy to be included among the expositions of the first class, it should not weary the visitors by making them walk long distances from point to point. In spite of its magnitude, it should have a kind of intimacy.

Choice of Material

There were certain dangers that the builders of the Exposition had to face. One of the most serious was that buildings erected for temporary use only might look tawdry. It was, of course, impracticable to use stone. The cost would have been prohibitive, and plaster might have made the gorgeous palaces hardly more than cheap mockeries.

Under the circumstances it was felt that some new material must be devised to meet the requirements. Already Paul E. Denneville had been successful in working with material made in imitation of Travertine marble, used in many of the ancient buildings of Rome, very beautiful in texture and peculiarly suited to the kind of building that needed color. He it was who had used the material in the Pennsylvania Station, New York, in the upper part of the walls. After a good deal of experimenting Denneville had found that for his purpose gypsum rock was most serviceable. On being ground and colored it could be used as a plaster and made to seem in texture so close to Travertine marble as to be almost indistinguishable. The results perfectly justified his faith. As the palaces rose from the ground, making a magnificent walled city, they looked solid and they looked old and they had distinct character. Moreover, through having the color in the texture, they would not show broken and ragged surfaces.

The Color Scheme

For the color-effects it was felt that just the right man must be found or the result would be disastrous. The choice fell on Jules Guerin, long accepted as one of the finest colorists among the painters of his time. He followed the guidance of the natural conditions surrounding the Exposition, the hues of the sky and the bay, of the mountains, varying from deep green to tawny yellow, and of the morning and evening light. And he worked, too, with an eye on those effects of illumination that should make the scene fairyland by night, utilizing even the tones of the fog.

The Planting

There was no difficulty in finding a man best suited to plan the garden that was to serve as the Exposition's setting. For many years John McLaren had been known as one of the most distinguished horticulturists in this part of the world. As superintendent of Golden Gate Park he had given fine service. Moreover, he was familiar with the conditions and understood the resources and the possibilities. Of course a California exposition had to maintain California's reputation for natural beauty. It must be placed in on ideal garden, representing the marvelous endowment of the State in trees and shrubs and plants and flowers and showing what the climate could do even with alien growths.

The first step that McLaren took was to consult the architects. They explained to him the court plan that they had agreed on and they gave him the dimensions of their buildings. Against walls sixty feet high he planned to place trees that should reach nearly to the top. For his purpose he found four kinds of trees most serviceable: the eucalyptus, the cypress, the acacia and the spruce. In his search for what he wanted he did not confine himself to California. A good many trees he brought down from Oregon. Some of his best specimens of Italian cypress he secured in Santa Barbara, in Monterey and in San Jose. He also drew largely on Golden Gate Park and on the Presidio. In all he used about thirty thousand trees, more than two-thirds eucalyptus and acacia.

Preparing the Landscape

Two years before the Exposition was to open McLaren built six greenhouses in the Presidia and a huge lath house. There he assembled his shrubs, his plants, and his bulbs. In all he must have used nearly a million bulbs. From Holland he imported seventy thousand rhododendrons. From Japan he brought two thousand azaleas. In Brazil he secured some wonderful specimens of the cineraria. He even sent to Africa for the agrapanthus, that grew close to the Nile. Among native flowers he collected six thousand pansies, ten thousand veronicas and five thousand junipers, to mention only, a few among the multitude a flowers that he intended to use for decoration. The grounds he had carefully mapped and he studied the landscape and the shape and color of the buildings section, by section.

The planting of trees consumed many months. The best effects McLaren found he could get by massing. He was particularly successful with the magnificent Fine Arts Palace, both in his groupings and in his use of individual trees. About the lagoon he did some particularly attractive planting, utilizing the water for reflection. There was a twisted cypress that he placed alone against the colonnade with a skill that showed the insight and the feeling of an, artist. On, the water side, the Marina, he used the trees to break the bareness of the long esplanade. And here and there on the grounds, for pure decoration, he reached some of his finest effects with the eucalyptus, for which he evidently had a particular regard. As no California Exposition would be complete without palm trees, provision was made for the decorative use of palms along of the main walks.

About two weeks before the opening, the first planting of the gardens was completed, the first of the three crops to be displayed during the Exposition. The flowers included most of the spring flowers grown here in California or capable of thriving in the California spring climate. In June they were to be re-placed with geraniums, begonias, asters, gilly-flowers, foxglove, hollyhocks, lilies and rhododendrons. The autumn display, would include cosmos and chrysanthemums and marguerites.

The Hedge

As the work proceeded, W. B. Faville, the architect, of Bliss and Faville, made a suggestion for the building of a fence that should look as if it were moss-covered with age. The result was that developing the suggestion McLaren devised a new kind of hedge likely to be used the world over. It was made of boxes, six feet long and two feet wide, containing, a two-inch layer of earth, held in place by a wire netting, and planted with South African dew plant, dense, green and hardy and thriving in this climate. Those boxes, when piled to a height of several feet, made a rustic wall of great beauty, Moreover, they could be continuously irrigated by a one-inch perforated line of pipe. In certain lights the water trickling through the leaves shimmered like gems. In summer the plant would produce masses of small purple flowers.

McLaren found his experiment so successful that he decided to build a hedge twenty feet high, extending more than a thousand feet. He also used the hedge extensively in the landscape design for the Palace of Fine Arts.

The Sculptors

The department of sculpture was placed under the direction of one of the most distinguished sculptors in the country. Karl Bitter, of New York, whose death from an automobile accident took place a few weeks after the Exposition opened. He gathered around him an extraordinary array of co-operators, including many of the most brilliant names in the world of art, with A. Stirling Calder as the acting chief, the man on the ground. Though he did not contribute any work of his own, he was active in developing the work as a whole, taking special pains to keep it in character and to see that, even in it its diversity, it gave the impression, of harmony.

Calder welcomed the chance to work on a big scale and to carry out big ideas. With Bitter he visited San Francisco in August, 1912, for a consultation with the architectural commission. Minutely they went over the site and examined the architectural plans. Then they picked the sculptors that they wished to secure as co-operators.

In December, 1912, Bitter and Calder made another visit to San Francisco for further conferring with the architectural commission, bearing sketches and scale models. Bitter explained his plans in detail and asked for an appropriation. He was told that he should be granted six hundred thousand dollars. The amount was gradually reduced till it finally reached three hundred and seventy-five thousand dollars.

It was at this period that Calder submitted his plan for the Column of Progress. He had worked it out in New York and had the scale models made by MacNeil and Konti. It won the approval of McKim, Mead & White, who declared that it made an ideal feature of the approach from the bay side to their Court of the Universe, then called the Court of the Sun and Stars.

The next few months of preparation in New York meant getting the sculptors together and working out the designs. The first meeting of the sculptors took place in January, 1913, in Bitter's studio, with a remarkable array of personages in attendance, including D. C. French, Herbert Adams, Robert Aitken, James E. Fraser, H. A. MacNeil, A. A. Weinman, Mahonri Young, Isidore Konti, Mrs. Burroughs and several others. In detail Bitter explained the situation in San Francisco and outlined his ideas of what ought to be done. Already Henry Bacon had sent in his design for his Court of the Four Seasons and sculptors were set to work on its ornamentation, Albert Jaegers, Furio Piccirilli, Miss Evelyn Beatrice Longman and August Jaegers, a time limit being made for the turning in of their plans.

Developing the Sculpture

In June, 1913, Calder returned to San Francisco to stay till the Exposition was well started. On the grounds he established a huge workshop. Then he began the practical developing of the designs, a great mass, which had already been carefully sifted. Hitherto, in American expositions the work had been done, for the most part, in New York, and sent to its destination by freight, a method costly in itself and all the more costly on account of the inevitable breakage. San Francisco, by being so far from New York, would have been a particularly expensive destination. From every point of view it seemed imperative that the work should be done here.

In a few weeks that shop was a hive of industry, with sculptors, students of sculpture front the art schools, pointers, and a multitude of other white-clad workers bending all their energies toward the completion on time of their colossal task. A few of the sculptors and artisans Calder had brought from New York. But most of the workers he secured in San Francisco, chiefly from the foreign population, some of them able to speak little or no English.

The modeling of the replicas of well-known art works were, almost without exception, made in clay. Most of the original work was directly modelled in plaster-staff used so successfully throughout the Exposition. For the enlarging of single pieces and groups the pointing machine of Robert Paine was chosen by Calder. It was interesting to see it at work, under the guidance of careful and patient operators, tracing mechanically the outlines and reproducing them on a magnified scale. For the finishing of the friezes the skill of the artist was needed, and there Calder found able assistants in the two young sculptors, Roth and Lentelli, who worked devotedly themselves and directed groups of students.

In all the sculpture Calder strove to keep in mind the significance of the Exposition and the spirit of the people who were celebrating. With him styles of architecture and schools were a minor consideration, to be left to the academicians and the critics. He believed that sculpture, like all other art-forms, was chiefly valuable and interesting

as human expression.

The Decorative Figures

Less successful on the whole than the blending of sculpture and architecture were the individual figures designed to be placed against the walls. Some of them were extremely well done. Others were obvious disappointments. The unsophisticated judgment, free from Continental bias, might have objected to the almost gratuitous use of nudity. For a popular exhibition, even the widely-traveled and broad-minded art lover might have been persuaded that a concession to prejudice could have been made without any great damage to art.

In the magnificent entrance to the grounds it was deemed fitting that the meaning of the Exposition should be symbolized by an elaborate fountain. So in the heart of the South Gardens there was placed the Fountain of Energy, the design of A. Stirling Calder, the athletic figure of a youth, mounted on a fiery horse, tearing across the globe, which served for pedestal, the symbolic figures of Valor and Fame accompanying on either side. The work, as a whole suggested the triumph of man in overcoming the difficulties in the way, of uniting the two oceans. It made one of the most striking of all the many fountains on the grounds, the dolphins in the great basin, some of them carrying female figures on their backs, contributing to an effect peculiarly French.

The Column of Progress

The Column of Progress, suggested by Calder and planned in outline by Symmes Richardson, besides being beautiful symbol and remarkably successful in outline, was perhaps the most poetic and original of all the achievements of the sculptors here. It represented something new in being the first great column erected to express a purely imaginative and idealistic conception. Most columns of its kind had celebrated some great figure or historic feat, usually related to war. But this column stood for those sturdy virtues that were developed, not through the hazards and the excitements and the fevers of conquest, but through the persistent and homely tests of peace, through the cultivation of those qualities that laid the foundations of civilized living. Isidore Konti designed the frieze typifying the swarming generations, by Matthew Arnold called "the teeming millions of men," and to Hermon A. MacNeil fell the task of developing the circular frieze of toilers, sustaining the group at the top, three strong figures, the dominating male, ready to shoot his arrow straight alit to its mark, a male supporter, and the devoted woman, eager to follow in the path of advance.

The Aim of the Sculptors

It was evidently the aim of the sculptors to express in their work, in so far as they could, the character of the Exposition. And the breadth of the plans gave them, a wide scope. They must have welcomed the chance to exercise their art for the pleasure of the multitude, an art essentially popular in its appeal and certain to be more and more cultivated in our every-day life. Though this new city was to be for a year only, it would surely influence the interest and the taste in art of the multitudes destined to become familiar with it and to carry away more or less vivid impressions.

The sculpture, too, would have a special advantage. Much of it, after the Exposition, could be transferred elsewhere. It was safe to predict that the best pieces would ultimately serve for the permanent adornment of San Francisco - by no means rich in monuments.

Mural Painting

It was felt by the builders of the Exposition that mural decorating must be a notable feature.

The Centennial Exposition of '76 had been mainly an expression of engineering. Sixteen years later architecture had dominated the Exposition in Chicago. The Exposition in San Francisco was to be essentially pictorial, combining, in its exterior building, architecture, sculpture and painting.

When Jules Guerin was selected to apply the color it was decided that he should choose the mural decorators, subject to the approval of the architectural board. The choice fell on men already distinguished. all of them belonging to New York, with two exceptions, Frank Brangwyn of London, and Arthur Mathews, of San Francisco. They were informed by Guerin that they could take their own subjects. He contented himself with saying that a subject with meaning and life in it was an asset.

In New York the painters had a conference with Guerin. He explained the conditions their work was to meet. Emphasis was laid on the importance of their painting with reference to the tone of the Travertine. They were instructed, moreover, to paint within certain colors, in harmony with the general color-scheme, a restriction that, in some cases, must have presented difficult problems.

The preliminary sketches were submitted to Guerin, and from the sketches he fixed the scale of the figures. In one instance the change of scale led to a change of subject. The second sketches were made on a larger scale. When they were accepted the decorators were told that the final canvases were to be painted in San Francisco in order to make sure that they did not conflict with one another and that they harmonized with the general plan of the Exposition. Nearly all the murals were finished in Machinery Hall; but most of them had been started before they arrived there.

Painting For Out-Doors

Some concern was felt by the painters on account of their lack of experience in painting for out-of-doors. There was no telling, even by the most careful estimate, how their canvases would look when in place. Color and design impressive in a studio might, when placed beside vigorous architecture, become weak and pale. Besides, in this instance, the murals would meet new conditions in having to harmonize with architecture that was already highly colored. Furthermore, no two of the canvases would meet exactly the same conditions and, as a result of the changes in light and atmospheric effects, the conditions would be subject to continual change. Finally, they were obliged to work without precedent. It was true that the early Italians had done murals for the open air, but no examples had been preserved.

That the painters were able to do as well as they did under the limitations reflected credit on their adaptability and good humor. The truth was they felt the tremendous opportunity afforded their art by this Exposition. They believed that in a peculiar sense it testified to the value of color in design. It represented a new movement in art, with far-reaching possibilities for the future. That some of them suffered as a result of the limiting of initiative and individuality, of subordination to the general scheme, was unquestionable. Some of the canvases that looked strong and fine when they were assembled for the last touches in Machinery Hall became anaemic and insignificant on the walls. Those most successfully met the test where the colors were in harmony with Guerin's coloring and where they were in themselves strong and where the subjects were dramatic and vigorously handled. The allegorical and the primitive subjects failed to carry, first because they had little or no real significance, and secondly because the spirit behind them was lacking in appeal and, occasionally, in sincerity.

In one regard Frank Brangwyn was more fortunate than the other painters. His murals, though intended to be displayed in the open air, were to hang in sequestered corners of the corridors running around the Court of the Ages, the court, moreover, that was to have no color. Besides, there were no colors in the world that could successfully compete against his powerful blues and reds.

The Lighting

The lighting of the Exposition, it was determined, should be given to the charge of the greatest expert in the country. Several of the leading electric light companies were consulted. They agreed that the best man was Walter D'Arcy Ryan, who had managed the lighting at the Hudson-Fulton Celebration and at the Niagara Falls Exposition. Mr. Ryan explained his system of veiled lighting, with the source of the light hidden, and made plain its suitability to an Exposition where the artistic features were to be notable, and where they were to be emphasized at night, with the lighting so diffused as to avoid shadows. After his appointment as director of illuminating he made several visits to San Francisco, and a year before the opening of the Exposition, he returned to stay till the close. His plan of ornamenting the main tower with large pieces of cut glass, of many colors, to shine like jewels, created wide-spread interest on account of its novelty. It was generally regarded as a highly original and sensational Exposition feature.

Watching the Growth

As the building went on the San Franciscans gradually became alive to the splendor. Each Sunday many thousands would assemble on the grounds. About a year before the date set for the opening an admission fee of twenty-five cents brought several thousands of dollars each week. On the Sundays when Lincoln Beachey made his sensational flights there would often be not less than fifty thousand people looking on.

The Walled City

If there were any critics who feared that the walled city might present a certain monotony of aspect they did not take into account the Oriental luxuriance of the

entrances, breaking the long lines and making splendid contrast of design and of color. Those entrances alone were worth minute study. Besides being beautiful, they had historic significance. Furthermore, the long walls were broken by artistically designed windows and by groups of trees running along the edge. Within the walls, in the splendidly wrought courts, utility was made an expression, of beauty by means of the impressive colonnades, solid rows of columns, delicately colored, suitable for promenading and for protection against rain.

From the hills looking down on the bay the Exposition began to seem somewhat huddled. But the nearer one approached, the plainer it became that this effect was misleading. On the grounds one felt that there was plenty of room to move about in. And there was no sense of incongruity. Very adroitly styles of architecture that might have seemed to be alien to one another and hostile had been harmoniously blended. Here the color was a great help. It made the Exposition seem all of a piece.

The War

In the summer of 1914 the Exposition received what for a brief time, looked like a crushing blow in the declaration of war. How could the world be interested in such an enterprise when the great nations of Europe were engaged in what might prove to be the most deadly conflict history?

The directors, in reviewing the situation, saw that, far from being a disadvantage in its effect on their plans, the war might be an advantage. In the first place, it would keep at home the great army of American travelers that went to Europe each year. With their fondness for roaming, they would be almost certain to be drawn to this part of the world. And besides, there were other travelers to be considered, including those Europeans who would be glad to get away from the alarms of war and those South Americans who were in the habit of going to Europe. Furthermore, though the Exposition had been designed to commemorate the services of the United States Army in building the Panama Canal, it was essentially dedicated to the arts of peace. It would show what the world could do when men and nations co-operated.

The Department of Fine Arts

Meanwhile, the war was upsetting the plans for the exhibits, notably the exhibit of painting and sculpture.

When John E. D. Trask, for many years director of the Pennsylvania Academy of Fine Arts, was appointed Director of the Fine Arts Department at the Panama-Pacific International Exposition, he had made a careful survey of the field he had to cover. It virtually consisted of the whole civilized world. After arranging for the formulation of committees in the leading cities of the East and the Middle West to secure American work, he made a trip to Europe, visiting England, France, Holland, Sweden, Germany, Hungary, Austria and Italy. With the exception of England and Germany, the governments were sympathetic. The indifference of those two countries was at the time was not quite comprehensible. There might have been several explanations, including the threat of war. There were also those who said that England and Germany had entered into a secret alliance against this country for the purpose of minimizing the American influence in commerce, soon to be strengthened by the

opening of the Panama Canal. Wherever the truth lay, the fact remained that both countries maintained their attitude of indifference. Individual English and German artists and organizations of artists, however, showed a willingness to co-operate.

Through emissaries, mainly unofficial, Americans of influence, Trask drew on the resources of all Europe. He also entered into negotiations with China and Japan, both of which countries, with their devotion to art, as might have been expected, co-operated with enthusiasm. The display at the Fine Arts Palace promised to make one of the greatest international exhibits in history, if not the greatest.

At the outbreak of the war it looked as if the whole of Europe might become involved and it might be impossible to secure anything that could properly be called a European art exhibit. Meanwhile, the space reserved for the European exhibitors must he filled. It happened that, at the time, Trask was in the East. He quickly put himself into personal communication with the New York artists, who had been invited to send three or four works, and he asked them to increase the number. He also arranged with his committee for the securing of a much larger number of American pictures. Under the circumstances he was bound to rely on the discretion of his juries. The result was that he had to take what came. It included a large number of excellent works and others of doubtful merit.

An Emissary to France and Italy

Meanwhile, during the few months after the outbreak of war, the art situation in Europe began to look more hopeful. It seemed possible that some of the nations concerned in the war would be persuaded to participate. Captain Asher C. Baker, Director of the Division of Exhibits, was sent on a special mission to France, sailing from New York early in November. The United States collier "Jason" was then preparing to sail from New York with Christmas presents for the children in the war zone, and the secretary of the navy had arranged with the Exposition authorities that, on the return trip, the ship should be used to carry exhibits from Europe. The first plan was that the exhibits should come only from the warring nations; it was later extended to include other nations.

In Paris Captain Baker found the situation discouraging. The first official he saw told him that, under the circumstances, any participation of France whatsoever was out of the question: France was in mourning, and did not wish to celebrate anything; if any Frenchman were to suggest participation he would be criticised; furthermore, Albert Tirman, at the head of the French committee that had visited San Francisco the year before to select the site of the French Pavilion, had come back from the front in the Vosges and was hard at work in the barracks of the Invalides, acting as an intermediary between the civil and military authorities.

Then Captain Baker appealed to Ambassador Myron T. Herrick. Although the ambassador was enthusiastic for the Exposition, he said that, in such a crisis, he could not ask France to spend the four hundred thousand dollars set apart for use in San Francisco. Captain Baker said: "Don't you think if France came in at this time a wonderfully sympathetic effect would be created all over the United States?" The ambassador replied, "I do." "Wouldn't you like to see France participate?" The

ambassador declared that he would. "Will you say so to Mr. Tirman?" The ambassador said, "Willingly."

A week later Baker and Tirman were on their way to Bordeaux to see Gaston Thomson, Minister of Commerce. They made these proposals: The exhibits should be carried by the Jason through the canal to San Francisco; the building of the French Pavilion should be undertaken by the Division of Works of the Exposition, on specification to be cabled to San Francisco of the frame work, the moulds for the columns and architectural ornaments to be prepared in France and shipped by express; the French committee of organization was to work in France among possible exhibitors; a statement was to be made to the ministry of what each department of the government could do in sending exhibits and what exhibits were ready; a statement should come from the Minister of Fine Arts as to how much space he could occupy and how many paintings could be secured for the Palace of Fine Arts; a complete representation of the Department of Historical Furniture and Tapestries, known as the Garde Meuble, was to be made for the pavilion.

In the interview with the Minister of Commerce Baker argued that, without France, an Exposition could not be international, and that the participation of France at this time, with her flag flying in San Francisco, would be like winning a battle before the world. It would show the people of the United States France's gratitude for the money sent the wounded and the suffering, and would warm the hearts of the American people.

Thomson responded with enthusiasm, and soon the government became enthusiastic. Several thousand dollars were spent in cabling; Henri Guillaume, the distinguished French architect, experienced in many expositions, was sent out. When the Jason stopped at Marseilles it took, on board one of the most remarkable collections of art treasures ever shipped to a foreign country, the finest things in one of the world's great storehouses of treasure, including even the priceless historical tapestries, and a large collection of French paintings for the Fine Arts Palace, gathered by the French committee after great labor, due to the absence of many of the painters in the war.

When Captain Baker left France he had accomplished far more for the Exposition than he realized himself. Reports of his success in securing French participation preceded him to Italy and helped to prepare the way. The Italians listened to his proposition, all the more willingly because France had been won over. Besides, he had a warm supporter in Ernesto Nathan, ex-Mayor of Rome, who had paid an extended visit to San Francisco and had become an enthusiastic champion of the Exposition. In a few days he had made arrangements that led to the collection of the splendid display of Italian art, shipped on the Vega, together with many commercial exhibits. Captain Bakers work in France and in Italy, accomplished within three weeks, was a triumph of diplomacy.

Foreign Participation in General

Germany was not to be completely over-shadowed by France notwithstanding previous indifference on the part of the government. German manufacturers wished

to be represented, and they actually received governmental encouragement. Austrians, not to be outdone by Italy, unofficially came in. In fact, despite the war, every country had some representation, England and Scandinavia and Switzerland included, even if they did not have official authority.

There are those who maintain that, in spite of criticism, the Fine Arts Department is now making a better showing than it could have made if there had been no war. American collectors, with rare canvases, were persuaded to help in the meeting of the emergency by lending work that, otherwise, they would have kept at home. It was thought that many of the Europeans would be glad to send their collections to this country for safe keeping during war time. But such proved not to be the case. A good deal of concern was felt about sending the treasures on so long a journey, subject to the hazards of attack by sea. Furthermore, from the European point of view, San Francisco seemed far away.

Looking for Art Treasures

A short time after Captain Baker sailed from New York another emissary went abroad for the Exposition, J. N. Laurvik, the art critic. A few weeks before Mr. Laurvik had returned from Europe, where he had represented the Fine Arts Department, looking for the work of the artists in those countries that were not to participate officially. At the time of the outbreak he was in Norway and he had already secured the promise of many collections and the co-operation of artists of distinction. His report of the situation as he left it persuaded the authorities that, in spite of the difficulties, he might do effective work.

When Laurvik arrived in Rome he found that Captain Baker had already prepared for his activities. Ernesto Nathan was devoting himself heart and soul to the cause. But the Italian authorities, for the most part, were absorbed in the questions that came up with the threat of war. Working with the committee, and aided by Ambassador Thomas Nelson Page, Laurvik quickly made progress. He secured magnificent canvases by the President of the French Academy in Rome, Albert Besnard, painted, for the most part, in Benares, with scenes on the Ganges, and a collection of pieces by the Norwegian sculptor, Lerche.

Notable Collections

From Rome Laurvik went to Venice, where he was greatly helped by the American consul, B. H. Carroll, Jr. Though the International Exhibit held in Venice every two years had closed several months before, many of the works of art were still there, their owners, either afraid or unable to take them away and yet concerned about their being so close to the scene of war. It was the general concern that enabled Laurvik to secure some of his finest material. Together with the Italian work, he arranged to have shipped here on the Jason, Norwegian and Hungarian paintings and fifty canvases by the man regarded as the greatest living painter in Finland, Axel Gallen-Kallela. He also made a short journey from Venice to the home of Marinetti, the journalist, poet and leader of the. Italian Futurist painters, who, after much persuading, promised to send fifty examples of the work done by the ten leaders in his group.

On leaving Venice Laurvik started for Vienna. In spite of the war, he was promised support by the Minister of Art. Unfortunately, the art societies fell to quarreling, and gave little or no help. Then Laurvik appealed to the artists themselves. In Kakosha, one of the best known among the Austrian painters, he found an ally. The collection he made in Vienna included several of Kakosha's canvases, lent by their owners, and a large number of etchings.

The Hungarian Collection

In Hungary Laurvik had a powerful friend in Count Julius Andrassy, a man, of wealth and influence, the owner of one of the newspapers published in Budapest. From, his own collection of Hungarian art Andrassy made a large contribution and he inspired other collectors to do likewise. The getting together of the material was full of difficulties. Much of it had been taken away for safekeeping. The museums were all closed and some of their treasures were buried in the ground. Already the Russians, during their raid on the Carpathian Mountains, had possessed themselves of rare art works, some of the best canvases cut from the frames and carried off by the officials. Among the sufferers was Count Andrassy himself, who lost valuable heirlooms from one of his country estates, including several Titians. In spite of that experience, Andrassy, refused to hide his possessions. He preferred the risk of losing them to showing fear, perhaps helping to start a panic.

The Hungarian collection came near missing the Jason. It was mysteriously held up in the train that carried it through the Italian territory to Italy, arriving in Genoa three days after the Jason was scheduled to so sail from there. But the Jason happened to be delayed three days, too.

By the German steamer, the "Crown Princess Cecilie," it happened that an interesting collection of German Paintings, after being exhibited in the Carnegie Institute of Pittsburgh, was started on the way to Germany; but the war caused the ship to return to an American port. After a good deal of negotiating the canvases were secured for the Exposition and taken off the ship.

On the opening day of the Exposition it was found that the Palace of Fine Arts, far from having too little material, had too much. Not only were China and Japan and several of the European nations well represented, but on the way were many art works that there would not be room for. The consequence was that a new building had to be erected. It was finished in July and it became known as the Fine Arts Annex.

I

The View From the Hill

"The best way to see the Exposition, in my opinion," said the architect, "is to stand on the top of the Fillmore Street hill and look down. Then you will find out what the architects were up to. The finest point of observation would be at the corner of Divisadero Street and Broadway."

The next day, as we stood at that point, the Exposition stretched out beneath us like a city of the Orient.

"When the architects first discussed the construction they knew it was to be looked at from these hills. So they had to have a scheme that should hide the skylight and avoid showing lack of finish on top and that should be pictorial and impressive from above. One of the problems was to make the roof architectural. Now as we look down, see how stunning the effect is - like a Persian rug."

"And the color helped there, too, didn't it?"

"Of course. And notice how skilfully the architecture and the coloring harmonized. As the Exposition was to be built on low, flat ground, it had to be lifted up. One way was by using the domes. The central portion of each of those palaces was lifted above the main surface of the roof to introduce a row of semi-circular windows to light the interior like a church. And the domes, besides being ornamental in themselves, gave spring to the towers. The big tower provided scope for the splendid archway that served as an approach and set the standard for the other arches."

It was plain enough that the top of the Exposition had not received the praise it deserved. "Think how crude that scene would have been if it had presented a straggling mass of roofs. And even as it is, with its graceful lines, if it were lacking in color it would seem crude. Perhaps it will help us to realize how unsightly most of the roofs of our houses are, and how unfinished. There's no reason in the world why they should be. The Greeks and the Romans had the right idea. They were very sensitive to lack of finish. They felt the charm of decorated roofs. See that angel down there that keeps recurring at the points of the gables. What a pretty bit of ornamentation. The Greeks used it to suggest the gifts of the gods coming down from heaven. 'Blessings on this house.' I suppose the wreath in the hand used here was meant to suggest the crowning of the work. It explains why the figure is called "Victory." By the way, it has an architectural value in giving lightness and grace to the roofs."

The builders, we could see, had cleverly adapted their plans to the conditions. "The effect might so easily have been monotonous and cold, and it might have been flat and dreary. It was a fine idea to lift the central portion of each of those main palaces above the surfaces of the roofs to introduce the semicircular windows in the domes. It helped to infuse the scene with a kind of tenderness and spirituality. And see how the two groups on top of the triumphal arches, the Orientals and the Pioneers, contribute to the soaring effect and to the finish at the same time. The Romans disliked bareness on the top of their arches. They wanted life up there, the more animated the better. So they put on some of their most dramatic scenes, like their chariot races."

The expert proceeded to point out the architectural balance of the buildings. The severe and mighty Palace of Machinery, impressive in its long sweep of line, at one side made a dramatic contrast with the delicately imagined and poetic Palace of Fine Arts on the other. In front of the walled city, between the long stretch of garden, stood two harmonious buildings, the Palace of Horticulture, with its glorious roof of glass, and the Festival Hall, closely related in outline, and yet very different in detail. And the garden itself, with its dark, pointed trees standing against the wall, and with its simplicity of design, made an agreeable approach to the great arched entrance under the Tower of Jewels. "Those banners down there, shielding the lights, are a

stroke of genius, both in their orange color and their shape. And those orange-colored streamers, how they add to the spirit of gaiety. The trees have been placed against the wall to keep it from seeming like a long and uninteresting stretch. And observe the grace in line of the niches between the trees. Even from here you can feel the warmth of the color in the paths. The pink effect is made by burning the sand. Only a man like Guerin, a painter, would have thought of that detail. I wonder how many visitors down there know that the very sand they walk on has been colored."

Around the Tower pigeons were flying, somehow relieving the mechanical outlines. Was the disproportion between the great arch, forming a kind of pedestal, and the outlines above due to mathematical miscalculation or to the interference of the ornamentation? We finally decided that the proportions had probably been right in the first place. But they had been changed by the Exposition authorities' cutting the Tower down one hundred feet, thereby saving $100,000. A matter of this kind could be reduced almost to an exact science. Besides, though the ornamentation interfered with the upward sweep of line, the effect of flatness was made by those horizontal blocks which seemed to be piled up to the top. If the outline had been clean, it would have achieved the soaring effect so essential to an inspiring tower, creating the sense of reaching up to the sky, like an invocation.

Thomas Hastings had a sound idea when he made that design. He wanted to do something Expositional, exactly as Guerin did when he applied the coloring. Now there were critics who said that the coloring was too pronounced. It reminded them of the theater. Well, that was just what it ought to remind them of. It had life, gaiety, abandon. The critic who said that the orange domes provided just the right tone, and that this tone ought to have been followed throughout, didn't make sufficient allowance for public taste. He wanted the Exposition to be an impressionistic picture in one key. But one key was exactly what Guerin didn't want. His purpose was to catch the excitement in variety of color as well as the warmth, to stimulate the mind. He succeeded in adapting his color scheme to architecture that had breadth and dignity. At first he expected to use orange, blue, and gold, carefully avoiding white. He did avoid white; but he expanded his color scheme and included brown and yellow and green. But, in that tower, Hastings did something out of harmony with the architecture, something barbaric and crude.

Here and there the bits of Austrian cut glass were sparkling on the tower like huge diamonds. "At times the thing is wonderfully impressive. There's always something impressive about a mass if it has any kind of uniformity, and here you can detect an intention on the part of the architect. There are certain lights that have a way of dressing up the tower as a whole, giving it unity and hiding its ugliness. And at all times it has a kind of barbaric splendor. It might have come out of an Aztec mind, rather childish in expression, and seeking for beauty in an elemental way. I can imagine Aztecs living up there in a barbaric fashion, their houses piled, one above another, like our uncivilized apartment houses."

In studying the Tower of Jewels in detail, we decided that it was not really so crude as it seemed on first sight. Much might be done even now by a process of elimination. And the arch was magnificent. "In its present condition the tower unquestionably

provides a strong accent. It has already become a dominating influence here. But it's an influence that teaches people to feel and to think in the wrong way. It encourages a liking for what I call messy art, instead of developing a taste for the simplicity that always characterizes the best kind of beauty, the kind that develops naturally out of a central idea."

From the Tower of Jewels we turned our attention to those other towers, the four so charming in design and in proportion, Renaissance in feeling, their simplicity seeming all the more graceful on account of the contrast with the other tower's over-ornamentation. "I wonder what the world would have done without the Giralda Tower in Seville? It has inspired many of the most beautiful towers in the world. It helped to inspire McKim, Mead and White when they built the Madison Square Tower, and the Madison Square Tower might be described as a relative of our own Ferry Tower, which is decidedly one of the best pieces of architecture in San Francisco. And it's plain enough that these four towers and the Ferry Tower are related. The top of the four towers, by the way, has a history. It comes from the Choragic Monument of Lysicrates, the little temple in Athens that was built by one of the successful chorus-leaders in the competitive choral dances of the Greeks, who happened to be a man of wealth. Afterward, when a chorus-leader won a prize, which consisted of a tripod, it was shown to the people on that monument."

"Some critics," I said, "have complained of the coloring and the pattern on those towers."

"They can't justify themselves, however. Though this plaster looks like Travertine, it nevertheless remains plaster, and it lends itself to plastic decoration. The Greeks and the Romans often used plaster, and they did not hesitate to paint it whenever they chose. Kelham's four towers have been criticised on account of their plastic design, which has a good deal of pink in it. But that design provides one of the strongest color notes in the whole Exposition, a delightful note, too. It happens that makers of wallpaper have had the good sense to use a design somewhat similar. But this fact does not make the design any the less attractive or serviceable."

Between the houses on the hill we could catch glimpses of the South Gardens between the glass dome of the Horticultural Palace and Festival Hall. The architects rightly felt that in general appearance they had to be French to harmonize with the French architecture on either side. In the distance the Fountain of Energy stood out, like a weird skeleton that did not wholly explain itself. Stirling Calder, the sculptor, must have forgotten that the outline of those little symbolic figures perched on the shoulder of his horseman would not carry their meaning.

Now, before our eyes, the Exposition revealed itself as a picture, with all the arts contributing. It suggested the earlier periods of art, when the art-worker was architect, painter and sculptor all in one.

II

The Approach

"You see," said the architect as we started down the hill, "when the Exposition builders began their work they found the setting of the Mediterranean here. It justified them in reproducing the art of the Orient and of Greece and Rome which was associated with it, modified of course to meet the special requirements. Besides, they didn't want to be tied down to the severe type of architecture in vogue in this country."

First of all, he went on to explain, they had created a playground. There they appealed to the color sense, strong in the Italians and the Orientals, and weak among the people in this country, decidedly in need of fostering, and the appeal was not merely to the intellect, but to the emotions as well. Color was as much a part of architecture as of painting. So, in applying the color, Guerin worked with the architects. He never made a plan without taking them into consultation. Then, too, Calder, acting head of the Department of Sculpture, and Denneville, the inventor of the particular kind of imitation Travertine marble used on the grounds, were active in all the planning. In fact, very little was done without the co-operation of Guerin, Calder, Denneville and Kelham, chief of the Architectural Board. In getting the Exposition from paper to reality, they had succeeded in making it seem to be the expression of one mind. Even in the development of the planting the architects had their say. Here landscape gardening was actually a part of the architecture. Faville's wall, for example, was built with the understanding that its bareness was to be relieved with masses of foliage, creating shadows.

Before the Scott Street entrance we paused to admire the high hedge of John McLaren. We went close to examine the texture. The leaves of the African dewplant were so thick that they were beginning to hide the lines between the boxes.

"Faville realized the importance of separating the city from the rest of the world, making it sequestered. He knew that a fence wouldn't be the right sort of thing. So he conceived the idea of having a high, thick wall, modeled after an old English wall, overgrown with moss and ivy. As those walls were generations in growing, he saw that to produce one in a few months or even a few years required some ingenuity. He set to work on the problem and he devised a scheme for making an imitation hedge by planting ivy in deep boxes and piling the boxes on one another. When he submitted it to McLaren he was told that it was good except for the use of the ivy. It would be better to use African dew plant. Later McLaren improved on the scheme by using shallow boxes.

"Faville designed a magnificent entrance here," the architect went on, glancing up at the three modest arches that McLaren had tried to make as attractive as possible with his hedge. "It would have been very appropriate. But the need of keeping down expenses caused the idea to be sacrificed. However, the loss was not serious. As a matter of fact, in spite of the efforts of the Exposition to persuade visitors to come in here, a great many preferred to enter by the Fillmore Street gate. During the day this approach is decidedly the more attractive on account of leading directly into the gardens and into the approach to the court. The Fillmore Street entrance, with the Zone shrieking at you at one side, hardly puts you in the mood for the beauty in the courts. At night the situation is somewhat different. The flaring lights of the Zone

make the dimness of the court all the more attractive."

III

In the South Gardens

Though the arrangement of the landscape might be French, these flowers were unmistakably Californian. The two pools, ornamented with the Arthur Putnam fountain of the mermaid, in duplicate, decidedly French in feeling, were brilliant with the reflected coloring from both the flowers and the buildings.

The intention at first had been to make a sunken garden here; but the underground construction had interfered. Now one might catch a suggestion of Versailles, except for those lamp posts. "Joseph Pennell, the American etcher, who has traveled all over Europe making drawings, finds a suggestion of two great Spanish gardens here, one connected with the royal palace of La Granga, near Madrid, and the other with the royal palace of Aranjuez, near Toledo. They've allowed the flowers to be the most conspicuous feature, the dominating note, which is as it should be. Masses of flowers are always beautiful and they are never more beautiful than when they are of one color."

"And masses of shrubbery are always beautiful, too,", I said, nodding in the direction of the Palace of Horticulture, where McLaren had done some of his best work.

"There's no color in the world like green, particularly dark green, for richness and poetry and mystery. It's intimately related to shadow, which does so much for beauty in the world."

"The Fountain of Energy almost hits you in the face, doesn't it?" I said.

"Of course. That's exactly what Calder meant to do. In a way he was right. He wanted to express in sculpture the idea of tremendous force. Now his work is an ideal example of what is expositional. It has a sensational appeal. One objection to it is that it suggests too much energy, too much effort on the part, not only of the subject, but of the sculptor. The artist ought never to seem to try. His work ought to make you feel that it was easy for him to do. But here you feel that the sculptor clenched his teeth and worked with might and main. As a matter of fact, he did this piece when he must have been tired out from managing all the sculpture on the grounds. He made two designs. The first one, which was not used, seemed to me better because it was simpler in the treatment of the base. Even the figures at the base here are over-energized, the human figures I mean. Still, in their sportiveness and in the sportiveness of Roth's animals, they have a certain charm. And with the streams spouting, the work as a whole makes an impression of liveliness. But it's a nervous liveliness, characteristically American, not altogether healthy."

The Fountain of Energy and the Tower of Jewels, we decided, both expressed the same kind of imagination. Like the fountain, the tower gave the sense of overstrain. "It's pretty hard to see any architectural relation between those figures up there on the tower and the tower itself. See how the mass tries to dominate Kelham's four

Italian towers, but without showing any real superiority."

The heraldic shields on the lamp posts near by attracted us both by their color and by the variety and grace of their designs. How many visitors stopped to consider their historic character? They went back to the early history of the Pacific Coast. For this contribution alone Walter D'Arcy Ryan deserved the highest recognition. Only an artist could have worked out this scheme in just this sensitive and appropriate way.

We stopped at the vigorous equestrian statue of Cortez by Charles Niehaus at our right, close to the tower. "I always liked Cortez for his nerve. He didn't get much gratitude from his Emperor for conquering Mexico and annexing it to Spain. And what he got in glory and in money probably did not compensate him for his disappointment at the end. When he couldn't reach Charles V in any other way, he jumped up on the royal carriage. Charles didn't recognize him and asked who he was. 'I'm the man,' said Cortez, 'that gave you more provinces than your forebears left you cities.' Naturally Charles was annoyed. We don't like to be reminded of ingratitude, do we, especially by the people who think we ought to be grateful to them? So Cortez quit the court and spent the rest of his life in the country."

At our right we met another of the many Spanish adventurers drawn to the Americas by the discovery of Columbus, Pizarro, who presented his country with the rich land of Peru. It was doubtless placed here on account of the relation between Spain and California. "Civilization is a development through blood and spoilation," the architect remarked. "If Pizarro hadn't been lured by the gold of the Incas we might not be here at this moment."

The figures on the tower, insignificant when viewed from a distance, at close range took on vigor: the philosopher in his robes, the bearer of European culture of the sixteenth century to these shores; the Spanish priest, typical of the early friars; the adventurer, so closely related to Columbus; and the Spanish soldier. The armored horseman, by Tonetti, in a row all by himself, suffering from being rather absurdly out of place, might have won applause if he had been brought on a pedestal close to the ground. His being repeated so often up there made an effect almost comic. The vases and the triremes, the pieces of armor, with the battle-axe designs on either side, the Cleopatra's needles, and the richly-girdled globe on top, sustained on the shoulders of three figures, were all well done. The only trouble was that they had not been made to blend into one lightly soaring mass.

"It's curious that Hastings should have gone astray in the treatment of the tower. He must have known the psychological effect of parallel horizontal lines. When skyscrapers were first built in New York a few years ago they were considered unsightly on account of their great height. So the architects were careful to use parallel horizontal lines in order to diminish the apparent height as far as possible. Then people began to say that there was beauty in the sky-scrapers, and the architects changed their policy. They built in straight parallel lines that shot up to the sky. In this way they increased the apparent height."

The inscriptions on the south side of the tower's base reminded us of the Exposition's meaning, Conspicuously and properly emphasized here. The pagan note

in the architecture was indicated in the ornamentation by the use in the design of the head of the sacred bull. And Triumphant America was celebrated in the group of eagles.

The dark stains on the yellow columns made us see how clever Guerin had been in his application of the coloring. In most places he had applied one coat only, trusting to nature to do the rest. Most of all, he wished to avoid the appearance of newness and to secure a look of age. On these columns the smoke from the steam rollers had helped out. One might imagine that they had been here for generations.

Here the builders had used the Corinthian column, with the acanthus leaves varied with fruit-designs and with the human figure. "It was a lucky day for architecture when the column came into use. It doubtless got its start from a single beam used for support. Then the notion developed of making it ornamental by fluting it and decorating the top. In this Exposition three kinds of columns are used, the Doric, which the Greeks favored, with the very simple top or capital; the Ionic, with the spiral scroll for the capital, and the Corinthian, with the acanthus flowing over the top, and the Composite which uses features from all the other three."

"Do you happen to know how the acanthus design was made? Well, Vitruvius tells the story. Anyone that wants to get a line on this Exposition ought to read that book, or, at any rate, to glance through it and to read parts of it pretty thoroughly. It is called 'The Architecture of Marcus Vitruvius Pollio.' There's a good translation from the Latin by Joseph Gwilt. It has become the architect's bible. According to Vitruvius, the nurse of Corinthian girl who had died carried to the girl's tomb basket filled with the things that the girl had particularly liked. She left the basket on the ground near the tomb and covered it with a tile. It happened that it stood over the root of an acanthus plant. As the plant grew its foliage pressed up around the basket and when it reached the tile the leaves were forced to bang back in graceful curves. Callimachus, a Corinthian architect, noticed the effect and put it into use."

IV

Under the Tower of Jewels

When we entered the arch we looked up at the magnificent ceiling used by McKim, Mead & White, in panels, with a pictorial design beautifully colored by Guerin. "The blue up there blends into the deeper blue of the Dodge murals just beneath. Those murals are in exactly the right tone. They give strength to the arch. But they are weakened by being in the midst of so much heavy architecture. Their subjects, however, are in harmony with the meaning of the tower. Guerin was right when he told the mural decorators that a good subject was an asset. By studying these murals you can get a glimpse of all the history associated with California and with the Panama Canal. Dodge has made drama out of Balboa's discovery of Panama and out of the union of the two oceans, a theme worthy of a great poet. And Dodge is one of the few men represented in the art on the grounds who have made pictorial use of machinery. There's the discovery by Balboa, the purchase by the United States, the presentation of the problem of uniting the two oceans, very imaginative and pictorial,

the completion of the Canal, and the crowning of labor, with the symbolic representation of the resulting feats of commerce suggested by the want of the winged Mercury. Dodge is dramatic without being too individual. His murals don't call the attention away from their surroundings to themselves. They are a part of the architecture, as murals always should be."

On either side we found the columned niches designed by McKim, Mead and White, each ornamented with a fountain. The back wall made a splendid effect as it reached up toward the tower.

To the right we turned to view Mrs. Edith Woodman Burroughs' "Fountain of Youth," lovely in the girlish beauty of the central figure, and in the simplicity and the sincerity of the design as a whole. In some ways the figure reminded us of the celebrated painting by Ingres in the Louvre, "The Source," the nude girl bearing a jug on her shoulder, sending out a stream of water. There was no suggestion of imitation, however.

"The symbolism in the design," said the architect, "does not thrust itself on you, and yet it is plain enough. That woman and man pushing up flowers at the feet of the girl make a beautiful conception. The whole fountain has an ingenuousness that is in key with the subject. Across the way," he went on, turning to view the Fountain of El Dorado, by Mrs. Gertrude Vanderbilt Whitney, "there's a piece of work much more sophisticated and dramatic, fine in its conception and strong in handling. No one would say offhand that it was the work of a woman; and yet it shows none of the overstrain that sometimes characterizes a woman artist when she wishes her work to seem masculine."

In approaching the "El Dorado" we noted the skill shown in the details of the conception. "This fountain might have been called 'The Land of Gold,' in plain English, or 'The Struggle for Happiness,' or by any other name that suggested competition for what people valued as the prizes of life. When Mrs. Whitney was asked to explain whether those trees in the background represented the tree of life, she said she didn't have any such idea in her mind. What she probably wanted to do was to present an imaginative scene that each observer could interpret for himself. These two Egyptian-looking guardians at the doors, with the figures kneeling by them, suggest plainly enough the futility that goes with so much of our struggling in the world. So often people reach the edge of their goal without really getting what they want."

V

The Court of the Universe

Through the arch we passed into the neck of the Court of the Universe, which charmed us by the warmth of its coloring, by McLaren's treatment of the sunken garden, by its shape, by the use of the dark pointed cypress trees against the walls, and by the sweep of view across the great court to the Marina, broken, however, by the picturesque and inharmonious Arabic bandstand. We glanced at the inscriptions

at the base of the tower carrying on the history of the Canal to its completion. Then we stopped before those graceful little elephants bearing Guerin's tall poles with their streamers. "That little fellow is a gem in his way. He comes from Rome. But the heavy pole on his back is almost too much for him. He's used pretty often on the grounds, but not too often. After the Exposition is over we ought to keep these figures for the Civic Center. They would be very ornamental in the heart of the city."

As we walked toward the main court, the architect called my attention to the view between the columns on the other side of the Tower of Jewels, with the houses of the city running down the hills. "San Francisco architecture may not be beautiful when you study individual houses. But in mass it is fine. And, of a late afternoon, it is particularly good in coloring. It seems to be enveloped in a rich purple haze. That color might have given the mural decorators a hint. It would have been effective in the midst of all this high-keyed architecture. It's easy here to imagine that you're in one of those ancient Hindu towns where the gates are closed at night. You almost expect to see camels and elephants."

What was most striking in the Court was its immensity. "Though it comes from Bernini's entrance court to St. Peter's in Rome, it is much bigger. There are those who think it's too big. But it justifies itself by its splendor. The use of the double row of columns is particularly happy. The double columns were greatly favored by the Romans. In St. Peter's Bernini used four in a row. And what could be finer than those two triumphal arches on either side, the Arch of the Rising Sun and the Arch of the Setting Sun, with their double use of symbolism, in suggesting the close relation between California and the Orient, as well as their geographical meaning? They are, of course, importations from Rome, the Arch of Constantine and the Arch of Titus all over again, with a rather daring use of windows with colored lattices to give them lightness and with colossal groups of almost startling proportions used in place of the Roman chariot or quadriga."

Originally, the intention had been to use here the name of the Court of Sun and Stars. Then it was changed to the Court of Honor, and finally to its present name, to suggest the international character of the Exposition.

Those two groups represented by far the most ambitious work done by the sculpture department. From designs by Calder, they were made by three sculptors, Calder, Roth and Lentelli. They presented problems that must have been both difficult and interesting to work out. First, they had to balance each other. What figure in the Pioneer group could balance the elephant that typified the Orient? Calder had the idea of using the prairie schooner, associated with the coming of the pioneers to California, drawn by great oxen.

The Oriental group doubtless shaped itself in picturesque outlines much more quickly than the sturdy, but more homely Americans of the earlier period. The Orientals displayed an Indian prince on the ornamented seat, and the Spirit of the East in the howdah, of his elephant, an Arab shiek on his Arabian horse, a negro slave bearing fruit on his head, an Egyptian on a camel carrying a Mohammedan standard, an Arab falconer with a bird, a Buddhist priest, or Lama, from Thibet, bearing his

symbol of authority, a Mohammedan with his crescent, a second negro slave and a Mongolian on horseback.

The Nations of the West were grouped around that prairie wagon, drawn by two oxen. In the center stood the Mother of Tomorrow a typical American girl, roughly dressed, but with character as well as beauty in her face and figure. On top of the wagon knelt the symbolic figure of "Enterprise," with a white boy on one side and a colored boy on the other, "Heroes of Tomorrow." On the other side of the wagon stood typical figures, the French-Canadian trapper, the Alaska woman, bearing totem poles on her back, the American of Latin descent on his horse, bearing a standard, a German, an Italian, an American of English descent, a squaw with a papoose, and an Indian chief on his pony. The wagon was modelled on top of the arch. It was too large and bulky to be easily raised to that great height.

The architect was impressed by the boldness of the designs and to the spirit that had been put into them. "It's very seldom in the history of art that sculptors have had a chance to do decorative work on so big a scale. It must have been a hard job, getting the figures up there in pieces and putting them together. Some of the workers came near being blown off. Some of them lost their nerve and quit. I wonder, by the way, if that angel on top of the prairie wagon would be there if Saint Gaudens hadn't put an angel in his Sherman statue, and if he hadn't made an angel float over the negro soldiers in his Robert Gould Shaw monument in Boston. He liked that kind of symbolism. He must have got it from the mediaeval sculptors who worked under the inspiration of the Catholic Church."

Varying notes we found around the American group. Cleopatra's needle, used for ornamentation, suggested Egypt and the Nile. That crenellated parapet once belonged to military architecture: between those pieces that stood up, the merlons, in the embrasure, the Greek and Roman archers shot their arrows at the enemy and darted back behind the merlons for protection. In spite of its being purely ornamental it told its story just the same, and it expressed the spirit that still persisted in mankind. Nowadays it was even used on churches. But religion and war had always been associated. Besides, in an International Exposition it was to be expected that the art should be international. How many people, when they looked at Cleopatra's needle, knew how closely it was related to the newspapers and historical records of today? The Egyptians used to write on these monuments news and opinions of public affairs. The Romans had a similar custom in connection with their columns. On the column of Trajan they not only wrote of their victories, but they pictured victorious scenes in stone.

The little sprite that ran along the upper edge of the court in a row, the star-figure, impressed me as making an unfortunate contrast with the stern angel, repeated in front of each of the two arches. My criticism brought out the reply that it was beautiful in itself and had its place up there. "These accidental effects of association are sometimes good and sometimes they're not. Here I can't see that they make a jarring effect. In the first place, a Court of the Universe ought to express something of the incongruity in our life. Ideally, of course, it isn't good in art to represent a figure in a position that it's hard to maintain without discomfort. But here the outlines

are purely decorative and don't suggest strain. In my judgment that figure is one of the greatest ornaments in the court. It gives just the right note."

The two fountains in the center of the sunken garden were gaily throwing their spray into the air. The boldness of the Tritons at the base represented a very different kind of handling from the delicacy of the figure at the top of each, the Evening Sun and the Rising Sun, both executed with poetic feeling. In the Rising Sun, Weinmann had succeeded in putting into the figure of the youth life, motion and joy. Looking at that figure, just ready to spread its wings, one felt as if it were really about to sweep into the air. Though the Evening Sun might be less dramatic, it was just as fine. "It isn't often that you see sculpture of such imaginative quality," said the architect.

Those great symbolic figures by Robert Aitken, at once giving a reminder of Michael Angelo, impressed me as being perfectly adapted to the Court, and to their subjects, Earth, Air, Fire, and Water. But my companion thought they were too big. He agreed, however, that they were both original and strong. There was cleverness in making the salamander, with his fiery breath and his sting, ready to attack a Greek warrior, symbolize fire. Under the winged girl representing air there was a humorous reference to man's early efforts to fly in the use of the quaint little figure of Icarus. Water and earth were more conventional, but worked out with splendid vigor, the two figures under earth suggesting the competitive struggle of men. "I remember Aitken in his beginning here in San Francisco. Though he often did poor stuff, everything of his showed artistic courage and initiative. Even then anyone could see there was something in him. Now it's coming out in the work he has contributed to this Exposition. The qualities in these four statues we shall see again when we reach the fountain that Aitken made for the Court of Abundance. They are individual without being eccentric. Compare these four figures with the groups in front of the two arches, by Paul Manship, another American sculptor of ability, but different from Aitken in his devotion to the early Greek. When Manship began his work a few years ago he was influenced by Rodin. Then he went to Rome and became charmed with the antique. Now he follows the antique method altogether. He deliberately conventionalizes. And yet his work is not at all conventional. He manages to put distinct life into it. These two groups, the 'Dancing Girls' and 'Music,' would have delighted the sculptors of the classic period."

Under the Arch of the Rising Sun two delicate murals by Edward Simmons charmed us by their grace, their lovely coloring, by the richness of their fancy and by the extraordinary fineness of their workmanship. "There's a big difference of opinion about those canvases as murals. But there's no difference of opinion in regard to their artistic merit. They are unquestionably masterpieces. Kelham and Guerin, who had a good deal to do with putting them up there, believe they are in exactly the right place. But a good many others think they are almost lost in all this heavy architecture. You see, Simmons didn't take Guerin's advice as to a subject. Each of his two murals has a meaning, or rather a good many meanings, but no central theme, no story that binds the figures into a distinct unity. So, from the point of view of the public, they are somewhat puzzling. People look up there and wonder what those figures are doing. But to the artist they find their justification merely in being

what they are, beautiful in outline and in posture and coloring. You don't often get such atmosphere in mural work, or such subtlety and richness of feeling."

Both murals unmistakably showed the same hand. "There's not another man in the country who could do work of just that kind. That group in the center of the mural to the north could be cut out and made into a picture just as it stands. It doesn't help much to know that the middle figure, with the upraised arm, is Inspiration with Commerce at her right and Truth at her left. They might express almost any symbols that were related to beauty. And the symbolism of the groups at either end seems rather gratuitous. They might be many other things besides true hope and false hope and abundance standing beside the family. But the girl chasing the bubble blown out by false hope makes a quaint conceit to express adventure, though perhaps only one out of a million would see the point if it weren't explained."

The opposite mural we found a little more definite in its symbolism, if not so pictorial or charming. The figures consisted of the imaginary type of the figure from the lost Atlantis; the Roman fighter; the Spanish adventurer, suggesting Columbus; the English type of sea-faring explorer, Sir Walter Raleigh; the priest who followed in the wake of the discoverer, the bearer of the cross to the new land; the artist, spreading civilization, and the laborer, modern in type, universal in significance, interesting here as standing for the industrial enterprise of today.

"Those murals suggest what a big chance our decorators have in the themes that come out of our industrial life. They've only made a start. As mural decoration advances in this country, we ought to produce men able to deal in a vigorous and imaginative way with the big spiritual and economic conceptions that are associated with our new ideals of industry."

One feature of this court made a special appeal to the architect, the use of the large green vases under the arches. "They're so good they're likely to be overlooked. They blend perfectly in the general scheme. Their coloring could not have been better chosen and their design is particularly happy."

VI

On the Marina

Along one of the corridors we passed, enjoying the richness of the coloring and the beauty of the great lamps in a long row, then out into the wide entrance of the court to the Column of Progress.

"I wonder if that column would be there now," said the architect, "if Trajan had not built his column in Rome nearly two thousand years ago. The Christianizing of the column, by placing St. Peter on top instead of Trajan, is symbolic of a good deal that has gone on here. But we owe a big debt to the pagans, much more than we acknowledge."

When I expressed enthusiasm over the column the architect ran his eye past the frieze to the top. "In the first place, that dominating group up there ought at once to express the character of the column. But it doesn't. You have to look twice and you

have to look hard. One figure would have been more effective. But there is a prejudice among some sculptors against placing a single figure at the head of a column, though the Romans often did it. But if a group had to be used it could have been made much clearer. Now in that design MacNeil celebrated the Adventurous Archer in a way that was distinctly old-fashioned. He made the archer a superman, pushing his way forward by force, and by the dominance of personality. And see how comparatively insignificant he made the supporting figures. The relation of those three people implies an acceptation of the old ideals of the social organization. MacNeil had a chance here to express the new spirit of today, the spirit that honors the common man and that makes an ideal of social co-operation on terms of equality."

At the base we studied the figures celebrating labor. "Konti is a man of broad social understanding and sympathy," said my companion. "But picturesque as those figures are, they're not much more. They give no intimation of the mighty stirring among the laborers of the world, a theme that might well inspire the sculpture of today, one of the greatest of all human themes."

From the Column of Progress the Marina drew us over to the seawall. "The builders were wise to leave this space open and to keep it simple. It's as if they said: 'Ladies and gentlemen, we have done our best. But here's Mother Nature. She can do better.' "

To our right stood Alcatraz, shaped like a battleship, with the Berkeley hills in the distant background. To the left rose Tamalpais in a majestic peak.

When I mentioned that there ought to be more boats out there on the bay, a whole fleet, and some of them with colored sails, to give more brightness, the architect shook his head.

"The scene is typically Californian. It suggests great stretches of vacant country here in this State, waiting for the people to come from the overcrowded East and Middle West and thrive on the land."

Our point of view on the Esplanade enabled us to take in the sweep of the northern wall, with its straight horizontal lines, broken by the entrances to the courts and by the splendidly ornate doors in duplicate. Of the design above the doorway the architect said: "It's a perfect example of the silver-platter style of Spain, generally called 'plateresque,' adapted to the Exposition. Allen Newman's figure of the Conquistador is full of spirit, and the bow-legged pirate is a triumph of humorous characterization. Can't you see him walking the deck, with the rope in his hand? It isn't so many generations since he used to infest the Pacific. By the way, that rope, which the sculptor has made so realistic and picturesque at the same time, reminds me that a good many people are bothered because the bow up here, on the Column of Progress, has no string. The artistic folk, of course, think that the string ought to be left to the imagination."

In the distance, to the west, we commented on the noble outlines of the California Building, an idealized type of Mission architecture, a little too severe, perhaps, lacking in variety and warmth, but of an impressive dignity. The old friars, for all their asceticism, liked gaiety and color in their building.

As we were about to start back to the Court of the Universe the architect reminded me of the two magnificent towers, dedicated to Balboa and Columbus, that had been planned for the approach to the Court of Four Seasons and the Court of Ages from the bay side, but had been omitted to save expense. They would have given the Marina a far greater splendor; but they would have detracted from its present simplicity.

VII

Toward the Court of Four Seasons

"There are critics," I remarked, as we walked back to the Court of the Universe, on the way to the Court of Four Seasons, "who say that the entrance courts ought to have been placed on the other side that the Exposition ought to have been turned round."

"They don't understand the conditions that the architects had to meet. That plan was considered; but when it was pointed out that the strongest winds here blow from the south and southwest, it was seen that it would not be feasible. Besides, the present arrangement has the advantage of leading the people directly to one of the most beautiful bays in the world. The only bays at all like it that I know anything about are the Bay of Palermo and the Bay of Naples. The view of the Exposition from the water is wonderfully fine. It brings out the charm of the straight lines. All things considered, the architects did an uncommonly fine job in making the courts run from the Esplanade."

Under the star figures, among the sculptured flowers' surrounding the head of the sacred bull, birds were nestling. We wondered if those birds were really fooled by those flowers or whether, in these niches, they merely found a comfortable place to rest. "There's an intimate relation, by the way, between birds and architecture. It's said that the first architectural work done in the world consisted in the making of a bird's nest. Some critics think that architecture had its start in the making of a bird's nest. Have you ever watched birds at work on their nests? If you have, you must know that they go about the job like artists. In our profession we like to insist, you know, that there's a big difference between architecture and mere building. In its truest sense architecture is building with a fine motive. It's the artistic printing press of all ages, the noblest of the fine arts and the finest of the useful arts. I know, of course," the architect went on, "that there's another tradition not quite so flattering. It makes the architect merely the worker in the rough, with the artistic finish left to the sculptors. But the outline is nevertheless the architect's, the structure, which is the basis of beauty. Even now a good many of the great French buildings are roughed out in this way, and finished by the sculptors and the decorators."

Under the western arch, leading to the inner court that united the Court of the Universe with the Court of the Four Seasons, we found the two panels by Frank Vincent Du Mond. Their simple story they told plainly enough, the departure of the pioneers from the Atlantic border for the Far West on the Pacific. In the panel to the right we saw the older generation saying farewell to the younger, and on the other

side we saw the travelers arriving in California and finding a royal welcome from the Westerners in a scene of typical abundance, even the California bear showing himself in amiable mood. "That bear bothered Du Mond a good deal. He wasn't used to painting bears. It isn't nearly as life-like as those human figures."

What I liked best about the murals was their splendor of coloring, and their pictorial suggestiveness and vigor of characterization. Perhaps there was a little too much effort on the part of the painter to suggest animation. But why, I asked, had Du Mond made most of the faces so distinctively Jewish?

My question was received with an exclamation of surprise. Yes, the strong Jewish types of features were certainly repeated again and again. Perhaps Du Mond happened to use Jewish models. It hardly seemed possible that the effect could have been intentional.

When I pointed to one of the figures, a youth holding out a long bare arm, and remarked that I had never seen an arm of such length, my criticism brought out an unsuspected principle of art. "The Cubists would say that you were altogether too literal. They are making us all understand that what art ought to do is to express not what we merely see with our eyes, but what we feel. If by lengthening that arm, the painter gets an effect that he wants, he's justified in refusing to be bound by the mathematical facts of nature. Art is not a matter of strict calculation, that is, art at its best and its purest. It's a matter of spiritual perception. All the resources of the artist ought to be bent toward expressing a spiritual idea and making it alive and beautiful through outline and color."

"But how about the mixture of allegory and realism that we see in these murals and in so much of the art here? Don't you find it disturbing?"

"Not at all. There's no reason in the world why the allegorical and the real should not go together, provided, of course, they don't grossly conflict and become absurd. What the artist is always working for is the effect of beauty. If a picture is beautiful, no matter how the beauty is achieved, it deserves recognition as a work of art. In these murals Du Mond has tried to reach as closely as he could to nature without being too literal and without sacrificing artistic effect. He has even introduced among his figures some well-known Californians, a Bret Harte, in the gown of the scholar, and William Keith, carrying a portfolio to suggest his painting."

In that inner court we noticed how cleverly Faville had subordinated the architecture so that it should modestly connect the great central courts. McLaren was keeping it glowing on either side with the most brilliant California flowers. The ornamental columns, the Spanish doorways, and the great windows of simple and yet graceful design were all harmonious, and Guerin and Ryan had helped out with the coloring.

VIII

The Court of the Four Seasons

As we entered the Court of the Four Seasons the architect said: "If I were to send a student of architecture to this Exposition, I should advise him to spend most of his time here. Of all the courts, it expresses for me the best architectural traditions. Henry Bacon frankly took Hadrian's Villa for his model, and he succeeded in keeping every feature classic. That half dome is an excellent example of a style cultivated by the Romans. The four niches with the groups of the seasons, by Piccirilli, screened behind the double columns, come from a detail in the baths of Caracalla. The Romans liked to glimpse scenes or statuary through columns. Guerin has applied a rich coloring, his favorite pink, and McLaren has added a poetic touch by letting garlands of the African dew plant, that he made his hedge of, flow over from the top. See how Bacon has used the bull's head between the flowers in the ornamentation, one of the most popular of the Renaissance motives. And he has introduced an original detail by letting ears of corn hang from the top of the columns. Those bulls up there, with the two figures, carry the mind back to the days when the Romans made a sacrifice of the sacred bull in the harvest festivals. This Thanksgiving of theirs they called 'The Feast of the Sacrifice.' "

Crowning the half dome sat the lovely figure of Nature, laden with fruits, by Albert Jaegers. On the columns at either side stood two other figures by Jaegers, "Rain," holding out a shell to catch the drops, and "Sunshine," with a palm branch close to her eyes. At each base the figures of the harvesters carried out the agricultural idea with elemental simplicity in friezes that recalled the friezes on the Parthenon. Here, on each side of the half-dome, we have a good example of the composite column, a combination of the Corinthian and the Ionic, with the Ionic scrolls and the acanthus underneath, and with little human figures between the two.

What we liked best about this court was its feeling of intimacy. One could find refreshment here and rest. Much was due to the graceful planting by John McLaren. His masses of deep green around the emerald pool in the center were particularly successful. He had used many kinds of trees, including the olive, the acacia, the eucalyptus, the cypress, and the English laurel.

We lingered in front of these fountains, admiring the classic grace of the groups and the play of water over the steps. We thought that Piccirilli had been most successful with his "Spring." "Of course, it's very conventional work," said the architect, "but the conventional has its place here. It explains just why Milton Bancroft worked out those murals of his in this particular way. He wanted to express the elemental attitude of mind toward nature, the artistic childhood of the race."

When we examined the figures of the Piccirilli groups in detail, we found that they possessed excellent qualities. They carried on the traditions of the wall-fountains so popular in Rome and often associated with water running over steps. The figures were well put together and the lines were good. All of the groups had the surface as carefully worked out. In "Spring" the line of festooning helped to carry on the line leading to the top of the group. There was tender feeling and fine workmanship in "Summer," with the feminine and masculine hands clearly differentiated. "The men of today have a chance to learn a good lesson from Rodin," said the painter. "He is teaching them what he himself may have learned from the work of Donatello and

Michael Angelo, the importance of surface accentuation, the securing of the light and shade that are just as necessary in modelling as in painting. In these groups there is definite accentuation of the muscles. It makes the figures seem life-like. The work reminds me of the figure of The Outcast, by the sculpter's brother, Attilio Piccirilli, that we shall see in the colonade of the Fine Arts Palace. So many sculptors like to secure these smooth, meaningless surfaces that excite admiration among those people who care for mere prettiness. It is just about as admirable as the smoothing out of character lines from a photograph. But the Piccirillis go at their work like genuine artists."

Those murals we were inclined to regard as somewhat too simple and formal. "After all," said the architect, "it's a question whether this kind of effort is in the right direction. So often it leads to what seems like acting in art, regarded by some people as insincerity. At any rate, the best that can be said of it is that it's clever imitation. But here it blends in with the feeling of the court and it gives bright spots of color. Guerin has gone as close to white as he dared. So he felt the need of strong color contrasts, and he got Bancroft to supply them. And the colors are repeated in the the other decorations of the court. It's as if the painter had been given a definite number of colors to work with. In this matter of color, by the way, Bancroft had a big advantage over the old Roman painters. Their colors were very restricted. In this court they might have allowed more space for the murals. They're not only limited in size, but in shape as well. Bancroft used to call them his postage-stamps.

In the entrance court we found Evelyn Breatrice Longman's "Fountain of Ceres," the last of the three fountains done on the grounds by women, and decidedly the most feminine. "Mrs. Longman hasn't quite caught the true note," the architect remarked. "The base of the fountain is interesting, though I don't care for the shape. But the figure itself is too prim and modish. Somehow I can't think of Ceres as a proper old maid, dressed with modern frills. The execution, however, shows a good deal of skill. The frieze might be improved by the softening of those sharp lines that cut out the figures like pasteboard. And these women haven't as much vitality as that grotesque head down near the base, spouting out water." The architect glanced up and noticed the figure of "Victory" on one of the gables, so often to be seen during a walk over the grounds. "There's more swing to that figure than to the one here, and yet there's a certain resemblance between them. They both show the same influence, the Winged Victory of Samothrace. Of course, Miss Longman has purposely softened the effect on account of the mildness of her subject. But she might have been more successful with her draperies if she had followed the suggestions in the Winged Victory more closely. There the treatment of the draperies is magnificent. Both the Greeks and the Romans were very fond of this type of figure. And it's often found among the ruins of Pompeii, which kept so close to Rome in its artistic enterprise."

The need of separating the entrance to the Court of the Four Seasons from Ryan's display of scintillators on the imitation of Morro Castle at the edge of the bay, had given John McLaren a chance to create another of these deep green masses that surrounded the pool. It shut the court off from the rest of the world and deepened the intimacy, leaving, however, glimpses of the bay and the hills beyond.

IX

The Palace of Fine Arts From Across the Lagoon

In returning to the Court of the Four Seasons, we started along another of those inner courts, made charming by those Spanish doorways and by the twisted columns, a favorite of the Romans, evidently borrowed from the Orientals. "All through the Exposition," the architect remarked, "we are reminded of the Oriental fondness for the serpent. Some people like to say that it betrays the subtlety and slyness of the Oriental people. But they admired the serpent chiefly because, in their minds, it represented wisdom, the quiet and easy way of doing things, a little roundabout perhaps, but often better than the method of opposition and attack."

Before us, looking down as if from an eminence, stood, the Palace of Fine Arts. The architect reminded me of the clever planning that had placed this magnificent conception in so commanding a position, looking down into the courts, on what he called "the main axis."

"It's the vision of a painter who is also a poet, worked out in terms of architecture. Maybeck planned it all, even to the details. He wanted to suggest a splendid ruin, suddenly come upon by travelers, after a long journey in a desert. He has invested the whole place with an atmosphere of tragedy. It's Roman in feeling and Greek in the refinement of its ornamentation. That rotunda reminds one of the Pantheon in Rome. Those Corinthian columns, with the melancholy drooping of the acanthus and the fretwork and the frieze, by Zimm, are suggestive of Greece. Maybeck says that his mind was started on the conception, 'The Island of Death,' by Boecklin, the painting that the German people know so well as the 'Todteninsel,' and by 'The Chariot Race,' of Gerome."

The architect went on to say that the resemblance was remote and chiefly interesting as showing how a great artist could carry a suggestion into an entirely new realm. The Boecklin painting merely suggested the general scope of the work, and the chariot race gave the hint for that colonnade, which Maybeck had made so original and graceful by the use of the urns on top of groups of columns with the figure of a woman at each corner. He had used that somewhat eccentric scheme on account of its pictorial charm. All through the construction Maybeck had defied the architectural conventions; but he had been justified by his success.

My attention was directed to a group of columns at the end of the colonnade. "There's just a hint of the Roman Forum over there. Perhaps it's accidental. Perhaps it's developed from a picture way down in Maybeck's consciousness. However, the idea of putting two columns together in just that way comes from the French Renaissance. The great French architect, Perrault, used it in the Louvre. In the competition he won out over Bernini, who is living again in the Court of the Universe. It gives great architectural richness."

People had wondered what McLaren had meant to indicate by the high hedges he had made over there with his dew plant. He had merely carried out the designs put into his hands. Maybeck had intended the hedge to be used as a background for

willow trees that were to run up as high as the frieze, in this way gaining depth. Through those trees the rotunda was to be glimpsed. Willow trees, with overhanging boughs, were also to be planted along the edge of the lagoon, the water running under the leaves and disappearing.

In the lagoon swans were swimming and arching their long necks. "The old Greeks and Romans would have loved this scene, though they would, of course, have found alien influences here," said the architect. "They would have enjoyed the sequestration of the Palace, its being set apart, giving the impression of loneliness. The architects were shrewd in making the approach long and circuitous."

"They might have done more with the water that was here before they filled in," I said. "It offered fine chances."

"Yes, and they thought of them and some ambitious plans were discussed. But the expense was found to be prohibitive."

At that moment a guard, in his yellow uniform with brass buttons, came forward with a questioning lady at his side. They stood so close to us that we could not help hearing their talk.

"What are those women doing up there?"

The guard looked at the urns, surmounting the columns. "They're supposed to be crying," he said.

"What are they crying about?"

The guard looked a little embarrassed. "They are crying over the sadness of art," he said. Then he added somewhat apologetically, "Anyway, that's what the lecturer told us to say."

The lady appealed to us for information. "What this gentleman says is true," remarked the authority at my side. "The architect intended that those figures should express something of the sadness of life as reflected in art."

"Oh," said the lady, as if she only half understood.

Then she and the guard drifted away.

"Those people have unconsciously given us a bit of art criticism, haven't they? One of the most pictorial notes in this composition of Maybeck's is the use of these figures. But it's also eccentric and it puzzles the average looker-on who is always searching after meanings, according to the literary habit of the day, the result of universal reading. Perhaps the effect would have been, less bewildering if those urns were filled with flowers as Maybeck intended they should be. Then the women would have seemed to be bending over the flowers. The little doors were put into the urns so that the man in charge of the flowers could reach up to them. But this item of expense was included among the sacrifices."

The coloring of the columns had been a subject of some criticism. The ochre columns were generally admired; but the green columns were considered too atmospheric to give the sense of support. And that imitation of green marble directly

under the Pegasus frieze of Zimm's, near the top, had been found to bear a certain resemblance to linoleum. But in applying, the colors Guerin had worked with deliberate purpose. The green under the frieze was really a good imitation of marble, and the shade used on the column suggested the weather-beaten effect associated with age.

"There are columns that, in my opinion, have more beauty than those Maybeck used. But that's a matter of taste. In themselves those columns are fine and they blend into impressive masses. That altar under the dome, with the kneeling figure, only a great artist could have conceived in just that way. Ralph Stackpole, the sculptor of the figure, worked it out in perfect harmony with Maybeck's idea. To appreciate his skill one ought to get close and see how roughly it has been modeled in order that the lines should be clear and yet give an effect of delicacy across the lagoon. And those trees along the edge of the lagoon, how gracefully they are planted, in the true Greek spirit. The lines in front of the rotunda are all good, as they run down to the water's edge. And how richly McLaren has planted the lagoon. He has given just the luxuriance that Maybeck wanted."

The Western Wall

We turned to get the effect of the western wall looking out on this magnificence. "Faville has done some of his finest work there. All over the Exposition he has expressed himself; but as his name is not connected with one of the great courts we don't hear it very much. When he tackled the Western Wall he had one of the hardest of his problems. There was a big expanse to be made interesting and impressive, without the aid of towers or courts. It was a brilliant idea to break the monotony with those two splendid Roman half-domes."

The figure of "Thought" on the columns in front of the Dome of Plenty and repeated on the Dome of Philosophy started the architect talking on the subject of character and art. "Only a sculptor with a very fine nature could have done that fellow up there. In that design Stackpole shows the qualities that he shows in the kneeling girl at the altar in the rotunda across the lagoon and in his figure of the common laborer and the little group of artisans and artists that we shall see on the doorway of the Varied Industries. They include fineness and cleanness of feeling, reverence and tenderness. This particular figure is one of three figures on the grounds that stand for virtually the same subject, Rodin's "Thinker," in the courtyard of the French Building, and Chester Beach's "Thinker," in the niches to the west and east of the tower in the Court of the Ages. They are all different in character. Stackpole's gives the feeling of gentle contemplation. That man might be a poet or a philosopher or an inventor; but a man of the kind of thought that leads to action or great achievement in the world - never. You can't think of him as competing with his whole heart and soul in order to get ahead of other men. However, it would be an achievement just to be that type and it's a good type to be held up to us for our admiration, better than the conventional ideal of success embodied in the Adventurous Bowman, for example."

The proportions of the domes we could see at a glance had been well worked out. Earl Cummings' figure of the Youth had a really youthful quality; but there was some

question in our minds as to the wisdom of repeating the figure in a semi-circle. "After all," the architect remarked, "in this country art owes some concession to habit of mind. We are not trained to frankness in regard to nudity. On the contrary, all our conventions are against it. But our artists, through their special professional training, learn to despise many of our conventions and they like to ignore them or frankly show their contempt for them."

That elaborate Sienna fountain was well adapted to the Dome of Plenty, though it was by no means a fine example of Italian work, with its design built up tier on tier. "It's the natural expression of a single idea that leads to beauty, isn't it? The instant there's a betrayal of effort, the charm begins to fade."

There was no criticism to be made, however, of the Italian fountain in the Dome of Philosophy, the simplest of all the fountains, and one of the most beautiful, the water flowing over the circular bowl from all sides. "It makes water the chief feature," said the architect approvingly, "which is the best any fountain can do. Is there anything in art that can compare for beauty with running water? This fountain comes from Italy and these female figures, above the doorway, with books in their arms, are by one of the most interesting of the sculptors represented here, Albert Weinert. We'll see more work of his when we get to the Court of Abundance."

At sight of the curious groups in the niches I expressed a certain disappointment. It seemed to me that, in the midst of so much real beauty, they were out of key. But the architect had another point of view. "They are worth while because they're different," he said. "They ought not to be considered merely as ornaments. They have an archaeological interest. They are related to those interesting studies that Albert Durer used to make, and they are full of symbolism. When Charles Harley made them he knew just what he was doing. The male figure in 'The Triumph of the Fields' takes us back to the time when harvesting was associated with pagan rites. The Celtic cross and the standard with the bull on top used to be carried through the field in harvest time. The bull celebrates the animal that has aided man in gathering the crops. The wain represents the old harvest wagon. That head down there typifies the seed of the earth, symbol of the life that comes up in the barley that is indicated there, bringing food to mankind. The woman's figure, unfortunately, is too small for the niche, 'Abundance.' The horn of plenty on either side indicates her character. She's reaching out her hands to suggest her prodigality. The head of the eagle on the prow of the ship where she is sitting, gives the idea an American application, suggesting our natural prosperity and our reason for keeping ahead in the march of progress. In one sense, those figures represent a reactionary kind of sculpture. Nowadays the sculptors, like the painters, are trying to get away from literal interpretations. They don't want to appeal to the mind so much as to the emotions."

X

The Palace of Fine Arts at Close Range

The path leading to the northern end of the colonnade attracted us. It brought us to the beautiful little grove of Monterey cypress that McLaren had saved from the old

Harbor View restaurant, for so many years one of the most curious and picturesque of the San Francisco resorts, one of the few on the bay-side. Though the architect frankly admired Paul Bartlett's realistic "Wounded Lion," the pieces of sculpture set out on the grass bothered him somewhat. He couldn't find any justification for their being there. He wanted them, as he said, in a setting. "I think I can see what the purpose was in putting them here, to provide decoration that would be unobtrusive. But some of these pieces, like Bartlett's, stand out conspicuously and deserve to be treated with more consideration. Besides, there's always danger of weakening a glorious conception like Maybeck's by putting too many things into it, creating an artistic confusion."

We began to see how the colonnade in Gerome's painting had worked its influence. It was easy to imagine two chariots tearing along here, between the columns, after the ancient fashion. And those bushes, to the right, rising on the lower wall, between the vases, surely had the character of over-growth. They carried out Maybeck's idea of an abandoned ruin.

The architect pointed to the top of the wall: "The little roof-garden on the edge of the upper wall gives the Egyptian note in the architecture that many people have felt and it is emphasized by the deep red that Guerin has applied, the shade that's often found in Egyptian ruins."

Above the main entrance of the palace we saw Lentelli's "Aspiration," that had been the cause of so much criticism and humorous comment during the first few weeks of the Exposition. "Lentelli had a hard time with that figure. It drove him almost to distraction. Perhaps a genius might have solved the problem of making the figure seem to float; but I doubt if it could have been solved by anyone. The foot-rest they finally decided to put under it didn't help the situation much."

Directly in front of "Aspiration," on its high pedestal, stood Charles Grafly's monumental statue of "The Pioneer Mother." "I suppose the obvious in sculpture has its place," the architect remarked, "and this group will appeal to popular sentiment. Its chief value lies in its celebrating a type of woman that deserves much more recognition than she has received in the past. Most of the glory of the pioneer days has gone to the men. The women, however, in the background, had to share in the hardships and often did a large part of the work. It's a question in my mind whether this woman quite represents the vigorous type that came over the plains in the prairie schooner. However, just as she is, she is fine, and she has a strong hand that looks as if it had been made for spanking. I wonder why the sculptor gave her that kind of head-covering. She might have appeared to better advantage bare-headed. The children are excellent. Observe the bright outlook of the boy and the timid attitude of the girl. There's a fine tenderness in the care the girl is getting from her mother and from the boy, too, suggesting dawning manhood. Altogether, the group has nobility and it's worthy of being a permanent monument for San Francisco. By the way, there's the old Roman idea of the decorative use of the bull's head again, at the base of the group. It has a very happy application here. It reminds us of the oxen that helped to get the Easterners out to California in the old days before the railroads. A good many of them must have dropped in their tracks and left their skulls to bleach in

the sun."

The other ornamental design we found very appropriate and direct, as we studied the pedestal. There was the ship that used to go round the horn, with the torches that suggested civilization, and, at the back of the pedestal, the flaming sun that celebrated the Golden Gate.

In the rotunda we found Paul Bartlett, represented again by the equestrian statue of Lafayette, in full uniform, advancing sword in the air. It unquestionably had a magnificent setting, though it suffered by being surrounded by so many disturbing interests. "The director of the Fine Arts Department cared enough about this figure to have it duplicated for the Exposition. It's a good example of the old-fashioned heroic sculpture, where the subjects take conventional dramatic attitudes."

The ceiling of the rotunda displayed those much-discussed murals by Robert Reid. Up there they seemed like pale reflections. "You should have seen them when they were in Machinery Hall. Then they were magnificent. But the instant they were put in place it was plain that the effect had been miscalculated. At night, under the lighting, they show up better. Judged by themselves, apart from their surroundings, they are full of inspiration and poetry. Only a man of genuine feeling and with a fine color-sense could have done them. But in all this splendor of architecture they are lost."

On examining them in detail we found that they covered an extraordinarily wide range of fancy, graceful and dramatic, even while, save in one panel, they showed an indifference to story-telling. One group celebrated "The Birth of European Art," with the altar and the sacred flame, tended by a female guardian and three helpers, and with a messenger reaching from his chariot to seize the torch of inspiration and to bear it in triumph through the world, the future intimated by the crystal held in the hands of the woman at the left. Another, "The Birth of Oriental Art," told the ancient legend of a Chinese warrior who, seated on the back of a dragon, gave battle to an eagle, the symbol relating to man's seeking inspiration from the air. "Ideals in Art" brought forward more or less familiar types: the Madonna and the Child, Joan of Arc, Youth and Beauty, in the figure of a girl, Vanity in the Peacock, with more shadowy intimations in two mystical figures in the background, the tender of the sacred flame and the bearer of the palm for the dead, and the laurel-bearer ready to crown victory. "The Inspiration in All Art" revealed the figures of Music, Architecture, Painting, Poetry and Sculpture. Four other panels glorified the four golds of California, gold, wheat, poppies and oranges, a happy idea, providing opportunities for the splendid use of color.

"It's a pity those murals couldn't have been tried out up there and then taken down and done over," said the architect. "But sometime they will find the place where they belong, perhaps in one of our San Francisco public buildings. They're too good not to have the right kind of display."

"The Priestess of Culture," by Herbert Adams, one of the best-known of American sculptors, eight times repeated, we felt, had its rightful place up there and blended into the general architectural scheme. But some of the other pieces of statuary might

have been left out with advantage.

Through the columns we caught many beautiful vistas. And those groups of columns themselves made pictures. "What is most surprising about this palace is the way it grows on you. The more familiar you are with it the more you feel the charm. Maybeck advises his friends to come here by moonlight when they can get just the effect he intended. In all the Exposition there's no other spot quite so romantic. It might have been built for lovers."

XI

At the Palace of Horticulture

At the Palace of Horticulture the architect said: "Here is the Mosque of Ahmed the First, taken from Constantinople and adapted to horticulture and to the Exposition. It has a distinct character of its own. It even has temperament. So many buildings that are well proportioned give the impression of being stodgy and dull. They are like the people that make goodness seem uninteresting. But here is use that expresses itself in beauty and adorns itself with appropriate decoration."

When I mentioned that some people found this building too ornate, the architect replied:

"There's an intimate and appropriate relation between the ornament and the architecture. Personally I shouldn't care to see just this kind of building in the heart of the city, where you'd have it before your eyes every day. But for the Exposition it's just right. And how fitting it is that the splendid dome should be the chief feature of a building that is really an indoor garden and that the most prominent note of the coloring should be green, nature's favorite and most joyous color. Some joker," he went on, "says that this Exposition is domicidal. He expresses a feeling a good many people have here, that there are too many domes. But I don't agree. The domes make a charming pictorial effect, and they harmonize with the general spirit of the architecture. And as for this dome, it is one of the greatest in the world. See how cleverly the architects, following the spirit of the French Renaissance, have used those ornamental shafts. The only criticism that can be made on them is that they serve no architectural purpose, which ought, of course, always to be intimately associated with use. Instead of growing from the nature of the building, they are put on from outside. Now, in the mosque they were very important in their service. They were the minarets where the Muezzins used to stand in order to call the faithful to prayer. Those minarets up there, carrying on the dome motive, on the corners of the walls of the main palaces are much closer to the old idea."

Our talk turned to the subject of domes in general. The idea had come from the bees, from the shape of their hives. Prehistoric man used for a dwelling-place a hut shaped like a hive, as well as an imitation of a bird's nest. In formal architecture, the dome showed itself early. The Greeks knew it; but they didn't use it much. The greatest users of the dome were the Byzantines. It was all dome with them. The first important dome was built in Rome in the second century, to crown the Pantheon. Of

all the domes in the world the most interesting historically was St. Peter's, the work of several architects. It was the inspiration of the dome of St. Paul's in London, built by the English architect, Sir Christopher Wren. Architecturally the most interesting of the domes was Brunelleschi's, built for the Florence Cathedral in the fifteenth century, known throughout the world by the Italian name for Cathedral, the Duomo.

It was in connection with the Duomo that the architect reminded me of the celebrated story about Brunelleschi. When the Florentine church authorities decided to build the Duomo they were puzzled as to how so mighty a dome should be developed. So they invited the architects to appear before them in competition, and to present their ideas. One architect, Donatello, explained that, if he secured the commission, he should first build a mound of earth, and over it he would construct his dome. But the authorities replied that there would be great labor and expense in taking the earth out. He said that he would put coins into the earth and, by this means, he would very quickly have the earth removed by the people. When Brunelleschi was asked how he would build his dome he said: "How would you make an egg stand on end?" They didn't know how, and he showed them, by taking a hard-boiled egg and pressing it down at one end, an idea like the one that occurred to Christopher Columbus about fifty years later.

The Palace of Horticulture as an illustration of French Renaissance architecture fascinated this observer, in spite of its overelaborateness. "It's marvelous to think of what the Renaissance meant throughout Europe," he said, "and how it showed itself in art through the national characteristics. French Renaissance and Italian Renaissance, though they have qualities in common, are very different. And you'll find marked differences even in the Renaissance art of the Italian cities, such as Rome and Florence and Venice. But the Renaissance showed that no matter how far apart the people of Europe might have been they were all stirred by a great intellectual and spiritual movement. It was like a vast moral earthquake. It meant the rediscovery and the joyous recognition of the relation of the past to the present and the meaning of the relation for mankind. It led to a new kind of self-emancipation and individualism. It created art-forms that have stamped themselves on the work all over these grounds. In a sense it was a declaration of artistic independence."

"Is there really such a thing as independence in art?" I ventured to ask.

The architect began to smile. "I'm afraid there isn't much independence. If there were this Exposition would not be quite so intimately related to Europe and the Orient. But wait till we get into Mullgardt's Court of the Ages. Then you'll find an answer to your question."

At this palace the architect found much to speculate on. "Here is one of the few buildings in the whole Exposition done in what might be called the conventional exposition spirit. I like it immensely as an exposition building, but I should hate it as a public building that I had to see every day. It's too fantastic. In this place it serves its purpose. But it might fit into a setting like the Golden Gate Park, where it would be close to nature. Now this Exposition is very different from most of the enterprises of the kind that have taken place in Europe. It is probably the most serious exposition

ever known, with the possible exception of the one in Chicago. If it were in a great European capital, for example, it would mainly express the spirit of gaiety. But the builders here, though they have been gay in their use of color, have been tremendously serious in purpose. They have worked largely for the sake of education."

The use of green on the building was unquestionably one of the most successful features of the coloring, particularly when it suggested, as it so often did, old copper. "To me the deeper green that Guerin uses is the more charming shade, far more charming, for instance, than the light green applied to Festival Hall. And the suggestion of green in the dome is altogether delightful. But it's a pity they didn't use another kind of glass. When people criticise Ryan for not doing more with his lighting effects-in this dome they evidently don't know that a mistake was made when the glass was sent and Ryan could do very little with it. In order to carry out his original plans Ryan would have to apply a coat of varnish to the interior of the dome, a rather expensive process. However, it may be done later."

Returning to the South Gardens

From where we stood we could get a good view of those green columns in the Tower of Jewels, occasionally criticised as being too atmospheric to give the sense of support. "Those columns were colored by Guerin to get an effect of contrast. That shade was one of the first of the shades he experimented with. He tried it out on the sashes in Machinery Hall. The French landscape painters used it a good deal in outdoor scenes, on trellises, for example. It made a pleasing effect against the deeper tones of the grass and foliage. The notion that it isn't suited to columns seems to me unwarranted. As a matter of fact, there are several kinds of green stone that have often been successfully used for columns in architecture, like malachite and Connemara marble. The Bank of Montreal has some magnificent Connemara columns. Of course, the use up there is theatrical, exactly as Guerin intended it to be. People seem to forget that Guerin got his earlier training as a scene painter. He was recognized as one of the greatest scene painters of his time. He deliberately undertook to make this Exposition a great spectacle, and he ought to be judged according to what he tried to do. It seems to me that his success was astonishing. He created a picture that was spectacular without being garish or cheap and that harmonized with the dignity and the splendor of the architecture. One explanation of his success lies in his being so fond of the Orient, where the architects have worked in color as far back as we can go. Every chance he makes a trip to the Orient and he comes back with a lot of Oriental canvases that he has painted there. Only a lover of the Orient would have dared to put that orange color on the domes. See what a velvety look he got, almost wax-like. He was careful not to apply, in most instances, more than one coat of paint. He wanted it to sink in and to become weathered. He knew that nature was the greatest of all artists, always trying to remove the shiny appearance of newness and to give seasoning."

As we looked up toward the center of the South Garden the white globes on the French lamp posts caught the architect's eye. "Don't you remember how cheap they looked on the first days?" he said. "The trouble was that they were too white. They

seemed cold and raw. So they were sprayed with a liquid celluloid to soften them into their present ivory hue. The change shows how important detail is, and how carefully Guerin's department has worked. While the construction was going on there was one remark that often used to be heard, 'It will never be noticed,' and a most foolish remark it was. It showed that the people who made it were lacking in imagination. Millions of eyes have been watching the details of this Exposition and very little has escaped notice."

A great crowd was pouring out of the afternoon concert in Festival Hall. The architect, as he looked on, remarked: "It's like being in Paris, isn't it? Or, perhaps, it's more like being in a lovely old French provincial city, where the theater is the chief architectural monument. It's hard for me to understand why the French have encouraged that kind of architecture for their theaters and opera houses. It seems so unrelated to sound, which ought to give the clue to the building. The use of the word festival here is a little old-fashioned and misleading. It doesn't mean what we usually consider festivity. It is essentially a concert hall, and the architecture ought to suggest concentration of sound by being built in a way that shall make such concentration inevitable. But this kind of building is obviously related to dissipation of sound. No wonder the acoustics turned out bad and the interior had to be remodeled."

XII

The Half Courts

In front of the Court of Palms we stopped to admire James Earl Fraser's "End of the Trail," the most popular group of sculpture in the Exposition. "It deserves all its popularity, doesn't it? It's finely imagined and splendidly worked out. The pony is excellent in its modeling and the Indian is wonderfully life-like."

At our side a man and a woman were standing, the man more than six feet tall, with broad shoulders and a face that had evidently seen a good deal of weather. "I've known fellers just like that Indian," we heard him say, "up in Minnesota. He might be a Blackfoot after a couple of days' tusselling with the wind and the rain in the mountains. I've seen 'em come into town all beat out. The man that made that statue knew his business. An' I guess he knew what he was doing when he called it 'The End of the Trail."'

When the visitor had passed, the architect said: "The symbolism gets them all, doesn't it; and the realism, too? But Fraser couldn't have expressed so much if he hadn't put a lot of heart into his 'Work. He really felt all that the Indian represented, as a human being and as a representative of a dying race."

"The Court of Palms" captured us both, by its shape, by the splendor of the Ionic columns, by the loveliness of its detail, by its coloring and by that charm of its sunken garden. "You can feel here the mind that developed those four Italian towers. It shows the same balanced judgment, and skill and taste. The two towers here, though they stand at either end of the court, and make a beautiful ornamentation, are really

a part of the wall. They help to give it dignity and variety. And how artistically the palms have been used here. They can be among the least graceful of plants; but here they are really decorative. And those laurel trees at the side of the main doorway make fine ornamental notes. The sculptured vases, too, are wonderfully graceful."

Above the doorways we found the three murals that gave further distinction to this court and enriched the coloring. In "Fruits and Flowers" Childe Hassam had done one of his purely decorative pictures, without a story, contenting himself with graceful pictures and delicate color scheme. Charles Holloway made "The Pursuit of Pleasure" frankly allegorical, the floating figure of the woman pursued by admiring youths. Over the main doorway Arthur Mathews had also painted an allegory, "Victorious Spirit," the Angel of Light, with wide-spread wings of gold, standing in the center and keeping back the spirit of materialism, represented by a fiery horse driven by his rider with brutal energy. "Observe how successfully Mathews has chosen his colors. These deep purples help to bring out the splendor of those golden tones. This canvas is unquestionably one of the best of all the murals. It shows that in Mathews San Francisco has a man of remarkable talent, one of the great mural painters of the country."

On the way to the second half-court we had a chance to see the South Wall at close range, with its rich ornamented doorways, its little niches and fountains devised to make it varied and gay. Those little elephant heads were another sign of Faville's careful attention to ornamental detail. And the coloring gave warmth to the background, contrasting with the deep green of the planting.

At the Court of Flowers we met Solon Borglum's "Pioneer, too old to be typical, different from the man in lusty middle age or in youth who came to California in the early days. But it justified itself by suggesting perhaps the greatest of the pioneers in old age, one who had grown with the community, the poet, Joaquin Miller. "It's Miller sure enough," said the architect, "even if the likeness isn't close. But why those military trappings on the horse? Like the rest of the pioneers, Joaquin was a man of peace."

The Court of Flowers we thought well named, both for its planting, McLaren at his best, and for its Italian Renaissance decoration, with that pretty pergola opening out on the scene, Calder's Oriental "Flower Girl" decorating the spaces between the arches. And those lions by Albert Laessle were a fine decorative feature. The fountain, "Beauty and the Beast," by Edgar Walter, of San Francisco, was one of the most original and decorative pieces of sculpture we had seen. The figure of the girl standing on the coils of the beast was remarkably well done and the water flowing over the bowl, with the pipes of Pan glimpsed underneath, made a charming picture. There was a whimsical and a peculiarly French suggestion in the use of the decorative hat and sandals on the nude figure. In detail those two towers at the end were slightly different from the other two. Like the others they served as a decoration of the wall, breaking the long lines."

XIII

Near Festival Hall

At close view we found the Festival Hall more interesting than it had seemed at a distance. It unquestionably had something of the elegance associated with the best French architecture. But, unlike most of the buildings here, it did not develop out of a central idea. Much of its ornamentation seemed put on from the outside.

Of all the domes this dome impressed us as being the least interesting. It did not even justify itself as being a means of giving abundant light. "This kind of architecture doesn't really belong in this country; but it seems to be making its way. Observe the waste of space involved. However, the curving arches on either side are rather charming. And the architect has succeeded in putting into the whole structure a certain amount of sentiment. In fact, throughout the whole Exposition you feel that the architects haven't worked merely for money or for glory. They have appreciated the chance of doing something, out of the commonplace."

The sculpture by Sherry Fry was evidently executed with the idea of festivity in mind, the "Bacchus" and "The Reclining Woman" and two "Floras" decorated with flowers, and "Little Pan," and "The Torch-bearer" reproduced above each of the smaller domes. But, somehow, those figures did not quite indicate the real character of the building, intended for concerts and lectures and conventions, rather serious business. The coloring, too, of the statues, was disappointing, the dull brown being out of key with the light green of the domes.

"In the smaller concert room upstairs, Recital Hall," said the architect, "there is some very fine stained glass; two windows, and on the landing of the north stairway there's a third window, all done by the man who has been called the Burne-Jones of America, Charles J. Connick, of Boston. Instead of being hidden away there, they ought to have been put in the Fine Arts Building. They represent something new in the way of stained glass, and they have a wonderful depth and brilliancy."

As we drew near the Avenue of Progress we saw the magnificent doorway of the Varied Industries, overladen with ornamentation. "It was clever of Faville to put that doorway just in this spot where it would be seen by the crowds that entered by Fillmore Street. It comes from the Santa Cruz Hospital, in Toledo, Spain, built by the Spanish architect, De Egas, for Cardinal Mendoza, one of the most famous portals in Europe. The adaptation has been wonderfully done by Ralph Stackpole, with those figures of the American workman carrying a pick at either side and the semicircular panel just above the door and the group on top. That panel is one of the finest pieces of sculpture in the Exposition. It has tenderness and reverence. It's the kind of thing the mediaeval sculptors who worked on religious themes would have been enthusiastic over. See how simple it is, just a group of workers, with the emblems of their work, the women spinning with the lamb close by, the artist and the artisan, and the woman with the design of a vessel's prow in her hands, suggesting commerce. The single figure in the center is the intelligent workman who works with his hands and knows how to work, too. The group on top is a very pretty conception, the Old World Handing Its Burden to the Younger World, with its suggestions of the European people coming over here and raising American children."

XIV

The Palace of Machinery

On reaching the Avenue of Progress we found ourselves at the gayest corner of the Exposition, with two fine vistas of the two avenues. To our right stood the massive Palace of Machinery, one of the largest buildings in the world, so successfully treated by the architect that it did not give the faintest suggestion of being cumbersome or monotonous. "It's the Baths of Caracalla in Rome," said the architect, "adapted by a master. Those three gables above the main entrance are taken directly from the baths. See how simple the ornamentation is and yet how satisfying. The building as a whole is a perfect example of old Roman architecture, feeling its way toward the big architectural principles that are in vogue today, among others the economical principle involved in the counteracting of thrusts. If the Roman Emperor who was nicknamed Caracalla on account of the hooded military tunic that he made fashionable in his day hadn't built those baths we should probably not have the glorious Pennsylvania station in New York, that some of the architectural authorities consider the most important building of its kind built in this country. Although the work here is all concrete, Clarence Ward, the architect, says that with care, it could last hundreds of years."

Now we were struck by those vigorous-looking figures, by Haig Patigian, that stood on top of the Sienna columns all evidently designed to express the power of machinery. At the entrance the reliefs of the columns were in the same spirit and, as one might have surmised, by the same sculptor working out the meaning of the buildings in designs that kept the contour of the columns, strong and well-modeled.

"There's distinctive character in this building," said the architect. "It actually conveys the sense of tremendous energy, and by the simplest means. And inside, Ward has done something new and interesting."

When we entered we found the supports of the roof left bare. Instead of being unsightly, they had a kind of beauty and impressiveness. "Observe the magnificence of the spaces here on the floor and up to the ceiling. Some one asked Ward if all this height were necessary. He said it wasn't; but he wanted it for pictorial effect, to carry out the feeling of massiveness and splendor."

In the great figures that stood on the columns in front of the Palace of Machinery the architect found a theme for a discourse on the human figure as the chief inspiration of art. "It is possible that we shall change our minds on that subject," he remarked. "Already the world is showing a tendency to get away from the worship of the body. Ever since the Christian era, of course, the physical has been deprecated. We may come to see that the body is useful as it develops and serves the spiritual, that is, as it subordinates itself. The marvel is that the pagan tradition has persisted so long in spite of the Christian influence. This Exposition shows how strong it remains."

"But what would you have in place of the human figure as the inspiration of art?" I asked.

"Oh, there are plenty of things that might take its place. Flower themes are just as beautiful in decoration as the shapes of men and women. I can conceive of the time when it will be considered uninteresting and commonplace to have human bodies used as a means of aesthetic display. The self-glorification in it alone becomes wearying. We are gradually learning that the best we can do in life is to forget about ourselves and our old bodies. There are even those who go so far as to look forward to the time when we shall escape from our bodies altogether. It would be interesting, by the way, to get the point of view of a very spiritual Christian Scientist on the display here. I suppose that it would see good in the tendency to reach finer and nobler conceptions of art according to our present understanding."

Then the architect proceeded to discuss the artistic superiority of the Japanese. Though they used the human figure in their art, they did not play it up, after the habit of the Western world. They did not make it seem to be of supreme importance. They conventionalized and subordinated it to outline and color. The use of the nude they never cultivated. Their attitude toward the body was characterized by discretion and modesty, qualities that they showed in their dress. You would never see a Japanese woman, for example, wearing a dress that conspicuously brought out the lines of her figure.

"On the other hand," the architect went on, "there's no doubt we've become absurdly prudish in this country. We're afflicted with shame of the body which, in itself, is unhealthy. If art can help us to get back to a more normal attitude it will do a big service. All the more reason then why it should keep within reasonable bounds."

XV

The Court of the Ages

As we turned from the Avenue of Progress toward the Court of the Ages the architect said: "The workmen about here call this inner court 'Pink Alley,' not a bad name for it, though its real name is the Court of Mines. Throughout the Exposition Guerin shows that he is very fond of pink, probably on account of its warmth. He has been criticised for using it so much on the imitation Travertine for the reason that there is no stone of exactly this color. And yet there is pink marble. But even if there weren't any pink stone in the world, Guerin would be justified in his use of the color for purely decorative purposes, just as he was justified in using it on his four towers."

Inside the Court of the Ages the architect drew a long breath.

"In this court we architects feel puzzled. We think we can read new architectural forms like a book, and find that they are saying things repeated down the ages. But we can't read much here. In that lovely round arch there are hints of Gothic, and yet it is not a Gothic arch. Throughout the treatment there are echoes of the Spanish, and yet the treatment is not Spanish. The more one studies the conception and the workmanship the more striking it grows in originality and daring. Mullgardt has succeeded in putting into architecture the spirit that inspired Langdon Smith's poem 'Evolution,' beginning 'When you were a tadpole and I was a fish.' In the chaotic

feeling that the court gives there is a subtle suggestiveness. The whole evolution of man is intimated here from the time when he lived among the seaweed and the fish and the lobsters and the turtles and the crabs. Even the straight vertical lines used in the design suggest the dripping of water. When you study the meaning of the conception you find an excuse for Aitken in flinging his mighty fountain into the center of all this architectural iridescence. He caught the philosophy of Mullgardt without catching the lightness and gaiety of the execution. In that fountain he has brought out the pagan conception of the sun, and he has used the notion that the sun threw off the earth in a molten mass to steam and cool down here and to bring forth those competitions between human beings that reveal the working of the elemental passions. Aitken is material and hard, where Mullgardt is delicate and fine. How subtly Mullgardt has interwoven the feeling of spirituality with all the animal forces in man. That tower alone is a masterpiece. I know of no tower just like it in the world. From every side it is interesting. And at night it is particularly impressive from the Marina."

The architect went on to explain something of the court's history. "When Mullgardt started to work out his plans he must have had in mind the transitional character of an exposition. He knew that he could afford to try an experiment that might have been impracticable if the court had been intended for permanency. He evidently was determined to cast tradition to the winds and to strike out for himself."

"I should think most architects would like to work in that way."

"The usual process is very different. As soon as an architect decides to design a building. he first chooses a certain type of architecture; then he saturates his mind with designs that have already been done along that line. Out of the mass of suggestions that he receives he is lucky if he evolves something more or less new. Often he merely re-echoes or he actually reproduces something that he is fond of or that has happened to catch his fancy. The chances are that Mullgardt will go down into history for his daring here. It isn't often that a man takes a big biological conception and works it out in architecture with such picturesqueness. It's never intrusive and yet it's there, plain enough for anyone to see who looks close. It represented a magnificent opportunity and Mullgardt was big enough to get away with it."

Then the architect told me the human story behind all this beauty as we wandered back into the center of the court and stood there. "Notice the incline," he said, "from the entrances? It reminds me that Mullgardt had originally intended to have the floor of the court like a sunken garden. And remember that the name expresses the original idea. The Court of Abundance, that it is wrongly called, would have applied much better to the Court of Four Seasons. Well, after the notion came to Mullgardt to suggest in the court the development of man from the life of the sea to his present state as a thinking being, less physical than spiritual, he planned to build a court that should be the center of the pageants for the Exposition, where art should have its living representation in the form of processions and of plays, some of them written for the purpose. In the sunken garden there should be plenty of room for the actors to move about, using it as a stage. There should also be room for the sculptured caldron

that was to be an architectural feature and that later developed into Aitken's massive evolutionary fountain. For the base of the tower there was designed a gorgeous semi-circular staircase, which was to serve as an entrance for the actors. Around the court there was to run an ornamental balcony, covered with a great canopy in red and gold, making an effect of Oriental magnificence. The people were to watch the spectacles from the balcony and from between the arches. In addition to the main tower, very like the present tower, but to contain a great pipe organ, there were to be two others, in the corner at right angles, to be called echo towers. The music of the organ was to be transmitted to the echo towers by wires and the echoes were to serve as a sort of accompaniment. The effect, if it had been managed right, would have been stunning."

"Mullgardt has kept the spirit of the pageant in his court," I said. "Just as it is it would make an ideal setting, particularly for pageant with music, opera, for example."

"Of course," said the architect. "But the music ought not to come as it does now, from a band. It ought to come from the orchestra. Violins belong there. Put brass never!"

"Well, what happened to the pageant scheme?"

"Oh, when Mullgardt showed the preliminary sketches it was ruled out as too expensive. Then he removed the balcony and the staircase and, in place of the staircase, he introduced a cascade, keeping the rest of the court as it had been before. His idea was to use the water in the cascade only in a suggestive way. It was to be almost completely hidden by vines, after the manner of Shasta Falls, and to symbolize the mysterious appearance and disappearance of water that came from - one didn't know where. But that scheme was rejected, too, as too expensive. However, Mullgardt accepted the situation. He was so interested that he worked out himself many of the details that most architects would have left to subordinates. He really cared enough to make the whole effect as close to perfection as he could. Everything he did he had a reason for doing. Not one thing here did he use gratuitously. He evidently doesn't agree with the idea that, in architecture, beauty is its own excuse for being; he wants to make it useful, too."

Then I was initiated into the details of the workmanship. "Observe how the ideas in the structure of the walls of the court are carried on in the ornamental details and in the tower." The primitive man and primitive woman repeated in a row along the upper edge had been finely conceived and executed by Albert Weinert. And the nobility of outline in the tower was sustained by the three pieces of sculpture in front made by Chester Beach. That top figure some people believed to be Buddhistic in feeling. But it belonged to no particular religion. It stood for the Spirit of Intelligence. The ornamentation on the head was not an aureole, as bad been reported, but a wreath of laurel, symbolic of success. The group beneath was mediaeval, depicting mankind struggling for the light, expressed in the torches, through those conflicts that so pitifully came out of the aspirations of the soul, expressed in religion. The lowest group showed humanity in its elemental condition, related to the animal, close

to the beasts. So, to be followed in sequence, the groups ought to be studied from the lowest to the highest, and then the eyes should be able to catch the meaning of the lovely ornamentation, crowning the tower, the petals of the lily, emblem of spirituality, the arrow-like spires above expressing the aspirations of the soul.

On the sides of the tower the symbolism was consistently maintained, war and religion marking the progress of man toward the state indicated by the single figure of The Thinker.

"And, speaking of the soul," the architect went on, "Observe these great clusters of lights that illuminate this court and the approach on the other side of the tower. They look like stars, don't they? And the intention evidently is to use them for their star-like character. But there is history behind them. They are like the monstrance used in the Catholic Church, to hold the sacred host, the wafer that is accepted by the faithful as the body and blood of Jesus Christ. Since the sixteenth century it has been used by the church, a beautiful emblem, made of gold and designed to suggest the prayer of the sun, the Spirit of God in radiance. Its use here helps to give the court its ecclesiastical character."

As we made our way toward the Marina we noted how much the court gained by its general freedom from color. In the colonnade, to be sure, Guerin had been particularly successful with the shade of blue. But he would have done better if he had omitted the color, in fact all color, from the niches in the tower.

Viewed from the Marina, the entrance to the court proved to be a vision of loveliness. There was only one intrusive note to jar the harmony, the coarse sea figure by Sherry Fry, presumably Neptune's, Daughter, standing in the center, with a great fish at her feet, plainly out of place here, in spite of the court's celebration of the sea as the source of human life.

XVI

The Brangwyns

We lingered in the colonnade to view the eight mural decorations by Frank Brangwyn, of London. In front of The Bowmen we found a friend, a gifted woman painter, fairly bursting with enthusiasm. "What delights me in Brangwyn," she said, "is his artistic courage. He dares to put down just what he feels. This sturdy figure in the foreground, for example, peering through the trees, how many other painters would have allowed him to turn his back on the spectator? And yet how interesting he is and how alive."

"Some of those heads strike me as curious," I remarked. "That fellow closest to the center, just about to let his arrow fly, seems to have no head to speak of."

"Sometimes he's careless with his drawing. And yet he can draw magnificently, too. He evidently had a purpose in making so many of the heads in these murals almost deformed. He wanted to suggest that these types were in no way mental. They were wholly physical. Notice the care he has lavished on their muscular bodies, their great shoulders and legs."

"It doesn't seem like English work, does it?" said the architect.

"No, there's something almost Oriental about it both in the feeling and the coloring. And there's the Pagan love of the elemental life."

"But what a chance Brangwyn had to do something new with this magnificent subject," the architect went on. "At last, after centuries of effort, men are actually conquering the air. They've learned to fly. They've become birds. Now why didn't Brangwyn give us a pictorial expression of that miracle? Why didn't the artist have as much sense as the man of affairs who pays Art Smith to come out here and fly before the multitude?"

I argued that Brangwyn preferred to deal with antique themes - they were so much more pictorial.

The architect interrupted with some impatience. "But that's exactly what they're not. In my opinion Whistler was perfectly right when he said that if a mural decorator couldn't make modern life pictorial he didn't know his business. Flying through the air is only one of many wonders in the life of today that cry out for expression in art; but you scarcely catch a note of them here."

"For example?" said the painter.

"Industry - our great machines, the new power they bring into the world, the change in industrial relations and social and moral ideals. Now in these murals, Brangwyn has simply repeated himself and he hasn't by any means done his best work. And I question whether his observation is so accurate as you admirers of his try to make it appear. Look at the way those fellows are holding their bows - with the left hand, presumably for the pictorial effect of the composition. Well, let that point pass. One fellow has shot his arrow. The other is holding his arrow between the fore finger and the middle finger. Well, it won't go very far. The Indians know better. They let the arrow rest on the thumb to give it plenty of freedom to fly. One of those bows, by the way, has no string. Brangwyn probably thought it wouldn't be missed."

As we looked at the other panels the architect conceded that the points the painter raised for Brangwyn, the brilliant use of color; the dramatic grouping and the fineness of characterization, were true enough. "But he's too monotonous. Though his groups are of different periods, some of them ages apart, they're all essentially alike and the figures are even dressed alike. I'm perfectly willing to make allowance for artistic convention. But why should an artist limit himself unnecessarily when he has all the ages to draw on? Why should he neglect the present, the greatest of all the ages?"

"Ah, I'm afraid you're too literal said the painter. "You want to limit a genius to rules."

We turned from The Bowmen to study in detail the second illustration of Air, much more modern and yet charmingly old-fashioned, the windmill and the little mill high in the background, the group of naked boys flying kites, the toilers and their children, going home as fast as they could, fighting the wind, their picturesque draperies flying around them.

The architect was impressed. "He's caught the feeling of the thunderstorm, hasn't he?" he said.

"And he's brought out all the picturesqueness and the color and the majesty and even the humor," said the painter. "See how wonderfully be has composed the picture, what pictorial use he has made of every detail. The background of the clouds and the rain, the dark blues and the green and the pink; and the kites catching some of the color, and the lovely color of the mill and of the grass dried by the sun. And see that figure up there on the steps, all windblown and rushing under cover. It's all beautiful and yet there's not one face or figure there that would be considered beautiful by the painter who works for prettiness. He has no interest whatever in what the average mural decorator considers beautiful. And yet he sees beauty everywhere and he makes it felt. How pictorially he has used those purple flowers in the foreground at the base of the composition. And observe their relation to the purple clouds on top. And then what character he has put into those active figures, particularly in this queer little boy, naked except for the purple drapery flying from his waist. He has caught something of the fantastic spirit that you often see in children."

In nearing the two panels illustrating Water we had a chance to see how dexterously Brangwyn could manage his design without perspective, which would have made a hole in the wall. Those women with jars on their heads stood against a sky none the less lovely because it was flat. It was exquisite in its varieties of blue and white and green. That sturdy fellow lifting a heavy jar was actually working and working hard. "And how splendidly Brangwyn has modeled the figure with his back turned to us," the painter exclaimed. "What a stroke of genius it was that a yellow handkerchief of just that shade should hang from his neck. And the figures in the companion panel drawing their nets, they are putting their heart and soul into their work and they are having a good time, too. And this man here in the corner, with the purple shadows on his bare back, lifting his net, he's evidently had a big catch. He's holding the net in a way that shows it's heavy. And how decorative those men in the background are, with the baskets on their heads. Brangwyn loves to use figures in this attitude. They are interesting and picturesque and dramatic at the same time."

"But they're too conscious," the architect insisted, "too posed.

"Remember, they're not paintings," the painter insisted. "They're formal decorations."

In the panel representing the elementary use of Fire we were all struck by Brangwyn's daring and fine treatment of the ugly. Nearly every face was almost grotesque. And yet every face was appealing for the simple reason that it expressed attractive human qualities. Two, a man and a woman, had noses ridiculously large. The group of men in the center of the background, at the base, around the fire, had apparently started the fire by rubbing sticks together. One was intently leaning forward, as if in the act of blowing. Among the figures behind the group stood a man with an infant in his arms, vividly characterized by the unseeing eyes.

That infant was instantly singled out by the painter.

"Brangwyn is very wonderful in his observation of children. He has a quality that is almost maternal. Observe the difference between the expression in the face of that baby and the expression in the face of that little boy to the left of the fire-makers. How intently he is looking on as he leans against the brown jar. He shows all the interest of a boy just learning how to do things."

The kiln charmed us, too, though we regretted that it did not explain itself quite so spontaneously as most of the other panels. "But symbolism ought not to be too obvious, you know," the painter argued. "There's a certain charm in vagueness. It makes you feel your way toward a work- of art. The more you think about this panel the more you find there. To me it suggests the relation between fire and the abundance of the earth. See how cleverly, in each case of these two panels, Brangwyn has used smoke, first as a thin line, breaking into two lines as it goes up and interweaving, and then as a great flowing wreath, dividing the panel in two parts without weakening the unity."

For composition we decided that the two Earth panels were among the most remarkable of all. With satisfaction I heard Brangwyn compared by the painter to a great stage manager. "When I look at these groupings, I am reminded of Forbes-Robertson's productions of plays." Now we could see how brilliantly the decorator had planned in securing his effects of height by starting his group of figures close to the top of the canvas. And with what skill he had used trees and vines and vegetables and fruits, both for design and for coloring. "He has always been mad about apples and squashes," said that feminine voice. "In nearly every picture here you will find not one squash only, but several squashes. He loves them for their color and their shape. And how wonderful he makes the color of the grape. He suggests the miracle of its deep purple."

We admired the painter's pictorial use of shadow on those powerful and scantily draped figures and the animation he put into the bodies of the wine-pressers. And down there in a corner he had perfectly reproduced the attitude and facial expression of the worker at rest, holding out his cup for a drink. "There's another of those queer and interesting children. But oh, most wonderful of all is the opposite panel that ought to be called Abundance. See that mother, holding her lusty baby. The face is commonplace enough, but it has all motherhood in it. And the woman behind, she looks as if she might be a mother bereft or one of those women cheated out of motherhood."

The architect, though he still had his reservations on the subject of the Brangwyns, conceded that they were distinctly architectural. They blended into the spirit of the court.

The painter at once supported the opinion. "In these colonnades Guerin has done some of his finest coloring. The blue and the red are in absolute harmony with Brangwyn's rich tones. They must have been applied to fit the canvases. But the marvel is that the murals should show up so magnificently. Brangwyn painted them in London and he must have had second sight to divine just the right scheme. Do you realize," she went on enthusiastically, fairly losing herself in her enjoyment, "the

immense difficulties he had to contend with? In the first place, see how huge those canvases are. Their size created all kinds of problems. To view them right, to get a line on the detail, so to speak, would have meant, for the average painter, walking long distances. But, in his studio, Brangwyn could not have taken anything like accurate measurements."

"Perhaps he painted them out of doors," the architect suggested.

"I believe the explanation is that he thought them all out and he saw them in their places. From Mr. Mullgardt he had probably received a complete account, with drawings, of just what the court was going to be like. Then it lived before him and he made the murals live. His work shows that he begins in the right place, unlike so many people who paint from outside. He feels the qualities of the people he is going to paint. He really loves them. He loves their surroundings. He must be very elemental in his nature. They say he is a great, uncouth sort of a fellow. When he first went to London he was very contemptuous of the work done by the academicians. It must have seemed to him, a good deal of it, effeminate and trifling. Can't you see how those murals show that he is a man clear through? They are masculine in every detail."

"And yet they have a good deal of delicacy, too, haven't they?" said the architect. "See how atmospheric those backgrounds are. They actually suggest nature."

"Because they are unconventional and because they are true. And yet they are purely decorative. You wouldn't like to think of them as standing apart in a great frame. When you go close you will see that the colors are laid on flat. And they don't shine. For this reason they have great carrying power. Observe The Bowmen down there in the distance. Even from this remote end of the court it expresses itself as lovely in color and composition. Let us walk down and see how it grows on us as we approach."

Slowly we moved along the colonnade, the figures seeming to grow more and more lifelike as the painter indicated their technical merits. "They are of the earth, those men, aren't they? They are the antithesis of the highly civilized types used by so many of the painters today. They suggest red blood and strength of limb and joy in the natural things of life, eating, drinking, the open air, and simple comradeship. They make us see the wonder of outdoor living, the kind of living that most of us have missed. What a pleasure it is to find a worker in any kind of work trying to do a thing and actually doing it and doing it with splendid abandon. Now if Brangwyn hadn't entered into the feelings of those bowmen in the foreground, he couldn't have made the figure alive. And the life, remember, isn't merely brought out by the happy use of the flesh tints or by the play of the muscles. It's in the animating spirit. As Brangwyn painted those fellows, he felt like a bowman. So he succeeded in putting into his canvas the strength that each bowman put into his bow. He isn't pretending to shoot, that sturdy fellow in front. He is shooting, and he's going to get what he is after."

Before each of the four pairs of murals, the painter indicated to us the happy way in which, by the deft use of the coloring, each blended into the other, and she called my attention to the clearness of the symbolism. So often, she remarked, the mural

decorators used compositions that seemed like efforts to hide secrets, a childish way of working, sure to defeat itself. Brangwyn had no secrets. He was sincere and direct. He was happy over this work. He said that he had enjoyed doing it more than anything else he had ever done before. If these canvases had been found in the heart of Africa they would have been identified as coming from Brangwyn. No one else used color just as he did, with his kind of courage. His colors were arbitrary, too. He didn't follow nature and yet he always conveyed the spirit of natural things. Throughout his work he showed that he was a close and subtle observer. The sweep of rain through the air, the movement of figures and of draperies in the wind, the expression of human effort, how wonderfully he managed to suggest them all and to make them pictorial. But he wasn't interested in merely an activity. He loved repose. In nearly all of these eight canvases, so brimming with life, there were figures looking on serenely, calmly, conveying the impression of being absolutely at rest.

In every particular, according to the searching observer, Brangwyn was successful, with the exception of one, his treatment of birds. He evidently didn't know birds. If he had known them he would have loved them, and if he had loved them he would have entered into their spirit and he would have flown with them and he would have made them fly in his painting. Now they merely flopped. They were just about as much alive as the clay figures used in a shooting match. Even his highly decorative flamingoes weren't right. They did not stand firmly on the ground. They weren't alive. And the necks of the two flamingoes never could have met in the curves that Brangwyn gave them. This very failure, amusing as it was and hardly detracting from the effect of his work as a whole, was another proof that he was an instinctive painter, who relied for his guidance on feeling. But it was plain enough that he had chosen those flamingoes for their color, and a right choice it was.

We could not decide which of the eight murals we liked best. Perhaps, after all, they could not be considered apart. Though each was in itself a unity, the eight completely expressed a big conception. And in detail each canvas was full of delightful bits. If you closed your hand and peered between your thumb and your fingers, you could see how beautifully the color had been applied and how, throughout the whole surface, the workmanship sustained itself. Never was there the sense of faltering or of petering out. And everywhere there were expressions of fine understanding and sympathy, in the study of a peasant mother holding her babe, nude boys flying kites, a happy face with the lips blowing a pipe, a muscular figure lifting a jar, all conveying abundant life and rich coloring.

The painter finally ran away from us, apologizing for her enthusiasm.

In discussing her opinions, the architect said: "Well, I don't altogether agree. But she may be right. She sees from the inside, which is very different from seeing from the outside. There is a great deal of artistic appreciation that can be felt only by the artist, by the fellow-craftsman. No wonder we go so far astray when we criticise aspects of art that we're only related to indirectly or not related at all."

We walked to the Marina. From there we saw the sun, a great red ball, sinking behind the Golden Gate.

XVII

Watching the Lights Change

"There probably never was an Exposition in a more magnificent setting," said the architect. "The stretch from here to the Golden Gate makes one of the most splendid bits of scenery in the whole world. It was a good idea on the part of the Exposition people to build the little railway here so that visitors should get a glimpse of all the beauty. But, ideally, the view ought to be seen from a height. The curve from here to the Cliff House makes our foreign visitors gasp. It also makes them wonder why our boasting over San Francisco doesn't include some of the things we have the best excuse to boast about."

We stopped at one of the open-air restaurants, where we could eat and watch the fading light at the same time. Then we went to the lagoon, which the architect declared to be particularly interesting at this time of day.

The rotunda and the colonnade began to take on a deeper mystery. Across the surface of the water ran a faint ripple. In the background, over the Golden Gate, the sky was turning to flame. Delicate, gray cobwebs seemed to float in the air like veils, dusk and fog intermingled.

The light grew dim as we sauntered along the colonnade of the Palace. Through the columns we could see the Tower of Jewels, suddenly illuminated from inside, all in red, obscuring the sculptured figures and giving the lines greater unity and reach.

In the red glow the Italian towers fairly leaped into the air. "It's curious how the light makes them taller," said the architect.

Now the grounds were twinkling with a multitude of bulbs.

Presently the red light in the tower softened into white. Two of the Italian towers grew paler, the other two retaining their brilliancy. Ryan was putting on his colors like a painter, one over another.

We made our way back to the Marina, where the scintillators were soon to blaze. Before we arrived they informed us of their presence by the great feathered fan, of many colors, that rose into the sky.

"There was some opposition to the decorating of the Tower with jewels. The architects with conservative ideas very naturally felt that architecture which depended on its lines for beauty didn't need that kind of ornament. But Ryan has unquestionably justified himself. The feature has been talked about throughout the country more than any other. See how the light falls on the tower like a great shimmering robe. It gains by the contrast it makes with the subdued lighting beneath."

The group on the Column of Progress stood out against the sky.

The doorways were taking on the color of gold, becoming even more beautiful than they had been by day.

"What Ryan tried hardest to get," said the architect, "was evenness of lighting. He wanted to bring out clearly the details of the architecture and he succeeded."

XVIII

The Illuminating and the Reflections

That motionless steam engine, all in gray, harmonizing with the Travertine, was furiously at work. Into the air it sent clouds of steam that turned to red and blue and green under Ryan's magic. And up there, at the top of the Column of Progress, we saw the Adventurous Bowman and his companions in two groups, one reflected on the illuminated fog.

Through the smoke and the fog the bombs were shooting and breaking into great masses of liquid fire, golden and green and pink and yellow. "Someone says we're all children at heart," the architect remarked. "These fireworks get more attention than all the architecture and the art put together. But, after all, they're just about as beautiful as anything man can make and, in the way of color, they put the artists to shame."

We were part of the crowd that swept to the Court of the Universe, never so splendid as at night, with the columns reflected in the pool and Calder's star figures shining from the concealed electric bulbs. On reaching the court itself we stood at the end of one of the corridors and looked down. Great drops of light hung on the columns like molten gold. "Ryan has done something very artistic and unusual there," the architect remarked. "So far as I know nothing just like it has ever been done before. It suggests the tongues of fire mentioned in the Scripture that descended from Heaven."

In the sunken garden those two shafts, rising from the fountains, looking like stone by day, had become great candles, glowing from the base to the glass globe on top. "They're practically the sole means of illuminating this court. The other lights are merely ornamental. So far as I'm aware nothing just like these shafts has ever been tried in an Exposition or anywhere else. It's a novel Expositional effect. Some people don't like it; but most people admire it immensely. It symbolizes the gold that first drew the multitude to this part of the world. If the golden color had been used more extensively throughout the Exposition it would have helped a lot. Guerin gets it at night by means of the light that shines through the windows and Faville gets it in the light behind those wonderful doorways of his that haven't been praised half as much as they ought to be."

The Court of the Ages lured us along the dimly lighted inner court, the arches taking on an even more delicate beauty in the night light. Once within the court we found ourselves under the spell of Mullgardt's genius. The architecture, the cauldrons sending out pink steam, the flaming serpents, the torches on the tower, the warm lights from within the tower, the great ecclesiastical stars, brilliant with electricity, all carried out the idea of the earth, cast off by the sun.

In the entrance court we found the effects less magnificent but, in their way, just as beautiful. The lighting emphasized the refinement of the court, the rich delicacy of the ornamentation. "Mullgardt ought to go down into history for this contribution to the Exposition," said the architect. "He has shown that originality is still possible in architecture."

In the Court of the Four Seasons we watched the Emerald Pool turning the architecture into a mermaids' palace. The water flowing under the four groups of the seasons shone from an invisible light beneath, coloring it a rich green. "When Ryan promised to illuminate the water here without letting the source of the light be seen, it was thought by the people it couldn't be done." For a long time we sat in front of the lagoon where the swans were silently floating and, and the Palace of Fine Arts was reproduced with a deeper mystery. Now we could feel the relation between the colonnade and Gerome's chariot race. "It would please Gerome if he could know that he had helped to inspire so magnificent a conception," said the architect. "And if Boecklin could see this vision and hear that his Island of the Dead had started Maybeck's mind thinking of it he would probably be astonished and delighted at the same time. With his fine understanding of the influences operating in art he would see that his contribution did not in any way detract from Maybeck's originality. Down the centuries minds have been influencing one another and, in this way, adding to the sum of wisdom and beauty in the world. Now and then, as in this instance, we can plainly see the influences at work. Behind Boecklin and Gerome there were doubtless influences that led to their making those two pictures, inspirations from nature or from other artists, or both together. And this palace will doubtless inspire many another noble conception."

"When we apply that thought to the Exposition as a whole," I said, "we can see what a big influence it is likely to have on the art of the country."

"It has undoubtedly had a big influence already, even though we may not he able, as yet, to see it working. The very interest the Exposition has, aroused in the people that come here, whether they are artists or not, can't help being productive."

Seeing the Lights Fade

We went over to the South Gardens to see the lights change on the Tower of Jewels, passing the half-dome of Philosophy, the stained glass of the windows enveiling the background. They were still robing the tower in pure white, and the hundred thousand pieces of Austrian cut glass were shimmering. "They must have had a hard time getting those jewels fastened on the ornamentation of the upper tiers. The wind up there is very strong. Some of the men came near being blown off. It took pretty expert acrobatic work to hang the jewels out on the extreme edges.

Suddenly the lights on the tower glowed into red. The tower itself seemed to become thinner and finer in outline.

"There are people who don't like this color," said the architect. "It's fashionable nowadays to feel a prejudice against red. But it is one of the most beautiful colors in nature and one of nature's greatest favorites, associated with fire and with flowers. To me the tower is never so beautiful as it is when the red light seemed to burn from

a fire inside. See how it tends to eliminate the superfluous ornamentation. It brings out the grace of line in the upper tiers, like folded wings. With just a few eliminations the improvement in that tower would be astonishing."

Presently the lights in the tower went out altogether. The four Italian towers also grew dim. It was getting late. People were hurrying out. But we lingered. We wished to see this city of domes as it appeared without any lights at all, except for those that were kept burning to meet the requirements of the law.

For an hour we roamed about the deserted place. Here and there we would meet a belated visitor or a group of people from some indoor festivity.

The material had taken on a finer quality. It looked like stone. Wonderful as the Exposition was by day and in the evening, it was far more wonderful at this hour.

Now it was easy to imagine the scene as a city, with the inhabitants asleep in their beds. But just what kind of city it was I could not make up my mind. When I expressed this thought to the architect, he said:

"Have you ever seen David Roberts' big illustrated volumes, 'Travels in the Holy Land'? If you haven't, look them up. Then you will see what kind of a city this city is. It's a city of Palestine. It's Jerusalem and Jaffa and Akka all over again."

Features that Ought to be Noted by Day

The South Gardens

Hedge. Idea suggested by W. B. Faville, of Bliss & Faville, architects, of San Francisco, and developed by John McLaren, landscape gardener and superintendent of the Golden Gate Park, San Francisco, to give impression of old English wall. African dew plant grown in shallow boxes, two inches deep, covered with wire netting.

Design of entrance at Scott Street, by Joseph J. Rankin.

South Gardens, French in character, with suggestions of Spanish. Planting by John McLaren.

In center, "Fountain of Energy," by A. Stirling Calder, acting chief of sculpture; French influence. Expresses triumph of energy that built the canal. Youth on horseback, standing in stirrups, "Energy." Figures on shoulders, "Fame" and "Valor." Figures on globe, two hemispheres; Western, bull-man; Eastern, lioness-woman. Figures on base, sea-spirits. Upright figure on globe, Panama. Large figures in pool, the oceans: The Atlantic, a woman with coral in her hair, riding on back of armored fish; North Sea, an Eskimo hunting on back of walrus; Pacific, a woman on back of large sea lion; and South Sea, a negro on back of trumpeting sea-elephant. Sea-maidens on dolphins' backs, in pool.

To right and left, in front of Festival Hall, and Horticultural Palace, at ends of long pools, French fountain of "The Mermaid," figure, by Arthur Putnam, of San Francisco.

To right, large building, Festival Hall, by Robert Farquhar, of Los Angeles; French theatre architecture. Studied from the theatres of the Beaux Arts style of French

architecture. Details, French Renaissance developed from the Italian influence.

To right, Press Building, designed and built by the Exposition; Harris H. D. Connick, Director of Works.

To left, large building, Palace of Horticulture, Bakewell & Brown, architects.

To left, Young Women's Christian Association.

French light standards, by Walter D'Arcy Ryan and P. E. Denneville.

French ornamental vases, filled with flowers, by E. F. Champney.

The wall, by Faville, with ornamental Spanish entrances, runs around main courts and palaces, making the walled city. Tiled roofs suggesting mission architecture, associated with early California missions, a style developed from the Spanish.

Four smaller towers, two on either side of large tower, by George W. Kelham, of San Francisco; Italian Renaissance.

Sand on walks, selected by Jules Guerin for its pink color to harmonize with color scheme. Binds together buildings, its pink harmonizing with pink of walls. Grains of sand in walks translucent.

Flag poles, ornamented with gilt star, by Faville. Orange-colored streamers by Guerin.

Heraldic designs related to history of Pacific Coast, by Ryan.

Thoroughfare running along wall and lined with palms, Avenue of Palms.

Equestrian statue, to right of Tower of Jewels, by Charles Niehaus, "Cortez," conquerer of Mexico.

Equestrian statue, to left, by Charles Cary Rumsey, "Pizarro," conqueror of Peru. Fine in action and spirit.

Tower of Jewels

Main tower breaking southern wall, facing South Gardens, the Tower of Jewels, by Thomas Hastings, of Carrere & Hastings, New York. Developed from Italian Renaissance architecture, with Byzantine modifications, and designed to suggest an Aztec tower; 433 feet high; original intention to make it 100 feet higher.

Inscriptions on wall at base of tower chosen by Porter Garnett of Berkeley, explain steps that led to building of Panama Canal, celebrated by Exposition. On both sides of inscriptions Roman fasces denoting public authority. From left to right: "1501 Rodrigo de Bastides pursuing his course beyond the West Indies discovers Panama"; "1513 Vasco Nunes de Balboa crosses the Isthmus of Panama and discovers the Pacific Ocean"; "1904 the United States, succeeding France, begins operations on the Panama Canal"; "1915 the Panama Canal is opened to the commerce of the world."

Large Composite columns on base. Arched capitals with acanthus, ornamented with the American eagle, the nude figure of child, and ornamental design suggesting California fruits. Colored to resemble Sienna marble.

Corinthian columns at either side, eagles at corners of capital, human head above.

Figures by John Flanagan, of New York, represent types in early California history: Spanish adventurer of sixteenth century, who came to California and started Spanish influence; priest, who brought the Catholic religion to California Indians; philosopher, or scholar and teacher; and the Spanish warrior, the soldier of sixteenth century, who came to win territory for Spanish king. Above cornice of tower stand four figures on each of the four sides, twice life-size.

Between statues by Flanagan, square decorative panels; youthful figures with wreath, repeated on north of tower. Designed by Hastings, modelled by Newman and Evans, New York.

Armored horsemen on terrace, by F. M. L. Tonetti, type of Spanish soldier. Repeated four times on each side. Well modeled, but damaged in effect by being placed in row.

Rows of eagles on niches of tower, symbol of American initiative.

Decorative vase on wings of tower, Italian. Use of ram's head below bowl.

Wreaths of laurel under eagles, rewards of courage, suggesting triumph of building canal.

Prows of triremes, at corners on third lift, denoting worldwide commerce.

Ornamental use of niches, columns, vases, head-piece, breastplates, shields, the pagan bull, Cleopatra's Needle.

Human figures supporting globe, encircled with girdle, point of tower; suggest Atlas; ancient idea; somewhat like the group of the four quarters of the world by Jean Baptiste Carbeaux in the gardens of the Luxembourg.

Tower broken into seven stages. Horizontal lines have flattening effect; tower does not appear so high as it really is.

One hundred and thirty-five thousand jewels on tower, suspended to vibrate. Ruby, emerald, aquamarine, white, yellow. Made in Austria, of Sumatra stone.

Arch of Tower of Jewels, 110 feet high, 60 feet broad; fine example of Roman arch, like Arch of Constantine and Arch of Titus.

Figure of Minerva on centerpiece of arch, north and south.

Recessed or coffered panels in ceiling, richly colored, blue harmonizing with murals on east and west walls.

Murals by William de Leftwich Dodge, of New York. To west, "Atlantic and the Pacific," with the "Purchase" to right, and the "Discovery" to left. Opposite, "Gateway of All Nations," with "Labor Crowned" and the "Achievement" on sides. Tone of murals strengthens arch. Subjects related to history of California and the Panama Canal.

Fountains, one in each of the colonnades. To right, "Fountain of Youth," by Mrs. Edith Woodman Burroughs, of Flushing, New York. Figure of girl, simple and

well-modeled; panels at either side show boats, youth rowing the older people; eagle and laurel wreath at back, suggest that central figure is United States. One figure shows a woman with hand at ear, her attention turned toward the beauty and happiness of lost youth. To left, "Fountain of El Dorado," by Gertrude Vanderbilt Whitney (Mrs. Harry Payne Whitney), of New York. Panels at either side show human struggle for "land of gold," or "happiness," or "success." Portals ajar, but Egyptian guardians bar the way. Dramatic subject, vigorous handling.

View of San Francisco hills between the columns, one of the most beautiful views on the grounds.

Inscriptions on north of tower, by Garnett, discovery of California and union with United States. From left to right: "1542 Juan Rodriguez Cabrillo discovers California and lands on its shores." "1776 Jose Joaquin Moraga founds the Mission of San Francisco de Asis"; "1846 the United States upon the outbreak of war with Mexico takes possession of California"; "1850 California is admitted to the Union as a sovereign State."

Forecourt of Court of Universe; coloring good, graceful planting of cypress.

Trees in niches under tower; contrast of colors, dark green, blue and pink.

Court of the Universe

Elephant poles, Roman, by McKim, Mead & White; streamers by Guerin.

Bear fountains, in walls of Palaces of Liberal Arts and Manufactures, north of Tower of Jewels. Three on each wall. Colors, pink, dark blue, light green.

Largest court in Exposition. By McKim, Mead & White, architects, of New York. Inspired by Bernini's entrance to St. Peter's, in Rome.

Area of court, seven acres; 650 feet wide from arch to arch; 1200 feet from Tower of Jewels to Column of Progress.

Palaces around court: northeast, Transportation; northwest, Agriculture; southwest, Liberal Arts; southeast, Manufactures.

Sunken Garden, planted by John McLaren.

Height of Arches of Rising Sun and Setting Sun, 203 feet from base to tip of sculpture.

East, Arch of Rising Sun; Arch of Setting Sun, in west. Suggested by arches of Constantine and Titus in Rome; modified by use of green lattices, Oriental, and by colossal sculptural groups, the East and the West, in place of Roman chariot or quadriga.

Columns in front of arches; composite, mingling of Ionic and Corinthian; female figure used as decoration.

"Angel of Peace," by Leo Lentelli, on each side of arches on Sienna columns, repeated four times. Sword is turned down, but not sheathed, a commentary on modern peace.

"Pegasus," in triangular spaces above arch, by Frederick G. R. Roth, repeated on the other side.

Medallions, right and left sides of arches. Female figures suggesting Nature, by Calder; male figures suggesting Art, by B. Bufano, of New York.

Above medallions on frieze, decorative griffons.

Quotations on Arch of Rising Sun, west side, facing court, chosen by Garnett. Panels from left to right: "They who know the truth are not equal to those who love it," from Confucius, the Chinese philosopher; "The moon sinks yonder in the west while in the east the glorious sun behind the herald dawn appears; thus rise and set in constant change those shining orbs and regulate the very life of this, our world," from "Shakuntala" by Kalidasa, the Indian poet; "Our eyes and hearts uplifted seem to gaze on heaven's radiance," from Hitomaro, the Japanese poet.

Quotations on Arch of Rising Sun, east side, facing Florentine Court. Panels from left to right: "He that honors not himself lacks honor wheresoe'er he goes," from Zuhayr, the Arabian poet; "The balmy air diffuses health and fragrance; so tempered is the genial glow that we know neither heat nor cold; tulips and hyacinths abound; fostered by a delicious clime, the earth blooms like a garden," from Firdausi, the Persian poet; "A wise man teaches, be not angry. From untrodden ways turn aside," from Phra Ruang, the Siamese poet.

Crenellated parapet on arches, note from military architecture. Archers used to shoot from behind.

Cleopatra's Needle repeated on edge of arches. Used by the Egyptians as historical records and public bulletins. Merely decorative.

Green jars, beautifully designed, in niches at base of Arches of Rising and Setting Sun, McKim, Mead & White. Eight in each arch.

Arch of the Rising Sun, surmounted by group representing types of Oriental civilization. "Nations of the East," designed by Calder, and executed in collaboration with Lentelli and Roth. From left to right: Arab sheik on horse, negro slave, Egyptian on camel, Arab falconer, Indian prince, Buddhist priest or lama from Thibet, Mohammedan with crescent, negro slave, and Mongolian on horseback.

Murals in arch by Edward Simmons, of New York. On north wall, from left to right, True Hope and False Hope, Commerce, Inspiration, Truth, Religion, Wealth, Family; in background Asiatic and American cities. On south wall: historical types, nations that have crossed the Atlantic; from left to right, "Call to Fortune," listening to the past, the workman, the artist, the priest, Raleigh the adventurer, Columbus the discoverer, the savage of lost Atlantis, the Graeco-Roman, and the Spirit of Adventure sounding the call to fortune. In background, ancient and modern ships.

Arch of Setting Sun. Statues, frieze, spandrels, parapet, identical with Arch of Rising Sun. Group on top, "The Nations of the West," designed by Calder, executed in collaboration with Lentelli and Roth. American figures grouped around prairie wagon, drawn by two oxen. Above wagon, "Enterprise"; in front, "The Mother of

Tomorrow," white boy on one side, colored boy on other; south, a French-Canadian, an Alaskan woman, a Spanish-American, a German; north, an Italian, British-American, squaw, American Indian.

Quotations on Arch of Setting Sun, chosen by Garnett. Panels from left to right, facing court: "In Nature's infinite book of secrecy a little I can read," from "Antony and Cleopatra," by Shakespeare, the English poet;

"Facing west from California's shores, Inquiring, tireless, seeking what is yet unfound,

I, a child, very old, over waves, toward the house of maternity, the land of migrations, look afar,

Look off the shores of my Western sea, the circle almost circled. from "Leaves of Grass," by Walt Whitman the American poet; "Truth, witness of the past, councillor of the present, guide of the future," from "Don Quixote," by Cervantes, the Spanish novelist.

Murals in Arch of the Setting Sun, by Frank Vincent Du Mond of New York. "Westward March of Civilization," beginning on north and continuing on south wall. Four groups in north panel, from left to right, Emigrants setting out for the west; two workmen and a woman holding child; symbolic figure of the Call to Fortune; types of those who crossed the continent, the driver, the Preacher, the Pioneer, the Judge, the Schoolmistress, the children; youth bidding farewell to parents; in background, New England home and meeting place. South wall: four groups in panel, from left to right; two Spanish-American soldiers and captain with a Spanish priest, suggesting Mission period; symbolical figure "Spirit of Enlightenment"; types of immigrants, the Scientist, the Architect, the Writer Bret Harte, the Sculptor, the Painter William Keith, the Agriculturist, the Laborer, women and children; California welcoming the easterners, figures of California bear, farmer, miner, fruit pickers; orange tree, grain and fruit, symbols of state.

Classic groups at head of steps in front of arches leading down into gardens by Paul Manship, of New York. North side, "The Dancing Girls"; south, "Music and Art."

Star-figure, along upper edge of court, by Calder. Repeated ninety times. Contrast with angel in front of arches.

Lion's head, on cornice below star-figure, repeated around court.

Gilt balls on the domes of all six pavilions. Represent an ornamental motive borrowed from the Byzantines and often used on synagogues. A feature of St. Mark's. Dr. Jacob Nieto, rabbi of the Temple Israel, of San Francisco, has an interesting theory as to their origin. "The ancients always had the greatest regard for the central star of each of the constellations that made tip the zodiacal signs. No doubt in their method of representation they would symbolize the central stars by a globe, as they also did the sun and the moon, looking upon them all as servants of the earth, and having, possibly, no idea that these other constellations might be separate solar systems."

Frieze on pavilions at corners of court, "Signs of the Zodiac," Atlas and fourteen daughters, seven Pleiades and seven Hyades twelve bearing plaques, by Herman A. MacNeil, of New York. On four sides of each of the six dome-covered pavilions. The third figure from the end on either side represents Electra. Sculptor, in modelling the form, put it on one side and then reversed it on the other side. The daughters of Atlas: only those representing signs of the Zodiac, have shields. On each shield is one of the signs of the Zodiac. What the sculptor has designed on the right is reversed on the left, securing absolute symmetry. The figures are finely done and merit special attention.

Lamps around sunken garden. Women; the Canephori, priestesses who carried baskets in ancient Greek religious festivals; men, suggestive of Hermes, used by Romans at ends of roads. Instead of baskets, they all carry jars.

"Fountain of the Rising Still." Ninety-foot column crowned by figure of Rising Sun, by Adolph A. Weinman, of New York. Reliefs at base of column, "Day Triumphant"; Time, Light, Truth, Energy, conquering Falsehood, Vice, and Darkness. Ornamental figures under upper bowl looking down into water, suggest Neptune, but are winged, "Spirit of the Waters."

"Fountain of Setting Sun." Column with figure of Setting Sun, a woman; called also "Descending Night." Reliefs at base of fountain, "Gentle Powers of Night," with Dusk covering Labor, Love, and Peace, followed by the Stars, Luna, Illusions, and Evening Mists.

Tritons in pools of Fountains of Rising and Setting Sun, by Weinman. Two statues; one, triton struggles with snake; in the other, with fish. Two duplicated in each pool.

Sheetlike appearance of water when full force of water is on; streams from figures in pool, overflowing from bowl, spouting from lion heads above frieze.

"The Elements," reclining figures at head of main stairs leading down to sunken gardens by Robert Aitken, of New York. In size and treatment, suggestive of Michael Angelo. Northeast, "Water," riding a wave, with his trident in one hand, sea weed in the other. Northwest, "Fire," a Greek warrior lies in agony, grasping fire and lightning, with Phoenix, bird of flame, at back, and the salamander, reptile of fire, under his right leg. Southeast, "Earth," a woman leaning against a tree, apparently sleeping; at back two human figures struggle to uproot tree, symbol of man's war with nature. Southwest "Air" woman holding star to ear; birds, symbol of air; Icarus, mythological aviator who fell into sea, tied to wings of woman, typifying man's effort to conquer the air.

Small lion fountains below "The Elements," by McKim, Mead & White.

Bandstand, Arabic; picturesque, but inharmonious; obstructs view through entrance court.

Four tigers at base of bandstand, facing pool; decorative.

Court leading from gardens to Column of Progress. Designs repeated in frieze and in jeweled lamps of shell design, McKim, Mead & White; fine detail.

Colonnades on either side of court leading to Marina. Large Roman hanging lamps. Stars in ceilings. Beauty in design, coloring and sweep of corridor.

Frieze around main doorway in colonnades, bird and conventionalized foliage; skilfully designed.

On the Marina

View from Marina: Extreme right, Berkeley and Oakland; in center of bay, Alcatraz Island, like a white citadel; left of Alcatraz, Angel Island; left of Angel Island, Belvedere; left, Marin County, including Sausalito and Mount Tamalpais, with military reservation facing the Golden Gate and looking across to the large military reservation, Presidio.

Column of Progress, celebrating the Progress of Man. Preliminary sketch by Calder. W. Symmes Richardson, architect. Reliefs at base, by Isidore Konti, of New York. Surmounting statue, by Hermon A. MacNeil, of New York.

Tablets on four sides of base, in commemoration of aerial advancement. To the west, the scientific phase, a tribute to Langley, who first solved the problem of flying. To the north, aerial achievement. To the east, aerial organization. To the south, history of flying.

Frieze at base on four sides celebrates beginning of progress. On south front, two women holding palm branches, symbol of victory, call mankind to achievement.

Wreath at base of column, reward of achievement.

Top of pedestal, ornamental garland, with figure of Sphinx at corners.

Spiral, winding around column, with ships in full sail, suggestive of upward progress of world. Similar spiral on Column of Trajan and Column of Marcus Aurelius, in Rome.

Circular frieze sustaining main group at top, "The Burden Bearers," by MacNeil.

Group on top, "The Adventurous Bowman," the Superman, representing moment of attainment. Three figures, the dominating male, with the male supporter steadying his arm, and the devoted woman ready to crown him with laurel.

First use of this kind of column for an idealistic conception. Prototypes of this column, like Trajan's Column, but to celebrate some warlike figure or feat.

Best place to view column, from north, near California Building.

Esplanade, straight northern wall, broken by Court of Four Seasons, Court of the Universe, and Court of the Ages. Northern facades of all four buildings, ornate doors in duplicate of Spanish plateresque doorways.

Main doorways, rich detail. Statues in niches, by Allen Newman, of New York. Center, "Conquistador," sixteenth century Spanish adventurer. Figure on either side in duplicate, Newman's "Pirate," who preyed on shore commerce of South America. Humorous touch in bowlegs.

Magnificent view from Marina of San Francisco back of the Tower Of Jewels. Like a painting by Cezanne.

Approaching the Court of Four Seasons From the Court of the Universe

Venetian Court.

Palaces on sides of court; to the north, Agriculture; to the south, Liberal Arts.

Quotation on Arch of Setting Sun, facing Venetian Court, chosen by Garnett. Panels from left to right: "The world is in its most excellent state when justice is supreme," from Dante, the Italian poet; "It is absolutely indispensable for the United States to effect a passage from the Mexican Gulf to the Pacific Ocean, and I am certain that they will do it. Would that I might live to see it. But I shall not," from Goethe, the German poet; "The Universe, an infinite sphere, the center everywhere, the circumference no where," from Pascal, the French philosopher.

Italian Renaissance architecture.

Colors rich and well harmonized; pink and green.

Picturesque lattice work in small doorways.

Lighting standards, by Faville.

Goats' heads at top of standards, just below the globe.

Arches on sides, coupled Corinthian columns. Endeavor to make them more interesting than formal type of fluted columns. Four designs. They add to richness of court.

Winged figures over arches, by Faville.

Blue medallions above arches, Faville. Italian adaptation of Byzantine, Ship of State, the Bison, the Twins holding garlands representing abundance, the horn of plenty and cadeucus, and tree.

Coloring under eaves, bright shades, blue and orange.

Planting, by McLaren, well-massed, in great profusion.

Court of Four Seasons

Court of Four Seasons, Henry Bacon, of New York, architect. Hadrian's Villa used as model for half-dome and columns in front of fountain. Italian Renaissance in feeling. Every detail in classic spirit. Gives impression of seclusion and peace.

Quotations on gateways chosen by Garnett. On the eastern gateway, "So forth issew'd the seasons of the yeare - first, lusty spring all dight in leaves and flowres - then came the jolly sommer being dight in a thin cassock coloured greene, then came the autumne all in yellow clad - lastly came winter cloathed all in frize, chattering his teeth for cold that did him chill," from "The Faerie Queene," by Edmund Spenser. On the western gateway, "For lasting happiness we turn our eyes to one alone, and she surrounds you now, great nature, refuge of the weary heart and only balm to breasts that have been bruised. She bath cool hands for every fevered brow and gentlest

silence for the troubled soul," from "The Triumph of Bohemia," by George Sterling.

Palaces around court: northeast, Agriculture; northwest, Food Products; southwest, Education; southeast, Liberal Arts.

Emerald pool. Surrounded by shrubbery. No sculpture. Architectural term, a "black mirror." Fine reflections.

Planting, by McLaren, simple and effective. Trees, olive, acacia, eucalyptus, cypress, laurel. All foliage, grey-green; banner poles same color.

Banners, by Ryan; no heraldic designs.

Best view of court from between columns of Fountains of Spring or Autumn.

Bulls at sides, above entrance to north court, "Feast of the Sacrifice," by Albert Jaegers, of New York. Youth and maiden leading bulls to harvest festival, suggested by great garlands.

Roman eagles below bulls on four corners of north court.

Bull's head with festoons, skull motive, at base of corner pavilions at four corners of north court, Roman.

Lion's head around cornice, designed by the architect, modelled by artisans of Exposition.

Bulls' heads above cornices between festoons of flowers around court. Roman motive.

Statue above south dome, "Harvest," by Albert Jaegers. Seated figure with horn of plenty. Fruits and grains on either side.

"Abundance," statue repeated four times over each gateway, by August Jaegers.

Vases repeated twenty-four times on balustrade around court; simple design, in harmony with classic plan of court.

Wreaths above cornice around court, harvest motive, wheat and grape.

Figures in triangular spaces over three arches of each gateway, repeated. By August Jaegers. Harvest motive.

In ceiling of east and west arches, faint relief, terra-cotta effect, Greek designs; coloring, orange, faint greens, and browns.

Signs of zodiac on gateways, Aries, Taurus, Gemini, Cancer, Leo, Virgo, Libra, Scorpio, Sagittarius, Capricornus, Aquarius, Pisces.

Half-dome to south, "Niche of Ceres." Rich coloring in vault, contrasted with light tones in arched section.

Figures on composite columns at right and left of half-dome, "Rain" and "Sunshine," Albert Jaegers. "Rain," a woman shielding head with mantle and holding shell; "Sunshine," woman shading head from sun with palm branch.

Capitals of columns of "Rain" and "Sunshine," agricultural figures, small harvesters. Modelled by Donnelly and Ricci after designs of the architect.

Pedestals at base of columns, agricultural scenes in low relief, modelled by Donnelly and Ricci after designs of the architect. Farmers going to work with women and children and dog.

In niches at corners of court, "Fountains of the Seasons," surmounted by statue groups representing seasons, Furio Piccirilli, of New York.

Delicate pink tinting of walls in niches, by Guerin, in imitation of pink marble.

Columns of colonnades, Ionic, with harvest suggestion in ears of corn hanging from capitals, flower at top.

Flower boxes, in walls of niches near top and at top; African dew plant hanging over edge; give note of age and break sharp outline of wall against sky, and contrast with color of background.

Southwest corner, "Spring," by Piccirilli. Young woman with floral garland, man adoring, Flora bringing flowers.

Northwest corner, "Summer," by Piccirilli. Group expresses fruition. Woman brings child to husband. Laborer with first sheaf from field.

Northeast corner, "Autumn," by Piccirilli. Young woman carrying wine jar, suggests fruitfulness. Harvest of fields and human race; one girl offers grapes, other a child.

Southeast corner, "Winter," by Piccirilli. Bare tree at back; laborer rests after tilling; one begins to sow, preparing for spring.

Murals in colonnades with fountains, by H. Milton Bancroft. Simple and obvious, in the pagan spirit.

Above doorway in southwest corner, Spring. "Spring" and "Seedtime."

Northwest corner, Summer. "Summer" and "Fruition."

Northeast corner, Autumn. "Autumn" and "Harvest."

Southeast corner, Winter. "Festivity" and "Winter."

Murals in half-dome to south, Bancroft. Coloring and arrangement of figures finer than in smaller panels.

On east wall under dome, "Art Crowned by Time." Father Time crowns Art; on one side, figures of Weaving, Jewelry Making, Glassmaking; on other Printing, Pottery, and Smithery.

"Man Receiving Instruction in Nature's Laws." Woman holds before a child a tablet inscribed "Laws of Nature." Nature's laws applied to Earth, Water, Fire, Love, Life, and Death.

North court, entrance to Court of Four Seasons. Wreaths, lion heads, bulls' beads, harvest design on capitals of columns, repeated.

"Ceres," by Miss Beatrice Evelyn Longman, goddess of agriculture, wreath of cereals and corn scepter. Figure conventional, prim and modish; flowing skirt.

Figures below "Ceres" on drum represent carefree nature. In deep relief. cameo-like. Figures of women, gracefully modeled, with garlands and tambourines.

Satyrs spout water into bowl of fountain.

Trees, yews in couples, on either side of walks and center of lawn; redwoods and eucalypti at sides of entrance to court.

Shiny-leaved dark green shrub, on borders in court, coprosma.

Mass of green, placed at end of court to hide Morro Castle. Deepens intimate note of court.

French lighting standards at north end of court, by Ryan and Denneville.

Aisle of Sunset

Aisle approaching the Palace of Fine Arts, leading from Court of Four Seasons, west to Administration Avenue, by Faville.

Central portal, Spanish Renaissance, with twisted Byzantine columns.

Globe above, symbolical of universal education.

Main sculptural group: "Education," by Gustave Gerlach, Weehawken, New Jersey. Tree of knowledge in background. Left, kindergarten stage. Center, half-grown children. Right, man working out problems for himself.

Below, open book of knowledge radiating light in all directions. Small figures draw aside curtains of darkness and ignorance. Hour-glass, "Time Flies." Crown, for seekers of knowledge.

Educational panels inlaid in wall over smaller entrances, by pupils of School of Sculpture of Beaux Arts Architects, and National Sculpture Society.

Woman teacher, by W. H. Peters.

Man teacher, by Cesare Stea.

"Victory," on gables of buildings, by Louis Ulrich, of New York; "Acroterium"; like "Victory of Samothrace."

Charm of green lattice-work in small doorways of palace.

Main doorway, Palace of Food Products, by Faville. Terra cotta effect on sides of door. Eagles above door, inspiration. Green lattice-work in doors.

Administration Avenue

West wall, magnificent; facing Palace of Fine Arts, broken by Aisle of Spring, and two large Roman half-domes in Palace of Food Products and Palace of Education.

Palaces facing avenue: from north to south, Food Products and Education; across lagoon, Fine Arts.

Greenery and niches in pink and blue prevent wall from being monotonous.

"Dome of Plenty," in Palace of Food Products, harmonizes with half-dome in Court of Four Seasons.

Fountain in dome; elaborate; Sienna design.

Man with oak wreath, repeated eight times above columns in portal representing strength, by Earl Cummings.

Great columns of imitation Sienna on either side of portal, surmounted by "Physical Vigor," by Ralph Stackpole.

Niches along wall, archaeological figures, by Charles Harley, of Philadelphia. "Triumph of the Field," man with harvest symbols, alternating with "Abundance," woman with horn of plenty.

Half-dome of Palace of Education, "Dome of Philosophy." Architecture as in "Dome of Plenty." Charm of background, ornamented ceiling, Corinthian columns with acanthus leaves.

Over doorways, beautiful use of stained glass.

Female figure repeated eight times above inner columns, by Albert Weinert; carries books; "Ex Libris," representing education.

Statue by Stackpole surmounting Sienna columns, reversed duplicate of figure before "Dome of Plenty," with different name, "Thought." Really represents vigorous man thinking.

Figures in niches repeated.

Roman fountain, "Dome of Philosophy," by Faville; simplest and one of the most beautiful of the fountains on grounds. Suggested by fountains in Sienna and Ravenna.

Palace of Fine Arts

Palace of Fine Arts, Bernard R. Maybeck, of San Francisco. Conception inspired by Boecklin's painting, "The Island of the Dead." Rotunda like Pantheon in Rome. Colonnade suggested by Gerome's "Chariot Race." Columns at end of colonnade, hint of Forum. Greek suggestion in Corinthian columns and fretwork and frieze around rotunda. Roof garden or pergola around edge of roof and the Egyptian red of wall gives Egyptian note. Suggestion of overgrown ruin; atmosphere of melancholy beauty. Originality of architectural design and treatment.

Curved hedge, obscuring view of floor of rotunda from opposite side of lagoon, by John McLaren. African dew plant, as in south hedge. Laurels and willows were originally planned to cover hedge and to reach to top of columns. Monterey cypress at north end of colonnade.

Kneeling figure on altar directly in front of rotunda, "Reverence," by Ralph Stackpole. Can be seen from across pool only.

Altar rock, planting grown down over edge gives effect of draped altar cloth.

Frieze on altar rock, below kneeling figure, by Bruno Louis Zimm, of New York. Represents "Source of Genius." In center, Genius; to left and right, mortals seeking to approach genius; lions guard the youth. Seen from across lagoon only.

Panels on exterior of rotunda just below dome, by Zimm, representing progress and influence of art.

Eastern panel, "Struggle for the Beautiful"; in center, Truth; at sides, Persistence and Strength, struggling with centaurs, symbols of materialism.

Panel to left, "Power of the Arts"; Genius taming Pegasus, inspiration in art; Wisdom inspiring Youth; Music with lyre; figures of Literature and Sculpture.

Panel to right, "Triumph of the Arts"; Apollo, patron of arts, in chariot; Fame, with olive branches; Ictinius, builder of Parthenon, leads procession of devotees.

Three panels, repeated on five sides of rotunda.

Decorative figure, man and woman alternating, between panels, repeated around rotunda.

Corinthian columns, ochre grouped with pale green ones; capitals of burnt orange.

Flower boxes by Ulric H. Ellerhusen; women at corners. Original plan was to have vines from boxes droop over, shoulders of women. Architect's purpose in attitude of women to suggest sadness of art.

Roman vases, eight or ten feet high around colonnade. Massive and graceful detail.

Sculpture Outside Fine Arts Palace Beginning at Northeast Corner of Lagoon

North of Lagoon

The Illustrious Obscure, by Robert Paine. (Fountain on island at north end of lagoon.)

Whaleman, by Bela L. Pratt.

Garden Group by Anna Coleman Ladd.

Dying Lion, by Paul Wayland Bartlett.

Garden Figure, Nymph, by Edmond T. Quinn.

Fragment of "Fountain of Time," by Lorado Taft. Representing the troubled generations.

Roadway to Right Before Entering Circle

Bird Fountain, by Caroline Risque.

The First Mother, by Victor S. Holm.

Circle at North End of Peristyle

Mother of the Dead, by C. S. Pietro. (Lagoon side of circle.)

Chief Justice Marshall, by Herbert Adams. (In walk.)

Destiny, by C. P. Dietsch.

Sundial, by Edward Berge.

Head of Lincoln, by A. A. Weinman.

Fountain Groups, by Anna Coleman Ladd. Sun-God and Python, Water Sprites, and Triton Babies. (To right.)

Sundial, by Gail Sherman Corbett.

Daughter of Pan, by R. Hinton Perry.

Boy Pan with Frog, by Clement J. Barnhorn,

Bondage, by Carl Augustus Heber. (Only feminist note in the grounds.)

Saki, Sundial, by Harriet W. Frishmuth. (In walk.)

Great Danes, by Anna Vaughan Hyatt.

Young Diana, by Janet Scudder.

Flower Urns, base of building along colonnade; Greek figures with garlands. Ulric H. Ellerhusen.

Wall of building facing colonnade, seventeen feet high. Acacia blooming there, suggesting over-growth, relieves severe lines of architecture. Broken by small doors, at corners decorated with spears. Doors suggest Greek design.

Corinthian columns and pilasters; harmony of color, smoked ivory and ochre, with shades of green in foliage.

Urns, on the wall on either side of the doorways and in the rotunda, designed by William G. Merchant. Suggested by urns in the Vatican, Rome.

North Peristyle (curved part colonnade north of rotunda).

Maiden of the Roman Campagna, by Albin Polasek. (To left.) Fountain: Duck baby, by Edith Barretto Parsons.

A Fawn's Toilet, by Attilio Piccirilli.

Apollo, by Haig Patigian. (To right.)

The Scalp, by Edward Berge. (To left.)

Primitive Man, by Olga Popoff Muller.

Youth, by Victor D. Salvatore. (To right.)

Soldier of Marathon, by Paul Noquet. (To left.)

Fountain: Fighting Boys, by Janet Scudder.

Garden Figure, by Edith Woodman Burroughs. (To right.)

L'Amour, by Evelyn Beatrice Longman. (To right.)

Returning from the Hunt, by John J. Boyle. (To left.)

Boy with Fish, by Bela L. Pratt. (To right.)

The Centaur, by Olga Popoff Muller.

The Sower, by Albin Polasek.

Beyond, by Chester Beach. (By main doorway.)

Aspiration, by Leo Lentelli. (Over main doorway.)

Pioneer Mother Monument, by Charles Grafly. (Before main doorway.)

Portrait of a Boy, by Albin Polasek. (Outside west archway.)

The Awakening, by Lindsey Morris Sterling. (Outside west archway.)

"Sculpture," relief on walls of west archway. Bela L. Pratt.

Rotunda, Entrance Through North Archway

William Cullen Bryant, by Herbert Adams. (At northwest archway.)

Lafayette, by Paul Weyland Bartlett. (Center of rotunda.)

The Young Franklin, by Robert Tait.

Princeton Student Memorial, by Daniel Chester French.

"Architecture," relief by Richard H. Recchia.

Commodore John Barry, by John J. Boyle.

"Architecture," relief by Richard H. Recchia.

Lincoln, by Daniel Chester French.

Thomas Jefferson, by Karl Bitter. (Outside southwest arch way.)

Murals in dome of rotunda, Robert Reid. Two series of paintings, four in each, "Birth and Influence of Art," alternating with "The Four Golds of California."

"Birth of Oriental Art," panel on west wall, toward main doorway. Man on dragon attacking eagle, heavenly bird of inspiration. China, man in bright robe. Japan, woman with parasol.

"Gold," panel to right, woman with wand; sits on horn of plenty pouring gold.

"Ideals of Art," panel to right. Greek ideal, nude. Religion Madonna and child. Heroism, Joan of Arc. Material youthful beauty, woman at left. Nature without inspiration or ideal, peacock. Figures with wreath and palm, rewards of art.

"Poppies," panel to right, second gold of California.

"Birth of European Art," panel to right. Altar with divine fire, guardian with torch. Mortal in chariot grasps torch of inspiration. Woman in lower corner with crystal globe, predicting future of art.

"Oranges," panel to right, third gold of California.

"Inspiration of Art," panel to right. Angels of inspiration above. Figures of Sculpture, Architecture, Painting, Music, and Poetry.

"Wheat," panel to right, fourth gold of California.

"Priestess of Culture," Herbert Adams, of New York; female figure surmounting columns within rotunda.

Coloring of dome, burnt orange, turquoise green, Sienna columns.

South Peristyle (curved colonnade).

Youth, by Charles Carey Rumsey. (To south of doorway.)

An Outcast, by Attilio Piccirilli. (To right.)

Idyl, by Olga Popoff Muller.

Dancing Nymph, by Olin L. Warner.

Boy and Frog, by Edward Berge. (To left.)

Eurydice, by Furio Piccirilli. (To right.)

Wild Flower, by Edward Berge.

Young Mother with Child, by Furio Piccirilli. (To right.)

Wood Nymph, by Isidore Konti.

Young Pan, by Janet Scudder, (To left.)

Michael Angelo, by Robert Aitken. (To right.)

Muse Finding the Head of Orpheus, by Edward Berge. (To left.)

Flying Cupid, by Janet Scudder.

Piping Pan, by Louis St. Gaudens.

Circle at South End of Peristyle

Bust of William Howard Taft, by Robert Aitken. (To right.)

Henry Ward Beecher, by John Quincy Adams Ward.

Bust of Halsey C. Ives, by Victor S. Holm. (To left.)

Seated Lincoln, by Augustus St. Gaudens.

South of Lagoon

Kirkpatrick Monument, by Gail Sherman Corbett, Indian pointing out spring to Jesuit priest. (To right on roadway running back of palace.)

American Bisons, by A. P. Proctor. (Sides of roadway.)

Peace, by Sherry E. Fry. (To left.)

Diana, by Haig Patigian.

Fountain: Wind and Spray, by Anna Coleman Ladd. (In lagoon, south end.)

The Scout, by Cyrus E. Dallin.

Sea Lions, by Frederick G. R. Roth.

Court of Palms

Court of Palms, by Kelham; opposite Palace of Horticulture, between Palaces of Education and Liberal Arts. Italian Renaissance. Sunken garden.

Palaces at sides of court: to the west, Education; to the east, Liberal Arts.

"The End of the Trail," equestrian statue at entrance, by James Earl Fraser. Exhausted Indian, suggests destiny of the American Indian race.

Italian Towers, Byzantine influence, by Kelham. Both sides of entrance to court; identical. Simpler than towers at Court of Flowers, to cast.

Coloring of towers, by Jules Guerin. Walls frankly treated, not as stone, but as plaster, after Italian method. Blue checkered border, pink and blue diaper design; turquoise columns on little towers above, in harmony with domes and columns of Tower of Jewels.

Design on top, repeated four times at corners, from choragic monument of Lysicrates, in Venice.

Sienna columns at entrances of towers. Effective contrast.

Reclining women, purely decorative, in triangular spaces above entrances to towers, by Albert Weinert.

Figures on side of shield over all portals, very graceful. Pink and turquoise.

"The Fairy," crowning Italian Towers, Carl Gruppe.

Female figures, the caryatides on wide frieze, above columns, by Calder and John Bateman, of New York. Flushed pink, against pink and blue background of imitation marble and terra cotta.

Festoons of fruit in panels, blues and reds.

Coupled Ionic columns, smoked. Effective against pink walls.

Vases, before entrances, by Weinert. Bacchanalian revels, low relief. Satyr handles.

Lighting standards on balustrade, designed by Ryan, modeled by Denneville.

"Pool of Reflections," no sculpture.

Italian cypresses, on sides of portals.

Balled acacias between columns on corridors.

Palms, in garden.

Corridors, pink walls, blue ceiling.

Lamp standards, smoked ivory globes. Designed by Kelham, modeled by Denneville.

Lamps in corridors Roman, hanging. Light pink, green, and cream; effective. By Kelham.

Murals, in corridors, at east, north, and west portals.

"Pursuit of Pleasure," east arch, Charles W. Holloway. Light touch, bright reds and blues in keeping with court; difficult use of floating figure.

"Victorious Spirit," north arch, Arthur F. Mathews. Spirit of Enlightenment protecting Youth from Materialism, symbolized by rampant horse and the rider, Brute Force. Arrangement good, coloring deep and beautiful.

"Fruits and Flowers," west arch, Childe Hassam. Early Italian.

Symbolism, obvious. Warmth of color.

Vista from south, graceful curve of court, view through north portal through Court of the Four Seasons, long colonnade, to purple bills and bay beyond.

Palace of Horticulture

Palace of Horticulture, Bakewell & Brown, architects, San Francisco.

Architecture dome and spires Byzantine, suggest mosque of Ahmed the First, in Constantinople. Ornamentation Renaissance, popular with modern French architects.

Basket on top of dome, 33 feet in diameter.

Dome, 186 feet in height, 152 feet in diameter steel construction St. Peter's, 137 feet, concrete. Pantheon, 142 feet, concrete.

Ornamental shafts, suggestive of minarets, in French style.

Semi-circular colonnade forming entrances, French lattice-work.

Hanging lamp, in entrances, flower basket design; elaborate.

Lamps, hanging along porches, simple design.

Female figures at base of spires, by Eugene Louis Boutier; purely ornamental.

Lavish decorations on building suggest variety and abundance of California horticulture. Floral designs; green wreaths with fruit motives and leaves; lamps; flowered shields over doorway; decorated columns; entrance under green lattice-work; great ornamental vases on sides.

Female figures used as columns supporting roof of porch, the caryatides, by John Bateman.

Building suggests festivity, done in exposition spirit.

Coloring, green, old copper. Green lattice-work in domes.

Along the South Wall, West of the Tower of Jewels

South Wall, by Faville. Spanish Renaissance. Domes, Byzantine.

Palaces facing Avenue of Palms, from west to east: Education, Palace Liberal Arts, Manufactures, and Varied Industries.

Vases beside doorways of Palace of Education, finely designed; pedestal of one, a Corinthian capital; of the other, an Ionic capital.

Main portals, Faville. Suggest Roman gateway. Coloring, pink, turquoise blue, and burnt orange; accentuates sculpture. Duplicated on Palaces of Manufactures and Liberal Arts.

Panel over doorway, by Mahonri Young, Ogden, Utah; figures of domestic life and industries, making of glass, metal work, statuary, textiles. Figures at side, to left, woman with spindle; to right, man with sledge-hammer.

Flat columns at side of portals, pilasters. Corinthian.

Lion, over centerpiece of arch.

"Victory," on gables by Louis Ulrich, like the winged figure used by the Greeks, " Blessings on this house."

Niches in wall, colored pink and blue. Heads of lions and elephants used as fountains, alternately by Faville.

Panel over niches, figures with garland, by Faville.

Festival Hall

Festival Hall, Robert Farquhar, of Los Angeles, architect. Modern French architecture, of the Beaux Arts style, Paris. Used in many French theatres; not a natural growth in this country, but growing in favor; building arrangement fine. Details from Le Petit and Le Grand Trianon. Coloring. light green, not so effective as on Horticultural Palace, popular with French architects.

Figure on corner domes, "The Torch Bearer," Sherry F. Fry, of New York.

Figures on sides of shield over big central arch, by Fry. Decorative. West entrance.

Reclining figures, above, on sides of entrance, by Fry. To right, Bacchus with grapes and wine-skin. To left, a woman listening.

Groups in front of ball, on sides of stairway, by Fry. "Flora," flower girl on pedestal, repeated. On left below pedestal, "Young Pan," seated on Ionic capital covered with fawn skin, his music arrested by sight of lizard. On right, young girl seated.

Greek drinking horns, rhytons, repeated around entrance, on cornice, suggest festivity.

Symbol of Music, the lyre, above entrance.

Recital Hall, on the second floor of Festival Hall, eastern end, contains fine stained glass windows. Designer and executor, Charles J. Connick, of Boston. Three windows, a small one or, the landing of the north stairway, and two larger ones on the west wall of the hall itself.

On the stairway. Figure of a young monk bearing a scroll inscribed with "Venite exultamus domin" ("Come, let us exalt the Lord").

In the hall, window to the left. In the large tipper section, a figure of St. Martha of Bethany. Below, Christ and three women, one kneeling.

In the hall, window to the right. In the large tipper section, figures of two men, the wise men, one watching the star, one seated reading; an owl and a lantern in the window also. In the small section below, a ship with a cross on the main sail; the cross is of the design used in the Crusades.

Court of Flowers

Court of Flowers, by Kelham. Italian Renaissance, Byzantine touches. Opposite Festival Hall, between Palaces of Varied Industries and Mines. Details different from Court of Palms; ornament richer.

Figure on tower, "The Fairy," by Carl Gruppe.

Palaces at sides of court: to the west, Manufactures; to the east, Varied Industries.

Italian towers, by Kelham, same feeling. Outlines on top different from those in Court of Palms.

"The American Pioneer," equestrian statue at entrance, by Solon Borglum, of New York. Patriarchal. Suggests Joaquin Miller. Warlike trappings of horse picturesque, but sixteenth century Spanish, out of place.

Spanish loggia around second story of court, southern in feeling, implying warm climate.

"Oriental Flower Girl," female figure in niches along loggia, by Calder.

Griffons around frieze on top of columns.

Corridors, pink walls, smoked olive columns with orange capitals.

Against wall, Corinthian coupled pilasters.

Roman banging lamps, by Kelham, suggest bronze, great weight. Bronze, pink, green, and cream. Italian bronze lanterns suggest blue eucalyptus.

Lamp standards between columns, globe half concealed, by Kelham. Charm of effect, improvement on those with globe wholly visible.

Conventionalized lions in pairs at portals, by Albert Laessle, of Philadelphia.

Fountain, "Beauty and the Beast," by Edgar Walter, of San Francisco. Sandals and hat on woman. Beast at her feet. Fauns and satyrs, piping, under circular bowl. Frieze outside edge of bowl, lion, bear, ape, and tiger repeated; playful. Designed for Court of Palms to be seen from above.

Lophantha trees, trimmed four feet from ground, branching out six feet across, along walks.

Vista through fairy-like Court of the Ages to Florentine Tower and blue sky beyond, from south entrance of Court of Flowers.

Along the South Wall, East of Tower of Jewels

Palaces facing Avenue of Palms, from east to west: Varied Industries, Manufactures, Liberal Arts, Education.

South facade of Palace of Varied Industries, by Faville. High walls, seventy feet in height, suggest eighteenth century California missions.

Green domes on corners, Byzantine, inspired by mosques of Constantinople.

Coloring of flags, cerulean blue, pastel red, and burnt orange.

Windows in corners, mosque design. Little hexagonal kiosks at corners below domes, Moorish.

Central portal, after portal of Santa Cruz Hospital, in Toledo, Spain. Sixteenth century Spanish Renaissance, plateresque. Lattice-work effect in doorway in harmony with lace-like silver-platter style. Niche walls pink, with ultramarine blue.

Pope Calixtus III sent for a Spanish goldsmith, Diaz, to do work for him in Rome. Diaz returned to Spain, carrying the influence of the Italian Renaissance. He met the son of the architect of the cathedral at Toledo, De Egas. To the son he imparted his knowledge and the son applied it to architecture, creating the plateresque style. Till then all Spanish cathedrals had shown the Gothic influence from the north.

Figures on large door by Stackpole. Upper figures, "Age Transferring His Burden to Youth," America. Figure in center piece of arch, "Power of Industry," the American workman. Figures in half circle above door, "Varied Industries," from left to right, Spinning, Building, Agriculture, Manual Labor, and Commerce. Figure repeated four times in lower niches, "Man with the Pick."

"California Bear" and "California Shield" on buttresses, or square columns supporting wall. Used in old mission buildings.

Avenue of Progress

Planting, some of the best landscape effects in Exposition. Against buildings, Monterey cypress; banked by Lawson cypress in front and between these, spruces and Spanish fir.

Machinery Palace, Ward & Blohme, of San Francisco, architects. Italian Renaissance, inspired by Roman baths. Like Baths of Caracalla. Largest building of its kind in world; three blocks long, seven acres in area.

Banners, by Ryan, heraldic designs of early Spanish explorers and soldiers.

Lophantha lawn, designed by John McLaren, trees trimmed off four feet above ground, and trained to grow flat alongside Palace of Varied Industries.

East facade of Varied Industries, made Italian to harmonize with Italian Machinery Palace.

Main portal, like gateways of old Roman walled cities.

"The Miner," in niches of gateway, by Albert Weinert of New York.

Small portals Italian, fine color effect; lattice-work, orange, blue, light green'.

Sculpture on Machinery Palace, by Haig Patigian, of San Francisco.

Large columns in front and in vestibule of half dome, imitation Sienna marble.

Small portals, orange columns at sides, pink niche, blue dome, orange above dome; pleasing tone,

Corinthian columns at sides of portals; eagles at corners of capitals, at top, symbolize inspiration.

Frieze around drums at base of columns "Genii of Machinery," by Haig Patigian; eyes closed, signifying Power of the spirit, or blind fate.

Figures in triangular spaces on either side above doorways, "Application of Power to Machinery," by Haig Patigian.

Figures on tall Sienna marble columns, "Power," "by Haig Patigian. "Steam Power," with lever. "Invention," carrying figure with flying wings, suggesting quickness of mind. "Imagination, eyes closed. Eagle bird of inspiration, about to fly. "Electricity," foot on earth, carrying symbol.

Eagles repeated on bar, the entablature, across front of domes; symbol of inspiration.

Coloring in vestibule of Machinery Palace: Finely harmonized; brown and brick-colored walls; orange and blue ceilings; green lattice work.

"Genius of Creation," group before court leading to Court of Ages, Daniel Chester French. Spirit above, a woman, creating life from shapeless mass of earth below. Man at left, courageous and enterprising; woman at right, timid, hesitating. Serpent, symbol of wisdom, coiled about mass.

Court of Mines, Leading to Court of Ages

Coloring, pink walls, pink streamers, by Guerin. Green shell lamp posts, by McKim, Mead & White, architects. Called "Pink Alley" by workmen during construction.

Palaces on sides of court: to the north, Mines; to the south, Varied Industries.

Lamp standards against walls, dark bronze, smoked ivory globes, by Faville.

Flat Ionic columns, called pilasters, against walls, by Faville.

Figure in niches, "The Miner," by Albert Weinert.

Court of the Ages

Court of Ages, Louis Christian Mullgardt, of San Francisco, architect. Most original of the courts. Faint influence of Spanish Gothic, Romanesque, French, Moorish. Richness and profusion. Suggests evolution of man.

Palaces around court: northeast, Mines; northwest, Transportation; southwest, Manufactures; southeast, Varied Industries,

Decorations on columns of archways around court, kelp, crabs, lobsters, and other sea animals. Vertical lines in columns suggest falling water.

Fairy lamps, two in each archway, delicately designed.

"Primitive Man and Woman," by Albert Weinert, repeated alternately above corridors around court. Man, a hunter, feeding pelican. Woman, the child-bearer.

Tower at north entrance, suggestive of French cathedral architecture, massive, but gives appearance of lightness. One of the great successes of the Exposition.

"The Rise of Civilization," groups of sculpture on tower, by Chester Beach. Central idea, evolution, Stone Age, Mediaeval Age, and Present Age. "Primitive Man," lowest group, just above great reptiles in foreground. Man is holding child and protecting mate. "Mediaeval Age" directly above, Crusader in center, Priest and Warrior on sides. The candlesticks on sides of crusader, used in mediaeval churches, the light of understanding. On sides of altar, "Modern Man and Woman," struggling for freedom from the physical to the spiritual. "Spirit of Intelligence" enthroned above; on one side, child with book; on the other side, child with wheel of industry.

Chanticleer, repeated on highest pinnacles of court, at level with altar. Signifying dawn of Christianity.

"Thought," figure on east and west sides of tower. Candlesticks at sides.

Design on upper part of tower, suggested by the lily, emblem of purity.

Star clusters, at south end of court and in north court, by Ryan, modeled from snow crystal, and deepening the ecclesiastical character of the court by suggesting the golden monstrance, shaped like the rays of the sun, used in the Catholic church and, in the small glass-covered circle at the center, holding the sacred host.

"Water Sprites," by Leo Lentelli. Girl archers on top of columns at four corners of central court, launching arrow at sprites on base of columns. Originally designed as fountains.

Serpent cauldrons, around pool, designed by Mullgardt.

"Fountain of the Earth," by Robert Aitken, in center of court. Two Parts to fountain; large central one with globe representing earth, surrounded by panels showing life on earth; and on same pedestal to south, groups representing life before and after death. "Setting Sun," group at extreme south of pool, by Aitken. Man holding golden ball, Helios; serpent, heat of sun.

Figures on west side of southern group, "The Dawn of Life." Hand of Destiny giving life, pointing toward earth; Sleep of Woman before Birth; the Awakening; Joy of Life; Kiss of Life; Birth. Gap to central group represents time between peopling and history.

Panels around earth; South Panel; Vanity in center with handglass; man and woman with children, representing Fecundity, starting on earthly journey.

West Panel: "Natural Selection;" women turn to fittest male; one rejected suitor angry, other despairing.

North Panel: "Physical Courage" or "Awakening of War Spirit." Two men fight for possession of woman on left. Woman on right attempts to draw one aside.

East Panel: "Lesson of Life." Old woman gives counsel to young man and woman. Old man restrains an angry, jealous youth.

Right of south panel, "Lust."

East side of southern group: Greed, looking back on earth. Faith offering Immortality, symbolized by scarab, to Woman. Figures of man and woman sinking back into oblivion, "Sorrow" and "Sleep." Hand of Destiny drawing mortality to itself.

Hermae, pillars with head of Hermes, god of boundaries, separating panels around earth.

Reptilian and fishy forms above panels of central mass of fountain.

Corridors, walls red, blue vault above, arches of smoked ivory, lines of blue on wall. Illumination by half-globes in cups on inner side of columns.

Murals, by Frank Brangwyn, of London, representing Elements. Best placed of all murals. At corners of court in corridors.

Northeast corner, "Fire." "Primitive Fire," figures around fire nursing it, or feeding it. "Industrial Fire," use of fire in service of man.

Southeast corner, "Water": Fishermen dragging in net, carriers with baskets on backs, "The Net." Women and men filling jars at a spring, flamingoes in water, luxuriant growth, clouds, "The Fountain."

Southwest corner, "Air": Men shooting arrows through trees, birds in flight, "The Hunters." Huge mill, children flying kites, clouds, grain blown by wind, "The Windmill."

Northwest corner, "Earth": Men high in trees and on ground, "The Fruit Pickers." Figures crushing juice out with feet, group in front with wine, "The Dancing of the Grapes."

Planting in Court: Tall Italian cypress before arches; orange trees; balled acacia; denseness of growth along colonnades; heavy and rank, suggesting tropical flora.

Large cauldrons, at side of steps leading down to sunken gardens, designed by Mullgardt.

North Entrance to Court of Ages

"Daughter of Neptune" or "Aquatic Life," large female figure in north Court of Ages, by Sherry E. Fry.

Planting: eucalyptus, acacia, laurel.

Features that Ought to be in Noted by Night

Illumination

Three kinds of light used; white arc lamps, extensively behind banners and shields to flood facades of outer walls and Court of Four Seasons; warmer light of Mazda lamps in clear and colored globes; and searchlights concealed on tops of buildings trained on towers and on high groups of sculpture.

Lighting scheme and scope completed long before buildings were up; made possible by advance in illuminating engineering, developed under name of science of lighting and art of illumination.

Chief of Department of Illumination, Walter D'Arcy Ryan, of the General Electric Company, Schenectady, New York; field assistant, A. F. Dickerson.

Ornamental details of lighting standards and fixtures, designed by J. W. Gosling; designs made at Illuminating Engineering Laboratories, Schenectady.

Keynote of lighting scheme - life and gaiety, without garishness.

Lighting kept subordinate to architecture; walks shaded to throw emphasis on brilliantly lighted facades and to bring out architecture, landscape and flowers. Same lighting principle used throughout; but effect in different courts radically different.

Area of surface illuminated, 8,000,000 square feet; 2,000,000 of wall surface, and 6,000,000 of ground surface.

Number of searchlights used: 373 arc searchlights, in diameter from 13 to 36 inches; 450 small searchlights, called the "Mosquito Fleet"; 250 incandescent projectors for flag lighting.

Fillmore Street Entrance

South facade of entrance, outline illumination, with bare electric lights following outlines of architecture; used elsewhere only in Zone.

Inside Fillmore Street entrance, Zone to right; brilliant lighting, outline illumination, more or less refined; exaggerated effects prohibited.

Zone, element of festivity in arches crossing street at short intervals, ribbons of turkey red suspended from each lamp give warmth and action.

Contrast of Zone lights with illumination in other parts of Exposition.

To left, Service Building, administration offices; coloring, Pinks and blue; ceiling of porch, intense blue, deepest used on grounds.

Corner of Avenue of Palms and Avenue of Progress: lights banners, towers, facades of buildings, walks, flood lights, spots of light and color.

Fairy-like effect of Avenue of Palms: towers look luminous; in early evening Italian Towers red hot, throbbing; glow stronger than Tower of Jewels; later, Tower of Jewels most brilliant spot on avenue.

Tower illumination, floods of light from searchlights; white light creates shadows, in turn illuminated by concealed colored light on various stages, on Tower of Jewels and Italian Towers.

Single light standards along Avenue of Palms, light yellow, dull points of light; contrast with white pearly light on tops of booths.

Avenue of Progress

Along Avenue of Progress: fine flag display; no direct sources of light; banners; beautiful scenes made by planting against walls and quality of green on lawn; daylight effect from luminous arcs which produce whitest artificial light in use.

Gas lights on tops of booths, emergency lights if electricity fails.

Banners and heraldic shields, designed by Ryan; banners, of early explorers and pioneers, heraldic shields related to history of California, Mexico, Central America, and Pacific Ocean.

Purpose of banners: to form beautiful lines of color, to screen eyes from direct light source, to reflect light toward buildings, and to suggest history of court.

Banners suspended, swung by wind, form moving spots of color.

Roman gateway, Palace of Varied Industries: faint light through small arches above doorway; delicate green lattice or grill work in door.

Light in doorways: appearance of life within, produced by reflectors inside palaces throwing light through glass of doors; new idea; contrast with dark windows of other expositions.

Arches of Machinery Palace: warm red glow in domes above; strong yellow through doors below.

Inner Court of Mines Leading From Palace of Machinery to Court of Ages

Illumination strongest on upper sections of wall; it becomes more subdued as it approaches flowers and lawns, and reaches lowest point on center of avenue; plan used on all avenues.

Green lattice work, filling entire main doorway, in harmony with lawns.

Single globe lamps placed against walls; only court with lights in this position.

Shell lamps, flooding walls with light, advanced method of wall illumination.

View of central fountain in Court of Ages: glow of red lights, faint shimmer in pools, steam rising to suggest the earth cooling after being thrown off by the sun.

Court of the Ages

Court of the Ages: mystery in blending of illumination from searchlights above; lack of direct illumination on court itself; steam cauldrons, with illumination, incandescent lights, gas torches in small serpent cauldrons, lanterns in arches of the arcade that burn around cloister.

Fountain of Earth in center of pool, carrying mind down the ages to correspond with architect's conception of court.

Steam rising from base of fountain; figures silhouetted in warm red glow; lighter tone of red at upper portion of ball; shimmering reflection of panels, with red background in pool at sides of fountain.

Serpent cauldrons, around edge of pool, to heighten weird effect, by the flickering of the gas lamps.

Large cauldrons at east and west entrances; effect of simmering molten liquid.

Steam used in court, obtained from twenty horse-power boiler under tower.

Main tower, only tower without direct light thrown on exterior; religious feeling, increased by candlesticks, two on each side; steam to suggest smoke drifting upward.

Reflection of tower in pool, to be seen from south.

Cathedral appearance of windows at sides of court, by illumination in warm orange tone from within.

Sunburst standards modelled in imitation of snow crystal, and resembling monstrance used in Catholic church; two at south of court; only large light sources in court; contrast with other illumination.

Two fairy lanterns in each arch around court.

Brangwyn murals lighted without glare by indirect diffusion from four corners.

Play of lights along colonnade; lighting on murals adds to apparent distance.

North Entrance to Court of the Ages

Similar treatment of lights, brighter than in central court; four star clusters, sixteen serpent cauldrons; effect heightened.

Tower, more beautiful from Marina side; note of refinement illumination in altar, shadow in two colors, created by red light illuminated by pale amber lights.

Star clusters convey to mind religious feeling in keeping with design; cathedral effect.

View of Italian Towers at sides of Court of Flowers, from north court, red glow and green columns of towers on either side of Mullgardt tower, vivid contrast.

To Court of the Universe, through Florentine Court.

Florentine Court

Florentine Court; only illumination, single lamp standards; contrast with intense light in Court of Universe, beyond.

Fine shadow effects against walls; vertical shadows of columns in arches contrasted with shadows of trees and shrubbery.

Court of the Universe

Arch of Rising Sun; light through latticed windows in arch to give faint spots of luminous color.

Illumination of main and side arches; curvature preserved and details thrown into relief by lights of different strengths and colors; concealed red light on one side and pale lemon light on other side thrown on arch. All main arches similarly accentuated.

Urns in side arches, effect heightened by lights thrown from sides, bring out lines; red on one side, on the other pale green.

Colonnade, illuminated by three translucent shell cups sunk into central groove of each column at rear; spear of light from each shell up the grooves or fluting; pleasant glow through shells from below. Effect of melted gold, suggesting the tongues of fire mentioned in the Scriptures.

Sculptural groups on Arches of Rising and Setting Sun, flooded with light from searchlights, creating black shadows, in turn illuminated by purple lights on top of arch. Figures thrown into relief.

Tower of Jewels, gradual illumination; early evening, faintly lighted; later, when searchlights are turned on, tower dominates southern wall; blaze of white light; jewels sparkle like diamonds; turquoise columns, faintly colored in bright light; statues, orange color.

Star figures around court above colonnades, jewelled; each has forty-two stones, illuminated by small searchlights on opposite side of court. Early evening, pretty effect; little jets of light from figures shoot across the court in clearly defined rays. Later, flood of lights from columns in court above the small rays.

Fountains of Rising and Setting Sun; columns, said to be strongest light sources ever created; aggregate 500,000 candlepower sufficient to illuminate 500,000 square feet of surface; fluting of columns glazed with special diffusion glass. For elimination of shadows caused by structure, there is diffusive glass inside. The glare from the light source is not excessive; brilliancy low; daring illumination of entire court.

Lights under water in pools of fountains; source and reflection concealed; yellow light diffused over surface.

Figures on pedestals of balustrades mark boundary of Sunken Garden; not for illumination, but for ornament merely.

Domes of corner pavilions, north of Tower of Jewels, fine contrasts in interior; delicate blue ceiling; orange at sides.

Bear fountains at sides of Palaces of Manufactures and Liberal Arts, north of Tower of Jewels; three on each wall in flat niches; coloring, pink wall, turquoise blue, green; lights concealed under water; when water is flowing, wavering light like heat waves; niches hardly noticeable when water is not flowing.

Tower of Jewels, interior of main arch, accentuated by lights at sides above columns; no illumination on murals.

In niches at either side, Fountains of Youth and El Dorado, flood-lighted from above; no colored lights; two single lamp standards in each court; reflection of fountain figures in pools.

On the Way to the Marina

Lighting of colonnades, vivid pinks and blues. Illumination in colonnade from lamps concealed in cups in one of the inner flutes of each column. Notice reflections of colonnade in pool.

Column of Progress; flood light on figures on top of column by searchlights.

On the Marina

North facade of buildings, tall dark-green planting against walls, black vertical shadows; shading of lawn; flood light standards, spots of dull orange light through translucent rigid shields. Spots of light from single globes along avenue, on water front, white lights on booths; glow from lamps at entrance to Court of Four Seasons.

Spanish doorway of Palaces of Food Products, Agriculture, Transportation and Mines, among most successfully illuminated portals on grounds; light pink walls in two shades, light blue vaulted ceiling, green edges; three arches; light green lattice work; dark shadows in niches of "Conquistador" and "Pirate."

"Adventurous Bowman," profile view of group from entrance to Court of Four Seasons; outlined against blue-black sky; stars, in sky about it, mere points of light. Group sometimes reproduced in the fog.

Venetian Court

Inner Court, between Court of the Universe and Court of Four Seasons.

Only illumination, single globe standards. Contrast of bright illumination in Court of Universe with more subdued light in Court of Four Seasons.

Coloring, pink walls in harmony with walls of corridors in Courts at either end.

Planting, low shrubbery, with tall trees massed in corners.

Court of the Four Seasons

Court of Four Seasons; flood illumination on the bulls at sides, glowing half-dome at south, figure of "Harvest" above dome, and twin Italian towers at sides.

Illumination of court in harmony with architecture, very quiet.

Charm of lighting in colonnades against Pompeian red walls; three half globes in cups at rear of plain columns.

Fountains of Four Seasons, illumination of red walls against intense blue of sky, in early evening like color in paintings by Maxfield Parrish. Concealed lights, red, orange, yellow and lemon, fall on walls and create interesting luminous shadows on fountain figures.

Water falling from cascades, a luminous green; not only are lights concealed, but also reflection of sources, an effect that, it was predicted, could not be achieved.

Figures on fountains reflected in green water.

Reflections in pool in center of court; from north, half dome and figure of "Harvest" above dome; from south, the bulls on the pylons.

View through north court toward bay, from half-dome, very interesting; intense white light of scintillator directly opposite court; statute of "Ceres," silhouetted against rays.

Banners in court, no heraldic designs.

Half dome in Court of Four Seasons; even distribution of light, ceiling lighted from base of dome, lights diffused through dome and softly graded down to floor by ten shell lamps up wall, back of vertical projection on each side.

Through Aisle of Spring to Administration Avenue, facing Palace of Fine Arts.

Along the Western Wall

Illumination: Yellow glow from single lamp standards along Administration Avenue. Searchlights on top of wall, flooding Palace of Fine Arts. Wall, lighted by reflection from shields; orange light through translucent portion of shields.

High wall flooded with light, in strong contrast with dark rippling surface of lagoon across the avenue.

Half-domes; warm golden glow; light from interior through stained glass windows in domes.

Planting, trees cast tall vertical shadows against wall; heavier shadows at base, from massed shrubbery.

Palace of Fine Arts

Illumination, "triple moonlight," three times the strength of the moon's rays. Searchlights flood the building; concealed yellow lights on cornices in rear of columns. Three effects; flood lighting, relief lighting, and combination of both. One night, flood light; next, combination.

View from Administration Avenue across lagoon; finest reflections on grounds; changing views; small sections of lagoon, mirror-like; others, rippled or wavering; entire colonnade and rotunda reflected.

Suggestion of ancient ruin, intended by architect, brought out by lighting. Great shadows, deepening toward base of columns.

Contrasted colors in colonnade, from across lagoon; pink walls, dark green doors, columns silhouetted against walls.

In the Colonnade, Entering From North

"Triple moonlight," bright rays across colonnade through columns, making intense shadows; when moon is shining the fainter rays cut weirdly through shadows; suggestion of moonlight coming from two directions.

Reflections in lagoon, from along colonnade, north of rotunda; West facade of walled city, with half domes of Palaces of Education and Food Products, and dim reflections of Italian towers. Changing reflections all along colonnade, and from rotunda.

Rotunda, on nights when relief illumination is used, lights on capitals of Corinthian columns; deep color effects in murals on dome.

View of palace from south across lagoon, with flood lights on rotunda and colonnade.

Avenue of Palms

Quality of light brings out color detail; fine display of flowers; massing of shrubbery at base of wall, and tall trees casting vertical shadows.

Elephant and lion fountains along south wall; colors, pink and blue; rippling of water causes light to wave.

Central doorway of Palace of Liberal Arts, rosetta or rose-window effect in semi-circular space above door; orange light through lattice work of door.

Court of Palms

Court of Palms, illumination of towers from searchlights. Only direct light, from single white globes painted to imitate Travertine, and Roman hanging lamps around in corridors; faint red shines through from below.

Reflections in circular and rectangular pools; north, east, and west portals; the columns, the colonnades at sides of entrances, the murals above doorways; pinks, blues, reds, orange.

Murals above east, west, and north doorways, best effect at night. Illumination at base of arches throws light on upper part of mural, shading softly and gradually down to base.

Palace of Horticulture

Dome of Palace of Horticulture; beams of light from concealed searchlights play through revolving lenses and color screens of green, orange, and red, fading slowly into each other in moving designs on glass dome.

Floral hanging lamps in east and north entrances; deep green of lattice work in domes above; hanging lamps along porches, pearl-white light.

South Gardens

French lighting standards, pale yellow light, hundreds of Travertined globes, soft light unique ivory color.

Clusters of lights- look like bunches of grapes.

Reflections in pools north of Young Women's Christian Association Building and Press Building

Flood lights on equestrian figure in Fountain of Energy.

Court of Flowers

No searchlights, no direct illumination; suggestion of dimness and seclusion.

Italian towers, glow of light through small doors above entrances; appearance of life inside; strong strong red shadows on first lift; turquoise columns on next lift, pink background.

Lamps in corridors, Italian and Roman; translucent, dull red light.

Floral lamp standards between columns in corridors, pale yellow light.

Flood light shields at south entrance to court; too bright necessarily.

Festival Hall

Reflection of Festival Hall in pool; Fountain of the Mermaid silhouetted against entrance window of hall; golden light through colored glass.

Warm pink illumination inside towers at corners of large dome; green coloring of dome, more effective than by day.

Blending of lines of building with planting against walls.